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INTRODUCTION

Urolithiasis (nephrolithiasis, kidney stone disease) is the abnormal formation and retention of solid phase inorganic and organic concretions in the lumen of the urinary tract. Kidney stone is not a diagnosis per se but the common manifestation of a variety of underlying causative and pathophysiologic factors. Although stones are localized to the urinary tract, urolithiasis is a systemic disease. Although the surgical treatment of urolithiasis has advanced over the years, the management extends beyond the mere removal of existent stones. The need to understand how stones form is of critical importance so that they can be prevented from recurring, which is a most important objective. Overall, the metabolic evaluation of urolithiasis is still not performed with enough frequency and depth in our opinion. In addition to the potential for uncovering treatable underlying diagnoses, the pathophysiologic definition of kidney stones also guides the selection and helps the monitoring of therapy.

EPIDEMIOLOGY

GENERAL POINTS

KIDNEY STONES IN ADULTS: 1976–1994

The prevalence of kidney stones has increased steadily over 4 decades,^{1,2} which is associated with a rise in both direct and indirect expenditures for this condition in the United States.³ The estimated economic burden in the United States for kidney stone disease management is estimated to be \$4.7 billion by 2030.⁴ This is in part because it is estimated that more than 44% of adults are expected to be obese in the United States by 2030.⁴ In 1994, data from the US National Health Examination Survey III showed a rise in the prevalence of self-reported history of kidney stones compared with the period 1976 to 1980 (from 3.8% to 5.2%). The increase was

greater in females than males and higher in the aging population.¹ African-Americans had a lower risk of urolithiasis compared with whites and Mexican Americans. Age-adjusted prevalence was higher in southern parts of the United States.¹ This is consistent with cross-sectional studies showing a higher prevalence of kidney stones in Southeastern parts of the United States.^{3,5}

Previous studies explored the model that environmental factors and systemic conditions, including quality of the water supply,⁶ climate,⁷ animal protein intake, and association with hypertension⁸ may be important in increasing the risk of kidney stone disease, in addition to geographic distribution. The contribution of mineral content and the quality of the water to the prevalence of kidney stones and its geographic distribution was questioned by a study in three Midwest regions of the United States, which showed no correlation between water calcium content and the prevalence of kidney stone disease⁹; the role of other minerals were not addressed. Prevalence data from 1988 to 1994 was not able to identify dietary factors that could explain the geographic variation in the prevalence of kidney stone diseases.¹ The lack of association between diet and kidney stone prevalence may be because of the cross-sectional nature of the studies and lack of temporal association between stone disease and time of the data collection.

KIDNEY STONES IN ADULTS: 2007 TO 2010

A more recent National Health and Nutritional Examination Survey (NHANES) cross-sectional study from 2007 to 2010 of 12,110 subjects with a self-reported history² has demonstrated a marked increase in the prevalence at 8.8%, compared with the aforementioned 1994 survey with a prevalence of 5.2%. The latest estimate shows that 1 in 11 individuals in the United States has a history of kidney stones, in contrast to a previous figure of 1 in 20 U.S. citizens.¹ The overall prevalence was 10.6% in male subjects and 7.1% in females compared with the previous study,¹ which showed 6.3% in men and 4.1% in women² (Fig. 38.1).

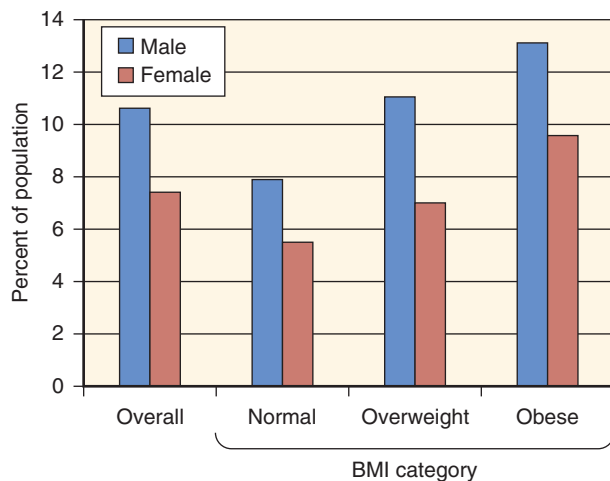
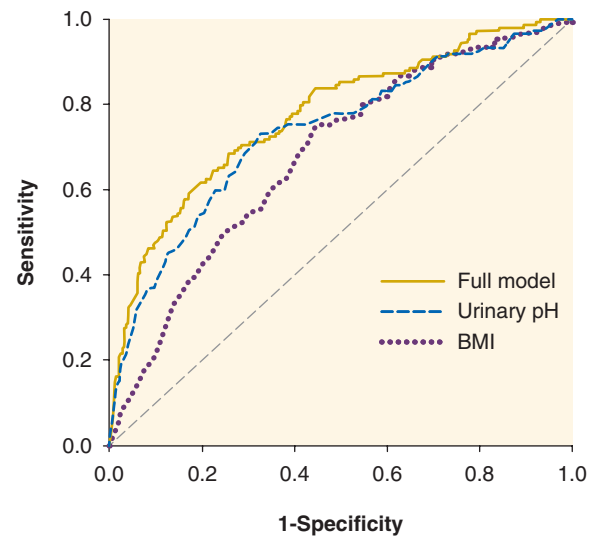


Fig. 38.1 Prevalence of urolithiasis plotted as a function of gender and body mass index (BMI). (Modified from Scales CD Jr, Smith AC, Hanley JM, Saigal CS. Urologic diseases in America. Prevalence of kidney stones in the United States. *Eur Urol.* 2012;62[1]:160–165.)

Aside from gender, race, age, ethnicity, and socioeconomic class, conditions associated with the metabolic syndrome are predictive of kidney stone disease.² Again, the likelihood of kidney stones was lower among black and Hispanic than white populations. Furthermore, obesity, diabetes, gout, and household income of \$19,999 or less were more likely to be associated with kidney stone disease. The association between features of the metabolic syndrome, such as obesity and diabetes, and prevalence of kidney stone disease is consistent with a prospective study showing that the risk of incident kidney stone disease increases with obesity and weight gain.^{10,11} A multivariate Cox regression model showed that obesity is the only strong predictor of stone recurrence in first-time stone formers.¹²

Using pooled data from 1965 to 2005 from seven countries, Romero and associates have shown a continuous increase in prevalence throughout the decades, with similar gender and age distribution profiles across countries within each time period.¹³ In general, the prevalence of kidney stone disease in adults is higher in the Western Hemisphere.¹⁴ However, the highest kidney stone prevalence was in Saudi Arabia.¹⁴ As discussed, factors such as the rise in prevalence in metabolic syndrome and dietary changes have been suggested to underlie this rise. Brikowski and coworkers¹⁵ took a different and interesting approach using global warming trends and modeled the northward expansion of the U.S. “stone belt” in the coming decades, which predicts a dramatic increase in stone prevalence in the northern United States. In one recent cross-sectional study of 63,994 patients who had kidney stone–related procedures in California from 2010 to 2012, the authors described the association between higher precipitation and temperature and an increased rate for kidney stone intervention.¹⁶ The authors concluded a larger trend in kidney stone prevalence with a warm wet climate than a warm arid climate.

Despite the increased global prevalence of kidney stones in recent decades, the association between changes in kidney stone composition, demographic characteristics, and biochemical profiles has not been fully explored. In



Model	AUC	95% Confidence interval
Full model	0.77	0.73–0.82
Urinary pH	0.74	0.68–0.78
BMI	0.68	0.64–0.72
Age*	0.65	0.61–0.70
Time periods*	0.57	0.52–0.61
Gender*	0.55	0.51–0.58
Creatinine clearance*	0.54	0.49–0.59

Fig. 38.2 Receiving operating characteristic curve and area under the curve (AUC) for clinical predictors of uric acid stone formation. The full multivariable model included urinary pH, body mass index (BMI), time period, gender, and creatinine clearance. Urinary pH and BMI were determined by univariable analysis; asterisks indicate variables that are not shown. (Modified from Xu LHR, Adams-Huet B, Poindexter JR, et al. Temporal changes in kidney stone composition and in risk factors predisposing to stone formation. *J Urol.* 2017;197[6]:1465–1471.)

one retrospective study of 1516 patients over 35 years with calcium or uric acid stones, the proportion of uric acid stones increased from 7% to 14%.¹⁷ The proportion of females with kidney stones also increased over time, but the increase in the female gender was more significant among calcium stone formers. Both age and body mass index (BMI) increased with time in both uric acid and calcium stone formers; however, uric acid stone formers were consistently older and had a higher BMI and lower urinary pH than calcium stone formers. Urinary pH, phosphorus oxalate, and sodium increased with time in calcium stone formers, but remained unchanged in uric acid stone formers. The most significant denominators for discriminating uric acid versus calcium stone was urinary pH¹⁷ (Fig. 38.2). This study was corroborated with another cross-sectional retrospective study performed in the United Kingdom in 2132 patients with kidney stone stratified to normal weight, overweight, and obese.¹⁸ In this study, both obese and overweight kidney stone formers had a greater risk, principally because of an increase in urinary promoters, including calcium, sodium, uric acid and lower urinary pH, but without any change urinary inhibitors of kidney stone formation, such as citrate and magnesium.¹⁸ Furthermore, the higher incidence of uric acid but not calcium stones was demonstrated in overweight kidney stone formers.

CALCIUM STONES

PREVALENCE OF CALCIUM STONES

In one study, kidney stone composition was evaluated using data from the National Veterans Administration Crystal Identification Center.¹⁹ A comparison with a previous study showed an increased percentage occurrence of calcium oxalate and brushite stones between 1996 and 2003. Furthermore, with each recurrent stone event, there was an increased percentage occurrence of calcium phosphate stones accompanied by a decrease in occurrence of calcium oxalate stones (Fig. 38.3).¹⁹ The underlying factors for the change in composition are not known. A retrospective evaluation of 1201 stone formers in the past 3 decades has shown that over time, an increased incidence of calcium phosphate stones coincides with increased urinary pH and the number of shockwave lithotripsy treatments.²⁰ Causality between stone risk and lithotripsy and urinary pH still remains to be determined. A study from France has shown an increased prevalence of calcium phosphate stone in men versus women from 1980 to 2004, from 9.4% to 26.6%,²¹ and the increase was associated with a gradual rise in BMI during this period.²¹

CALCIUM INTAKE

Several large studies have explored the association between dietary factors and the risk of kidney stone disease. The role of dietary calcium as a risk factor was assessed in a few prospective observational studies showing that low dietary calcium intake is associated with a higher risk of kidney stones in women and young men.^{22–25} On the contrary, calcium supplementation was associated with a higher risk of kidney stones solely in older women.^{24,26} The epidemiologic association of low dietary calcium with nephrolithiasis is consistent with the findings of a randomized controlled study in hypercalciuric men with calcium nephrolithiasis.²⁷ This study will be discussed later.

VITAMIN D INTAKE

Vitamin D deficiency is endemic worldwide, influencing approximately 1 billion people.^{28,29} Vitamin D deficiency is

highly prevalent in kidney stone formers, of which 20% are hypercalciuric.^{30,31} Thus, clinicians are reluctant to treat vitamin D deficiency in kidney stone formers because of the concern with increased kidney stone risk. The studies addressing the relationship between vitamin D supplementation and hypercalciuria are limited and contradictory.^{32–37} The controversy arose in part as a result of the small sample size of the retrospective studies, nonadherence to the vitamin D supplement, and dietary indiscretion of salt and calcium intake. However, given these shortcomings, and in the absence of prospective control trials, the consensus is that among kidney stone formers with vitamin D deficiency, a short course of vitamin D repletion does not increase urinary calcium excretion, except in the subset of hypercalciuric patients.^{31,35} Thus, vitamin D repletion should not be avoided in the kidney stone population, but urinary calcium excretion should be assessed during the repletion period.

OXALATE INTAKE

One epidemiologic study using a food frequency questionnaire in men, older women, and younger women has shown that oxalate intake does not correlate with kidney stone disease.³⁸ One has to exercise caution in drawing conclusions about such negative studies, given the limitations of questionnaires, the heterogeneity of cause among the stone formers, and the magnitude of the effect of dietary oxalate.

PROTEIN CONSUMPTION

Epidemiologic studies have demonstrated a positive relationship between animal protein consumption and kidney stone formation in men, but not as much in women.^{22,24,26} However, in a randomized controlled trial (RCT), consumption of a low-animal protein and high-fiber diet was not shown to reduce the relative risk of recurrent kidney stones in calcium oxalate stone formers compared with those instructed to maintain a high fluid intake.³⁹ Nonetheless, the pathophysiologic basis and metabolic studies supporting the relationship between dietary protein and stone risk is very solid, and one interventional study, where multiple variables, including dietary protein were manipulated, showed benefits from protein restriction.²⁷

URIC ACID STONES

PREVALENCE OF URIC ACID STONES

There are regional differences in uric acid stone prevalence. The highest prevalences are reported in the Middle East and a few European countries,^{40–42} whereas uric acid stones comprise only 8% to 10% of all kidney stones in the United States.⁴³ In the United States, uric acid stones among Chinese and Japanese descendants have been reported to be slightly higher than in the general population at approximately 15% to 16%.⁴⁴ The highest prevalence of uric acid stones and gouty arthritis has been demonstrated in the Hmong population of East Asian and Southeast Asian heritage living in Minnesota.^{44,45} The factors underlying these differences are not known. The high prevalence of uric acid stones in the Hmong population is unique, given that a low prevalence of reported uric acid stones is reported in Northeastern Thailand, which is geographically adjacent to Laos, from which the Hmong population originated.⁴⁶

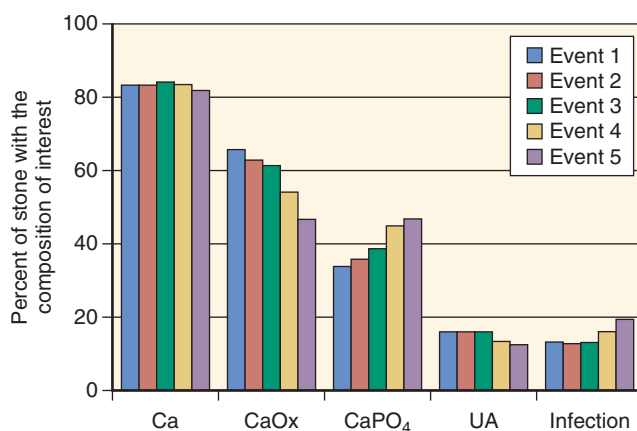


Fig. 38.3 Percentage occurrence of stone types. Calcium (Ca), calcium oxalate (CaOx), calcium phosphate (CaPO₄), uric acid (UA), and infection stones as function of event; that is, ranging from the first to the fifth stone. (Modified from Mandel N, Mandel I, Fryjoff K, et al. Conversion of calcium oxalate to calcium phosphate with recurrent stone episodes. *J Urol.* 2003;169[6]:2026–2029.)

GENETIC AND DIETARY FACTORS

The underlying causes for the global differences in uric acid stone prevalence remain unknown. However, a high prevalence of obesity, diabetes, and hypertension is commonly associated with uric acid nephrolithiasis in Western societies^{47–49} and has also been shown to be present in Hmong populations born in the United States. Therefore, it is likely that an interplay between genetic factors and diet have an impact on uric acid stone formation. Among stone formers in the United States and Europe, uric acid stones occur more frequently in subjects with type 2 diabetes mellitus (T2DM) and obesity than in nonobese, nondiabetic stone formers (Fig. 38.4).^{48–52}

DATA ON STONE INCIDENCE

Compared with prevalence, precise knowledge of kidney stone incidence among adult and pediatric populations is limited. In adult populations, one small study from Minnesota has shown an incidence rate of 101.8/100,000 patients per year.⁵³ Two studies using a questionnaire found an overall incidence rate among adult males and females of 306 and 95/100,000 person-years, respectively.^{22,26}

One study from South Carolina used data from pediatric emergency visits and deduced that the incidence of nephrolithiasis increased from 7.9 to 18.5/100,000 children (≤ 18 years of age) in a little over a decade.⁵⁴ It has been suggested that childhood obesity may play a key role in the rising kidney stone incidence of children in recent years. However, larger U.S. epidemiologic studies do not support this notion.⁵⁵

Dietary factors, including increased sodium consumption⁵⁶ and decreased calcium intake⁵⁷ from milk, which has been substituted by sugar drinks,^{58,59} have also been implicated to

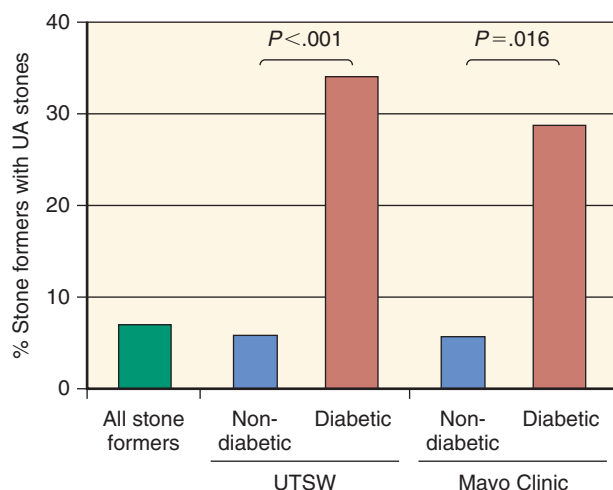


Fig. 38.4 Prevalence of uric acid urolithiasis in the kidney stone population. Mayo, Mayo Clinic; UA, uric acid; UTSW, University of Texas Southwestern Medical Center. (From Lieske JC, de la Vega LS, Gettman MT, et al. Diabetes mellitus and the risk of urinary tract stones: a population-based case-control study. *Am J Kidney Dis.* 2006;48[6]: 897–904.)

have a pathophysiologic role in the development of kidney stones in the pediatric population. Children are also consuming less water than in the past.⁶⁰ Data from Asia and Europe have also shown incidence rates consistent with those of pediatric populations in South Carolina.^{61,62} Like adults, calcium-containing stones are the most predominant kidney stones in pediatric populations.^{63,64} On the contrary, uric acid stones are less common, comprising only 2% to 3% of stones in this population.⁶⁵

HISTOPATHOLOGY

Although urinary biochemical stone risk factors can be studied in rodents, the histopathology of urolithiasis that genuinely resembles human disease, involving both urinary and epithelial factors, is rather unusual in rodents.⁶⁶ The main source of human histopathologic data has been from clinical surgical samples. In addition to urinary factors such as solutes that comprise the chemical components of stones, known inhibitors and promoters, there are epithelial factors that initiate and promote crystal adhesion, growth, and agglomeration.⁶⁷ This field was advanced significantly by the work of Borofsky and Evan and colleagues, who used human samples from intraoperative biopsies and provided histologic descriptions that enable the formulation of pathogenic models of stone formation.^{68,69}

RANDALL'S PLAQUE

In the 1930s, Randall examined the papilla of over 1000 pairs of cadaveric kidneys and observed a cream-colored area near the papillae in one of five kidneys. These plaques are mainly interstitial rather than luminal in locale and are associated with interstitial collagen and tubular basement membranes; they are composed of calcium, nitrogen, carbon dioxide, and phosphorus.⁷⁰ More advanced lesions intrude into tubular lumens. What differentiates these plaques from more normal variants was the finding of small stones attached to some (not all) of the plaques, projecting into the renal pelvis in more than 60 kidneys. Randall concluded that the attached stones were growing from the interstitial calcium plaque, rather than directly from the normal epithelium.⁷⁰ There was no clinical information on Randall's subjects but there is a reasonable chance that they were calcium oxalate stone formers.

These observations were confirmed by many surgeons in the decades to follow, and the Indianapolis-Chicago group of investigators shed light on the importance of Randall's plaques. The pathogenesis of Randall's plaques is still an enigma, but a degenerative stress model of induction of dedifferentiation and transdifferentiation followed by pathologic biomineralization has been proposed.⁷¹

IDIOPATHIC CALCIUM OXALATE STONES

Plaques form in the papillary tip in the basement membrane of the thin loop of Henle, with mineral and organic layers alternating in a concentric configuration (Fig. 38.5). Evan and coworkers have proposed that this is the final form of plaque composed of fused particles in close association with type 1 collagen, with the mineral phase covered by organic

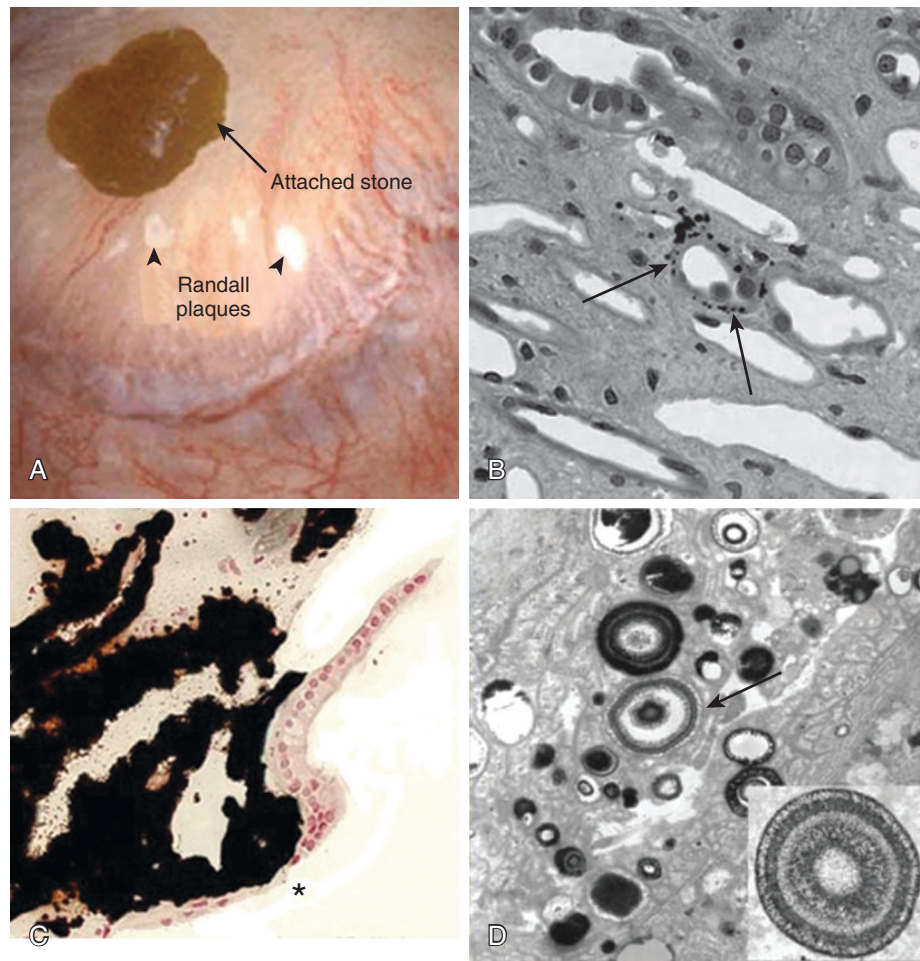


Fig. 38.5 Histopathology of Randall's plaques in calcium oxalate stone formers. (A) View during endoscopic surgery. One plaque (not visible) has an overlying attached calcium oxalate stone (arrow). Two Randall's plaques without stones are below (arrowheads). (B) Light microscopic image of the locale of the initial crystal deposits in the basement membrane of the thin loops of Henle (arrows). (C) Calcium phosphate (stained black) showing a more advanced lesion filling the interstitial space and extending (*), eventually protruding into the urinary space and corresponding to the cream-colored plaques in A. (D) The individual deposits (arrow) assume the morphology of multilaminated spheres of alternating layers of electron-lucent inorganic crystal and electron-dense matrix (inset). (From Evan AP, Lingeman JE, Coe FL, et al. Randall's plaque of patients with nephrolithiasis begins in basement membranes of thin loops of Henle. *J Clin Invest.* 2003;111[5]:607–616.)

matrix.⁷² In the absence of longitudinal biopsies, extrapolation from cross-sectional data has indicated that the particles move from the tubular basement membrane into the surrounding interstitium.

En bloc examination of stones with attached tissue⁷³ show loss of epithelial cells at the attachment site and that the plaque is exposed to the urinary lumen. On contact with urine, the plaque is overlaid with new matrix, and new crystals and matrix form in sequential waves. Osteoprotegerin is present in the stone and plaque, and Tamm-Horsfall protein, known to be restricted to the urine and cells of the thick ascending limb, is present only on the urine space side of the interface.⁷³ Coe and associates have proposed that the organic material overlaying the exposed plaque comes from urine molecules adsorbed initially onto the matrix of plaque. As new crystals nucleate in this urine matrix, the crystals themselves attract molecules with affinity for them, thereby perpetuating stone formation.⁷⁴

CALCIUM PHOSPHATE STONES AND STONES IN RENAL TUBULAR ACIDOSIS

Although Randall's plaques are observed in calcium phosphate (CaP) stone formers, they are fewer in number compared with calcium oxalate stone formers. CaP stone formers have apatite deposits in the lumen ducts of Bellini and inner medullary collecting ducts associated with massive ductal dilation⁷⁵ (Fig. 38.6). The dilated ducts seem to have lost epithelial cells and are surrounded by interstitial fibrotic tissue. The abundance of CaP crystals is correlated with higher urinary CaP supersaturation,²⁰ driven mostly by a high urine pH and, to a lesser extent, by hypocitraturia and hypercalciuria. Coe and coworkers have postulated that shock wave lithotripsy may injure the epithelium and impair local luminal acidification.⁷⁶ Interestingly, a similar pattern is observed in patients with CaP stones from primary hyperparathyroidism.⁷⁷ Luminal plugging and plaque can coexist. A scenario of

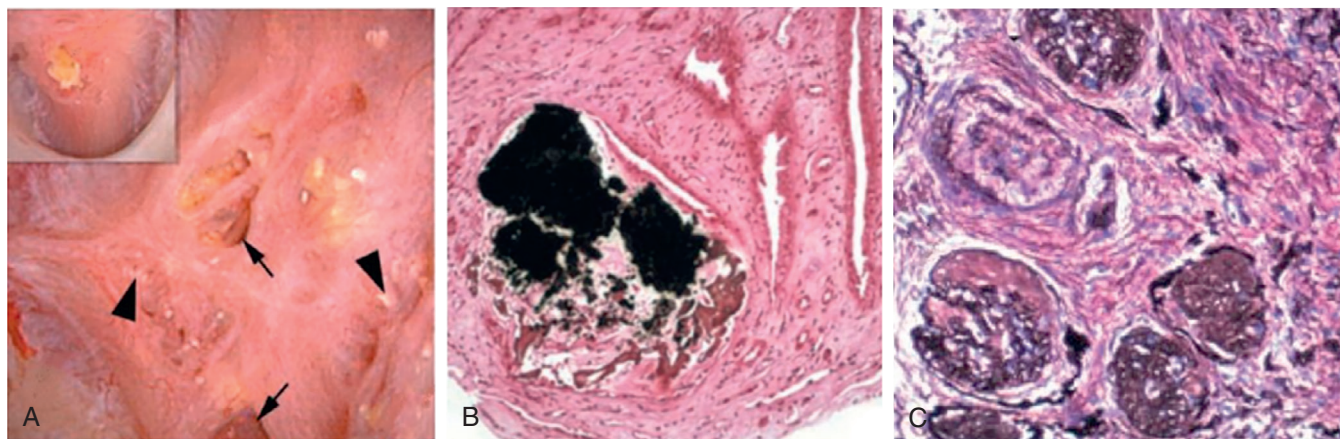


Fig. 38.6 Histopathology of calcium phosphate stone formers. (A) View during endoscopic surgery. Depressions near the papillary tips (arrows) are unique to calcium phosphate stone formers, which coexist with Randall's plaques. The papillae shows yellow crystalline deposits coming out of the ducts of Bellini (inset). (B) Micrograph of luminal deposits in an inner medullary collecting duct. The crystal deposits greatly expanded the lumen, and cell injury and necrosis were found. Interstitial inflammation and fibrosis surround the intraluminal crystal deposition. (C) Renal biopsy from a patient with calcium phosphate stones showing advanced glomerulosclerosis, tubular atrophy, and interstitial fibrosis. This is rarely seen in calcium oxalate stone formers. (From Evan AP, Lingeman JE, Coe FL, et al. Crystal-associated nephropathy in patients with brushite nephrolithiasis. *Kidney Int.* 2005;67[2]:576–591.)

persistently high pH is seen in CaP stone formers with distal renal tubular acidosis (dRTA) from congenital or acquired causes. Papillae show multiple dilated ducts of Bellini, with intraluminal CaP deposits. Distortion, flattening, and fibrosis are common. Atrophic remnants of nephron structures lie within the fibrotic fields of interstitium. It is plausible that the bulk bladder urine pH does not truly reflect the much higher luminal pH in damaged ducts.

STONES IN ENTERIC HYPEROXALURIA

In patients with enteric hyperoxaluria, crystallization is mainly driven by a high urinary luminal oxalate concentration. In patients after gastric bypass surgery, the epithelium appears rather normal, but calcium oxalate crystals are lodged in the lumen.⁷⁸ In hyperoxaluric calcium oxalate stone formers from small bowel resection (e.g., as a result of Crohn disease) the inner medullary collecting ducts contain crystal deposits associated with cell injury, interstitial inflammation, and structural deformity of the papillae, tubular atrophy, and interstitial fibrosis. Interestingly, Randall's plaques are observed, similar to those seen in idiopathic calcium oxalate stone formers. The inner medullary collecting duct deposits contain apatite, with calcium oxalate in some cases.⁷⁹ The birefringent thin crystalline material scattered on the inner medullary collecting duct (IMCD) cell membranes may be the initial crystal lesion.

In summary, based mainly on histopathologic findings and aided by clinical chemistry and imaging, Coe and colleagues have postulated three morphologic pathways of stone formation.⁷⁴ First is overgrowth on interstitial apatite plaque, as seen commonly but not exclusively in idiopathic calcium oxalate stone formers; similar lesions were observed in stone formers with primary hyperparathyroidism, ileostomy, and small bowel resection and in some brushite stone formers. Second is crystal deposits in renal tubules rather than the interstitium, which was seen in all stone formers other than the idiopathic calcium oxalate stone formers. Third is free

solution crystallization in the lumen, as seen in patients with cystinuria or enteric hyperoxaluria.

PATHOPHYSIOLOGY

PHYSICAL CHEMISTRY OF URINARY SATURATION

GENERAL CONCEPTS

Lithogenesis involves solutes entering a solid phase in urine, among other processes. Some definitions are in order here. Using calcium oxalate as an example, equilibrium is attained when the calcium and oxalate concentrations in solution and the amount of calcium oxalate crystals bathed by that solution is constant with time. The product of the free ionized calcium and oxalate concentrations in such a solution in equilibrium is the equilibrium solubility product. However, solubility is the concentration of the calcium oxalate complex (not the individual components) in solution, which is approximately 6.2 μM . Solubility for CaP (brushite; initial phase of complex formation) is about 0.35 μM and, for uric acid, is about 520 μM . Activity refers to the portion of the ion that is chemically active (not always 100%) and is related to its chemical concentration by the activity coefficient (activity = activity coefficient $\times$ chemical concentration). States of saturation are dependent on the activities of the component ions. When free ion activity product is below the solubility product, the crystals will dissolve; this solution is undersaturated. When free ion activity product is higher than the solubility product, it is in a supersaturated state, which causes crystals to grow (Fig. 38.7).

When soluble calcium and oxalate are added to a saturated solution at equilibrium to raise the ion activity product beyond the equilibrium solubility product, there will be growth of any preformed crystals, if such were present. However, in the absence of a preexisting solid phase, no new crystals appear, despite activity products sitting above the solubility product. A solution that will cause the growth of preformed

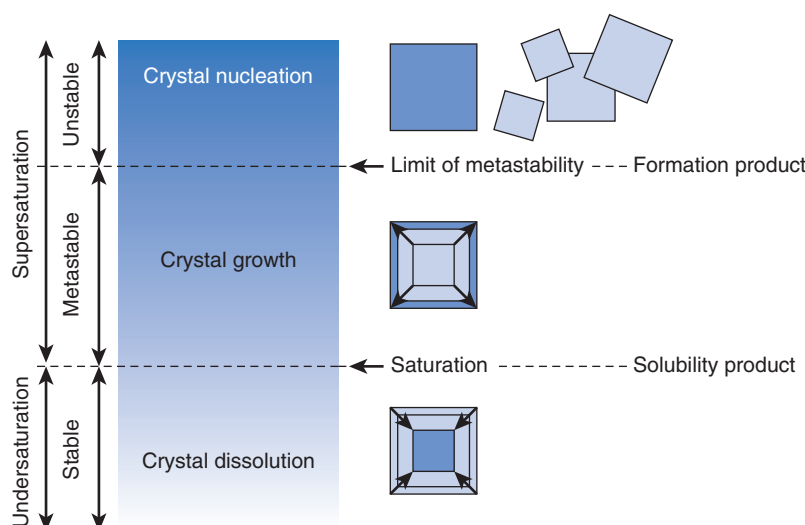


Fig. 38.7 Physicochemical parameters used in the assessment of kidney stone risk. These are shown in relation to the three states of crystal—dissolution, growth, and nucleation.

crystals but not the appearance of a new solid phase is supersaturated and metastable. If one raises the activity product yet higher, with further calcium and oxalate addition, new crystals will appear at some point without existing crystals (see Fig. 38.7). The product of the activities of the ions has now reached the formation product, which is also called the *upper limit of metastability*. Above the formation product, a solution changes from metastable to unstable, and crystallization is inevitable. Therefore, urine may be undersaturated, metastable, or unstable with respect to calcium oxalate or CaP crystals (see Fig. 38.7).

FACTORS INFLUENCING SATURATION

Rate of urinary excretion of the component species of the activity products (e.g., calcium, oxalate, phosphate, urate) and water (denominator) are the primary determinants of saturation. Additional factors, such as complexation and changes in urine pH, can all influence the free ion concentrations and are important in regulating saturation. Therefore, total concentration measurements may not provide full information of the actual activity product. For example, citrate readily complexes calcium, reducing the ionized calcium (activity) levels⁸⁰; a similar relationship exists for magnesium and oxalate.⁸¹ Changes in urine pH can drastically affect the monovalent to divalent phosphate and urate-to-uric acid ratio. For this reason, hypercalciuria, hyperoxaluria, hypocitraturia, unduly alkaline urine, and chronic dehydration all increase the risk of calcium stones, but the relationships are complex.

Thus, interpretation based solely on individual chemical concentrations can be misleading. An example is when a patient with idiopathic hypercalciuria is treated with thiazide, and the urinary calcium excretion decreases. However, a state of potassium deficiency can develop, causing secondary hypocitraturia. It is not easy for a clinician to look at the urinary calcium and citrate excretion rates and concentrations and determine whether the stone risk is reduced. Another scenario is a patient with CaP stones from distal renal tubular acidosis and profound hypocitraturia that is

treated with alkali. The urinary citrate level increases but the urine pH concomitantly increases as well, so it is not easy to determine whether the risk of CaP crystallization is reduced or increased. Several methods have been designed to address these questions.

URINE SATURATION MEASUREMENTS IN VITRO

Upper Limits of Metastability and Formation Product

Upper limits of metastability (ULM) and formation product (FP) is an empiric entity determined in the laboratory by raising the activity product. This is done by adding ligands and noting the activity products at which solid phases begin to appear spontaneously, which is usually detected optically (see Fig. 38.7). Because this is an empiric method, it takes the concentration of the lithogenic solutes and all the inhibitors and promoters into consideration.

Activity Product Ratio

Activity is the fraction of the total solute in free ionized form that is free to react. Crystals of interest are seeded to urine and incubated at 37°C with stirring at a constant pH until equilibrium, until the crystal mass has stabilized (neither increases nor decreases). The activity product ratio (APR) is defined as the ratio of the activity product in the original sample to that obtained after incubation and stabilization of the sample with crystals. If the crystals have grown after reaching equilibrium, the activity product will be lower than the original sample; thus the APR is less than 1, implying that the solution has high lithogenic propensity. If the crystals partially dissolved, the activity product at equilibrium is higher than the original sample and the APR is more than 1, implying that the solution is not lithogenic. Thus, the APR describes whether preexistent crystals, once formed, will grow (APR >1) or shrink (APR <1) while suspended in it.

Although the APR provides a good indication of crystal growth, it gives incomplete information about the ability of that particular urine sample to produce new crystals. In simple

salt solutions, the ULM for calcium oxalate was found to occur at an APR of 8.5 by Pak and Holt⁸² and at 10.0 by Robertson and associates.⁸³ The small difference is mainly methodologic in origin.

Concentration Product Ratio

Activities are important chemical parameters but are difficult to ascertain definitively. Pak and colleagues have used an empiric method to measure urine saturation by comparing urine chemical concentrations (instead of activities) of the components for a given crystal to their concentrations after reaching equilibrium.^{84,85} In their original assay, crystals of interest were seeded to an aliquot of urine and incubated at 37°C, with stirring at a constant pH for 2 days. At equilibrium, the crystal mass has stabilized. If the activity coefficients for calcium, oxalate, and phosphate remain stable throughout the incubation, the ratio of the concentration product at the start of incubation to the concentration product after incubation (equilibrium), must equal the APR, even though the concentration products themselves do not equal the activity products (the activity coefficients cancel out). If activity coefficients can be assumed to be stable, the empiric concentration product ratio (CPR) is a valid estimation of the APR, provided the calcium concentration is below 5.0 mM and the oxalate level is below 0.5 mM.⁸⁵ This assay has been simplified technically and was validated for the analysis of the propensity of calcium oxalate and CaP stone formation.^{86–88}

URINE SATURATION ESTIMATION IN SILICO

Physicochemical methods are informative, although empiric. An obvious drawback of physicochemical methods is the labor involved. Another option is to take clinical urine chemistry data and use a computer program to calculate urine free ion activities for calcium, oxalate, and phosphate from their measured chemical concentrations and their known tendencies to form soluble complexes with each other and with other ligands.^{83,85,89}

Relative Supersaturation Ratio

Using all the relevant association constants (K_a), one can calculate the free ion activity product. If one divides this by the corresponding equilibrium solubility product, one obtains the relative supersaturation ratio (RSR), which estimates the degree of saturation. An RSR above 1 connotes oversaturation and below 1, undersaturation. The validity has been confirmed by studies showing a correlation between the type of stone a patient forms and the prevailing supersaturations in two or three 24-hour urine samples, as estimated by the EQUIL2 program (a BASIC computer program for the calculation of urinary saturation).^{90,91}

Supersaturation Index

Another computer-based program, the Joint Expert Speciation System (JESS), uses the same principle as EQUIL2 but differs from it in that more thermodynamic constants are used to calculate mixed ligand speciation.^{92,93} One important difference is that JESS includes the calcium phosphocitrate complex, which is soluble, and whose formation is pH-dependent. Pak and coworkers have compared the two computer-based programs to empiric physicochemical methods and found a closer approximation of JESS than EQUIL2 to physicochemical methods.^{86–88} Although EQUIL2

provides relative supersaturation (RS) as a readout, JESS expresses the data as the solubility index (SI), which is calculated based on the same physicochemical principles as RS values generated by EQUIL, but the two programs use different speciation concentrations.

URINE SATURATION IN STONE FORMERS

Pak, Robertson, Marshall, and Weber and colleagues have independently provided evidence that urine from stone formers is more supersaturated than urine from non-stone formers.^{82,94–96} The results were similar, although absolute values differed for the three investigative groups, likely because of differences in methods. Stone formers have higher average values of urine saturation than normal subjects whether or not saturation is measured with respect to calcium oxalate, brushite, octocalcium phosphate, or hydroxyapatite. Weber and associates have shown that supersaturation for calcium oxalate is higher among hypercalciuric than normocalciuric patients.⁹⁶ In all studies, urine from non-stone formers is supersaturated with respect to calcium oxalate. Added crystals grow in urine from most normal persons.^{82,96} Hautmann and coworkers measured calcium and oxalate concentrations in tissues from the cortex, medulla, and papilla of seven human kidneys.⁹⁷ The calcium oxalate concentration product in the papillae ($1 \times 10^{-4} \text{ M}^2$) exceeded that of urine ($5 \times 10^{-7} \text{ M}^2$) and those of the medulla and cortex ($8 \times 10^{-7} \text{ M}^2$ and $6 \times 10^{-7} \text{ M}^2$, respectively). In addition, high CaP supersaturation is common in the tip of loop of Henle because both tubule fluid pH and the Ca^{2+} concentration are high because of water extraction on the descending limb.⁹⁸

The actual ULM for calcium oxalate and brushite in human urine samples from normal subjects and from hypercalciuric, normocalciuric, and hyperparathyroid stone formers are surprisingly variable.⁸² The ULM of calcium oxalate and brushite was measured in urine from calcium stone formers and gender- and age-matched control subjects^{99,100}; the gap between prevailing supersaturation of the urine and the ULM was reduced in stone formers, rendering it more likely for crystallization and stone formation to occur. The reduced ULM may represent deranged crystallization inhibition. Values of APR are close enough to the ULM for calcium oxalate and CaP that new crystal formation is expected. Most urine, even from normal persons, is metastable with respect to calcium oxalate, so growth of crystal nuclei is expected.

ASSESSMENT OF NUCLEATION

Saturation estimates the propensity to crystallize. Nucleation refers to the initial formation of a crystal nidus. Nucleation is followed by crystal growth, epitaxial growth, and aggregation. Homogeneous nucleation, the spontaneous formation of new crystal nuclei in a supersaturated metastable solution, is actually uncommon. Usually, debris particles, surface irregularities, or other existing crystals furnish a substrate on which crystal nuclei begin to form at a lower APR than what is required for true homogeneous nucleation. The metastable zone reflects the greater free energy required to create new nuclei than to simply enlarge preformed nuclei. Any surface that can serve as a substrate for ions in solution on which to organize may act as a heterogeneous nucleus, bypassing the energetically more costly process of creating a solid phase *de novo*. In other words, there is a lower apparent ULM for heterogeneous nucleation.

The efficiency of heterogeneous nucleation depends on the similarity between the spacing of charged sites on the preformed surface and in the lattice of the crystal that is to grow on that surface. This matching is referred to as “epitaxis,” and its extent is usually referred to as a good or poor epitaxial relationship.¹⁰¹ To achieve homogeneous nucleation, all potential heterogeneous nuclei must be excluded. This is an unreasonably difficult task when the urine formation product ratio (FPR) for any given crystal is conditioned by preformed heterogeneous nuclei, and the nuclei of other crystals that form as the APR is raised during the experimental determination itself.

A number of urine crystals have good epitaxial matching and behave toward one another as heterogeneous nuclei. Monosodium urate and uric acid are excellent heterogeneous nuclei for calcium oxalate,^{102,103} so uric acid or urate could, by crystallizing, lower the ULM for calcium oxalate. Heterogeneous nucleation may be the mechanism linking hyperuricosuria to calcium oxalate stones.^{104,105} The epitaxial overgrowth of calcium oxalate on a surface of uric acid has been experimentally documented.¹⁰⁶ Both brushite and hydroxyapatite can also nucleate calcium oxalate,^{107,108} which is likely the basis for Randall’s plaque as initiators of calcium oxalate stones.

Randall’s plaque is formed in the interstitium of the papilla but can erode through the papillary epithelium to be exposed to the urinary space, thus furnishing a nucleating site for the crystallization of calcium oxalate and anchoring site. This allows retained new crystals to have sufficient time to grow to a clinically significant size. Apatite is frequently found at the core of calcium oxalate stones,¹⁰⁹ and there is increased prevalence and severity of Randall’s plaques in stone formers as compared with non-stone formers.¹¹⁰ Evan, Coe, Kuo, and colleagues have extensively studied Randall plaque formation in patients with various types of nephrolithiasis.^{72,75,111–114} Their findings are presented in the relevant sections later.

ASSESSMENT OF CRYSTAL GROWTH AND AGGREGATION

Growth is the enlargement of a crystal, and aggregation is the coalescence of crystals. Once formed, crystals will grow if bathed in urine with an APR greater than 1. Growth and aggregation are pathogenic because microscopic nuclei are too small to cause disease. Crystals are regular lattices, composed of repeating subunits, and they grow by the incorporation of calcium and oxalate or phosphate, or uric acid, into new subunits on their surfaces. In metastable solutions at 37°C, growth rates of calcium oxalate and the stone-forming CaP crystal show appreciable changes in macroscopic dimensions over hours. The growth rate increases with supersaturation and is most rapid in urine samples with the highest APR. Small crystals aggregate into larger crystalline masses by electrostatic attraction from the charged surface of the crystals, which rapidly increase particle size, producing a crystal that can lodge in the urinary tract. Stone formers’ urine contains larger crystal aggregates compared with non-stone formers.¹¹⁵

Inhibition of Crystal Agglomeration

The inhibition of crystal agglomeration (ICA) can be measured by adding a fixed amount of solid calcium oxalate to a synthetic solution intently and metastably supersaturated with calcium oxalate (CaOx), or to a similar solution spiked with the test subject’s urine. ICA is expressed as the time to

reach half of the maximum decline in the [Ca] half-life ($t_{1/2}$) in synthetic solution. A higher $t_{1/2}$ indicates a higher ICA.¹¹⁶

CELL-CRYSTAL INTERACTIONS

Finlayson and Reid have proposed that crystals cannot grow or aggregate fast enough to anchor in the urinary tract during the normal urinary transit time.⁶⁷ This is analogous to throwing a handful of sand into a sink with the faucet turned on full; the sand will simply wash into the drain. Crystals must anchor to the renal tubule epithelium to delay its passage so that they can grow large enough to be of clinical significance. Although Randall’s plaques¹¹⁷ may offer both an anchoring or nucleating site, it is also clear that some stones can form in the absence of Randall’s plaques,¹¹⁰ so alternative mechanisms are present. CaOx crystals can adhere to cultured collecting duct epithelial cells.¹¹⁸ The in vitro adherence and uptake of crystals appear to be greater for CaOx than CaP.¹¹⁹ The crystals bind to anionic sites on the cell membrane¹²⁰ and can be inhibited by a variety of anionic compounds normally found in urine.¹²¹ Phosphatidylserine in the lipid bilayer appears to be a binding site, and enrichment of cell membranes with phosphatidylserine increases CaOx crystal binding by renal epithelial cells in culture, although the in vivo significance cannot be confirmed.¹²²

In addition for adhesion and crystal growth, it has been proposed that cell-crystal interactions result in cell injury and death,^{123,124} which constitutes an integral part of stone formation. Nano-sized crystals induce predominantly apoptotic cell death, leading to cell and nuclear shrinkage and phosphatidylserine ectropion. In contrast, micron-sized crystals primarily caused necrotic cell death, leading to cell swelling and plasma membrane and lysosomal disintegration.¹²⁵

Thus far, the evidence on the role of cell-crystal interaction has been mostly from cell culture experiments. The importance of the cell-crystal interaction in the pathogenesis of human kidney stone disease is not clear at this time.

INHIBITORS

CONCEPT

Stone constituents such as CaOx, CaP, sodium urate, or uric acid are often supersaturated in normal urine, yet precipitation does not occur.^{82,94–96} This suggests that there are inhibitors to stone formation present in normal urine.^{126,127} The ULM is lower in patients who form stones compared with matched controls, indicating deficiency in inhibitors in stone formers.¹¹⁴ Despite this phenomenologic finding, the exact role of inhibitors is far from fully understood. There has been an argument posed that the therapeutic efficacy of agents such as potassium citrate in reducing the rate of stone recurrence may be because of their effect on the supersaturation of urine with respect to CaOx and CaP, rather than to any increase in inhibitory activity¹²⁸; this remains unresolved. One can arbitrarily classify inhibitors into four groups: (1) multivalent metallic cations such as magnesium; (2) small organic anions such as citrate; (3) small inorganic anions such as pyrophosphate; or (4) macromolecules such as osteopontin and Tamm-Horsfall protein (Table 38.1).

MAGNESIUM

Magnesium has been touted as a kidney stone inhibitor for many years.¹²⁹ It is present in urine in millimolar

Table 38.1 Natural Inhibitors of Stone Formation

Inhibitor	Examples
Inorganic	Magnesium Pyrophosphate
Small organic anion	Citrate
Macromolecules	Bikunin Calgranulin FK binding protein-12 Glycosaminoglycans Nephrocalcin Osteopontin Prothrombin F1 fragment Lithostathine Matrix Gla protein Tamm-Horsfall protein Urinary trefoil factor-1

concentrations, and it readily binds to oxalate. Some have expressed the notion that magnesium is efficacious in preventing stone formation by binding or complexing oxalate in both the bowel and urine,^{130–132} by inhibiting CaOx crystallization,^{131,133,134} and by increasing urinary citrate when given as alkali salt.¹³⁵ In vitro studies have shown that magnesium is an inhibitor of CaOx crystal growth in artificial, rodent, and human urine, but these measurements were largely made at supraphysiologic concentrations.^{136–141} Magnesium inhibits both nucleation and growth of CaOx crystals.¹⁴² In a seeded crystal growth system of constant composition, magnesium was a weak inhibitor.¹⁴³

The clinical trial experience has not been uniform. Fifty-five stone formers were given 500 mg of Mg daily as Mg(OH)₂ and experienced a decrease in stone rate from 0.8 to 0.08 stones/year.¹²⁹ However, other controlled studies have suggested that magnesium does not alter the recurrence rates for CaOx stone formation, perhaps because of its poor absorption.¹⁴⁴ Another possibility is that the beneficial inhibitory effect of magnesium may be offset by its hypercalciuric effects because luminal Mg inhibits the calcium channel, TRPV5, in a pH-dependent manner.¹⁴⁵ Human metabolic studies have shown interacting effects of urinary Mg and pH on calcium excretion.¹⁴⁶ At any given urine pH, Mg marginally increases urine calcium levels. Provision of an alkali load significantly lowered urine calcium levels and saturation of CaOx at any given level of urinary Mg excretion. At present, magnesium supplement is not routinely recommended as antiurolithiasis therapy; Mg supplements can be used if there is evidence of Mg depletion in a stone former—potassium magnesium citrate should be used instead of Mg oxide.^{147,148}

CITRATE

Citrate is the best studied, understood, and tested stone inhibitor. It is also one that is amenable to use as therapy. In addition to chelating calcium, which reduces calcium's availability to bind oxalate and phosphate, citrate also inhibits three properties of calcium oxalate crystals: nucleation, growth, and aggregation.^{149,150} Citrate therapy directly inhibits crystallization^{135,151} and increases the upper limit of metastability by increasing both urine citrate and pH.¹⁵² Potassium citrate was shown, in most clinical studies, to reduce recurrent calcium nephrolithiasis in patients with idiopathic hypercal-

ciuria and distal renal tubular acidosis, even in stone formers considered normocitraturic.^{147,153–158} Citrate is the most important base equivalent in human urine. The principal regulator of urinary citrate levels is the acid-base status of the organism, particularly proximal tubule cell pH.¹⁵⁹ Because urinary citrate serves dual roles as a calcium chelator and the principal urinary base, this presents a conflict when base conservation is required, thus putting the urine at risk of calcium crystallization.¹⁶⁰ During systemic or proximal tubule cellular acidosis, there is an increase in both proximal citrate reabsorption and metabolism, reducing the amount of citrate excreted in the urine.¹⁶¹ A reduction of urinary citrate, because of an increase in the acid load generated from dietary protein ingestion or to intestinal base loss, can promote the formation of both CaOx and CaP stones.^{156,162}

PYROPHOSPHATE

The inhibitory role of pyrophosphate on calcium crystallization has been known for years,¹⁶³ and it plays important anticalcification roles outside the kidney.¹⁶⁴ Pyrophosphate binds to the surface of basic CaP crystals, including hydroxyapatite, and arrests or at least retards the crystal growth of both CaP and CaOx crystals.^{165–167} The average pyrophosphate concentration in bulk urine is sufficient to inhibit crystal growth significantly,¹⁶⁷ but the concentration in the micro-environment around the stone is not known and may be inadequate. There is a reduced pyrophosphate-to-creatinine ratio in 50% of stone formers, suggesting that a lack of pyrophosphate may predispose to nephrolithiasis.¹⁶⁶

MACROMOLECULES

Macromolecules are potent inhibitors of calcium oxalate crystallization.¹⁶⁸ Some of the macromolecules listed in Table 38.1 will be highlighted. These molecules are generally highly anionic and contain large amounts of acidic amino acid that undergo posttranslational modification, with negatively charged side chains. Bergsland and colleagues have compared the inhibitory proteins in urine from 50 stone-forming and 50 non-stone-forming, matched, first-degree relatives of calcium stone-forming patients¹⁶⁹; they found that profiles of inhibitory proteins were effective in discriminating relatives of stone formers from non-stone formers and were even more effective than conventional measurements of supersaturation. Clinical use of inhibitor profiles is hampered by the difficulty and expense of these measurements.

Osteopontin

Osteopontin, previously termed *uropontin*, is an acidic phosphoglycoprotein that was initially isolated from bone^{170–173} but is present in urine at about 4 mg/day.^{171,172} In bone, osteopontin inhibits hydroxyapatite formation during osteogenesis.^{174–176} It is expressed in cells of the thick ascending limb of Henle and distal convoluted tubules and secreted into the urine. Osteopontin inhibits nucleation, growth, and aggregation of CaOx stones in vitro.^{170,171,177} Osteopontin-deleted mice develop intratubular CaOx crystals when given ethylene glycol, a nephrotoxin that leads to metabolic acidosis and hyperoxaluria, at doses that do not induce crystal formation in the wild-type mice.^{170,178} Osteopontin is upregulated at sites of biomineralization in genetic hypercalciuric stone-forming rats,¹⁷⁹ and other models have suggested that it is increased in response to crystal formation.

Tamm-Horsfall Protein

Tamm-Horsfall protein (THP) is the most abundant protein found in human urine, with approximately 100 mg excreted/day. It is synthesized in the thick ascending limb of Henle's loop and, because of self-aggregation, is the principal component of urinary casts.¹⁸⁰ Also known as uromodulin (UMOD), THP is implicated and involved in a multitude of functions and pathobiologic effects, including ion transport, innate immunity, blood pressure, and acute and chronic kidney injury.^{181–183} UMOD/THP protein inhibits CaOx crystal aggregation but does not alter growth or nucleation.^{184,185} UMOD/THP-deleted mice spontaneously form calcium crystals in the lumen of tubules in the papilla and medulla.¹⁸⁶ When these mice are given ethylene glycol to induce CaOx stone formation, there is a marked induction of renal osteopontin at sites of crystal formation.¹⁸⁶ UMOD/THP also activates the calcium channel TRPV5¹⁸⁷ and can theoretically reduce calciuria. One study has described higher UMOD/THP excretion in recurrent stone formers with positive family histories but concluded that the UMOD/THP is aggregated and functionally defective in inhibiting CaOx aggregation in vitro.¹⁸⁸ In a urinary proteomic study, the administration of lime (calcium, oxide, hydroxide, carbonate) increased UMOD/THP, but the biologic effect on lithogenesis was unclear.¹⁸⁹

Urinary Prothrombin Fragment 1

Urinary prothrombin fragment 1 (UPTF1, crystal matrix protein) is a fragment of prothrombin made in the kidney.¹⁹⁰ It is a potent inhibitor of CaOx growth, aggregation, and nucleation¹⁹¹ and is present in kidney stones.¹⁹² Differences in UPTF1 levels and genetic alleles have been found between stone formers and non-stone formers.^{193,194}

Bikunin

This is the light chain of the inter-alpha-trypsin inhibitor, which inhibits both calcium oxalate growth and nucleation.¹⁹⁵ It is found in the proximal tubule and in the thin descending segment of the loop of Henle.¹⁹⁶ Bikunin, along with the heavy chains of the inter-alpha-trypsin inhibitor, have been isolated from kidney stones, suggesting that multiple fragments of this inhibitor may be active in preventing stone formation.¹⁹⁷ Bikunin was present only in the collecting duct apical membranes and loop cell cytoplasm of stone formers, colocalizing with osteopontin and inter-alpha-trypsin inhibitor heavy-chain 3, and extensive heavy-chain 3 was only present in stone formers and not in controls.¹⁹⁸ One study has shown differences in the electrophoretic mobility pattern between stone formers and non-stone formers.¹⁹⁹

Glycosaminoglycans

Glycosaminoglycans (GAGs) occurring in the urine of normal individuals typically has around 50% chondroitin sulfate, around 25% heparan sulfate, around 10% low-sulfate chondroitin sulfate, and around 5% to 10% hyaluronic acid (HA).²⁰⁰ Chondroitin sulfate retards nucleation; dermatan sulfate inhibits nucleation.²⁰¹ In rodents induced to precipitate CaOx, there is increased expression of heparan sulfate in tubules.²⁰² HA, which is in the extracellular matrices, is thought to bind at cell surfaces.²⁰³ Human tubular cells in primary culture bind crystals when damaged, and this is dependent on the expression of HA.²⁰⁴ Canine tubular cells increase

the synthesis of glycosaminoglycans to protect from toxic insults of CaOx crystals and oxalate ions in vitro.²⁰⁵ Human studies have shown decreased urinary GAG levels in stone formers,^{206,207} but this finding has not been universal.^{208,209}

Matrix-Gla Protein

Matrix-Gla protein (MGP) is a small (14-kDa) glycoprotein that was the first urinary protein found to have crystal inhibitory properties.^{184,210,211} MGP contains five γ -carboxyglutamic acid (Gla) residues, which have a high affinity for calcium and phosphate ions and hydroxyapatite crystals²¹² and inhibit crystal growth, nucleation, and aggregation. MGP protects the vasculature from calcification^{213,214} and may serve a similar role in the kidney. MGP mRNA expression is increased in renal tubular epithelial cells following exposure to CaOx crystals.^{215,216} Interestingly, multilaminated crystals formed in the injured tubules with lack of MGP expression.²¹⁷ In contrast, no crystal formation was seen in tubules with MGP expression. MGP genetic single-nucleotide polymorphism is associated with the individual susceptibility of nephrolithiasis.²¹⁸ MGP from some stone-forming patients lacks Gla and has diminished ability to inhibit nucleation and growth of CaOx crystals.^{184,185,219}

Urinary Trefoil Factor 1

An anionic protein was isolated with inhibitory activity against CaOx crystal growth and identified as human trefoil factor 1 (TFF1). Functional studies of urinary TFF1 have demonstrated that its inhibitory potency is similar to that of nephrocalcin. The inhibitory activity of urinary TFF1 was neutralized by anti-C-terminal antibody, consistent with a model in which the C-terminal glutamic residues of TFF1 interact with calcium. Concentrations and amounts of TFF1 in the urine of patients with idiopathic CaOx kidney stone were significantly less than those found in controls.²²⁰

CALCIUM STONES

CaOx is the most prevalent kidney stone worldwide, accounting for approximately 70% to 80% of kidney stones. CaP stones contribute to a smaller percentage (15%).^{221–225} However, the percentage of CaP stones appears to increase with each new stone event in individual stone formers.¹⁹ Risk factors for CaOx stone formation include hypercalciuria, hyperuricosuria, hypocitraturia, hyperoxaluria, and altered urinary pH.^{221,222} CaP stones share some common mechanisms with CaOx stones, including hypercalciuria and hypocitraturia but, in contrast to CaOx, unduly alkaline urine is characteristic among CaP stone formers but not in CaOx stone formers. An ambulatory evaluation in over 1270 patients with recurrent calcium nephrolithiasis has demonstrated hypercalciuria in 60% of the patients, whereas hyperuricosuria was found in 36%, hypocitraturia in 31%, hyperoxaluria in 8%, abnormal urinary pH in 10%, low urinary volume in 16%, and no metabolic abnormality in 4% of patients.²²³ These percentages are merely approximations because all the variables in question are continuous in their distribution. A study of 82 brushite stone formers has demonstrated hypercalciuria in 80%, alkaline urinary pH in 60%, and hypocitraturia in 50% of patients; hyperuricosuria and hyperoxaluria were uncommon, comprising 18% and 10%, respectively. Low urinary volume, however, was found in nearly 60%.²²⁴

HYPERCALCIURIA

Hypercalciuria is the most prevalent metabolic abnormality in patients with calcium nephrolithiasis, occurring in 60% of adults with calcium stones.^{224–229} The pathophysiologic mechanisms for hypercalciuria are protean and may involve increased intestinal absorption (absorptive hypercalciuria; Fig. 38.8), diminished renal tubular calcium reabsorption (renal leak hypercalciuria), and enhanced calcium mobilization from the bone (resorptive hypercalciuria).^{222,230–233} Intestinal hyperabsorption of calcium is the most common abnormality in this population,²³⁴ but all these physiologic derangements may coexist in individual patients.

INTESTINAL HYPERABSORPTION OF CALCIUM

Flocks first showed a link between hypercalciuria and nephrolithiasis.²³⁵ Subsequently, Albright and Henneman have used the term “idiopathic hypercalciuria” to highlight the unknown origin of hypercalciuria in this population.^{236,237} The metabolic basis of intestinal hyperabsorption of calcium was first described by Nordin and Peacock, who introduced the term “absorptive hypercalciuria” (AH).²³⁸ Balance studies have shown a negative calcium balance in patients with hypercalciuria compared with non-stone-forming control subjects.^{239,240} Compatible with the balance data was information from dietary recall, which also hinted that patients with hypercalciuria consume less calcium than they excrete.²⁴¹

Balance studies have shown low fecal calcium content,²⁴² and hypercalciuria was corrected by intestinal calcium binding with sodium cellulose phosphate.^{243,244} The characteristic features of absorptive hypercalciuria are hypercalciuria with normocalcemia, normal or suppressed serum parathyroid hormone (PTH), and/or urinary cyclic adenosine 3',5'-monophosphate (cAMP; surrogate marker of PTH bioactivity in the kidney), reflective of an increased calcium load on the system. Hyperabsorption is the most prevalent feature in patients with idiopathic hypercalciuria. Furthermore, it is likely

that absorptive hypercalciuria is a heterogeneous disorder, consisting of two broad subtypes, calcitriol-dependent^{245–249} or calcitriol-independent.^{234,242,250–252}

1,25-Dihydroxyvitamin D-Dependent Absorptive Hypercalciuria

Increased 1,25(OH)₂D concentration has been reported in four different studies in hypercalciuric nephrolithiasis patients.^{245–248}

The diagnosis of absorptive hypercalciuria was established by direct measurement of intestinal calcium absorption^{251,253} or indirectly²⁴⁷ by showing a lower serum PTH concentration. The underlying pathophysiologic mechanism responsible for an increased serum 1,25(OH)₂D level likely results from increased production rather than decreased metabolic clearance. This is based on studies using an infusion equilibrium technique in 9 patients with absorptive hypercalciuria (>4 mg/kg/day and/or >300 mg/day) on a defined calcium intake (Fig. 38.9).²⁴⁶ The diagnosis of absorptive hypercalciuria in this study was based on a high calciuric response after oral calcium load in association with a fasting normal or decreased PTH level or nephrogenic cAMP.²⁴⁶ Further support for 1,25(OH)₂D in enhancing urinary calcium excretion has come from a study in normal subjects who became hypercalciuric after receiving a high dose of 1,25(OH)₂D.²⁴⁵ In a separate study, net intestinal calcium absorption significantly increased during calcitriol treatment, and urinary calcium excretion in stone formers exceeded that of controls,²⁵⁴ demonstrating a negative calcium balance during calcitriol administration in stone formers. This was associated with an increase in the urinary hydroxyproline level, a marker of bone resorption, suggesting that elevated 1,25(OH)₂D in healthy men on a low-calcium diet enhances bone resorption. There is a direct correlation between serum 1,25(OH)₂D, calcium excretion, and calciuric responses, suggesting the pathogenetic role of excess 1,25(OH)₂D in increased intestinal absorption.²⁴⁵ It is still unknown whether the calcium in 1,25(OH)₂D-driven hypercalciuria originates from the bone²⁵⁴ or intestine.²⁴⁵

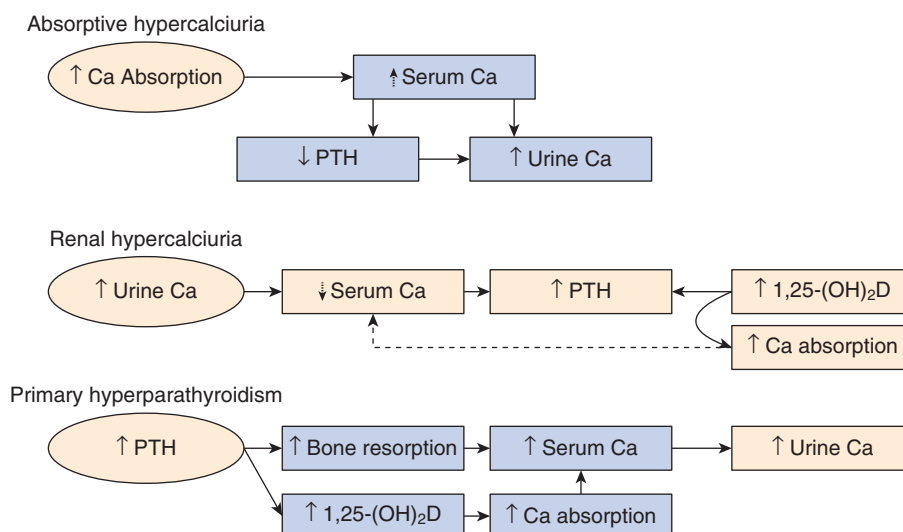


Fig. 38.8 Pathophysiologic mechanisms of hypercalciuria. PTH, Parathyroid hormone. (From Pak CY. Etiology and treatment of urolithiasis. *Am J Kidney Dis.* 1991;18[6]:624–637.)¹

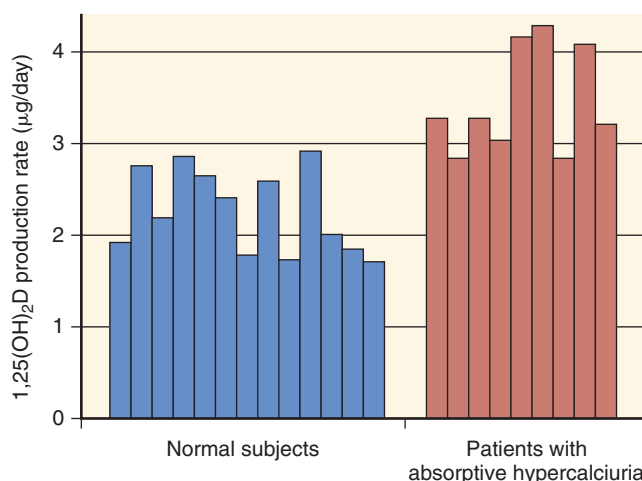


Fig. 38.9 Individual values for 1,25(OH)₂D production rates in normal subjects and in those with absorptive hypercalciuria. The mean 1,25-dihydroxyvitamin D production rate was significantly higher in patients with absorptive hypercalciuria than in normal subjects (3.4 ± 0.5 vs. 2.2 ± 0.5 µg/day; $P < .001$). (From Insogna KL, Broadus AE, Dreyer BE, et al. Elevated production rate of 1,25-dihydroxyvitamin D in patients with absorptive hypercalciuria. *J Clin Endocrinol Metab.* 1985;61[3]:490–495.)

The reason for elevated 1,25(OH)₂D production in an absorptive hypercalciuria population is unclear. PTH, one of the main regulators of 1,25(OH)₂D synthesis, and urinary cAMP, a marker of PTH bioactivity, have never been found to be increased in this population.^{223,225,234,247,251,252, 255–257}

There were no frank reductions in serum phosphorus concentration, the other main regulator of 1,25(OH)₂D production.^{245–247,258}

Some studies have found that certain patients with absorptive hypercalciuria may have a primary defect in renal tubular reabsorption of phosphorus (renal phosphorus wasting), which consequently stimulates 1,25(OH)₂D synthesis and enhances intestinal calcium absorption.^{247,259,260} However, the serum phosphorus concentration does not differ in hyperabsorptive and normal subjects under the same dietary regimen,^{257,261} and hypophosphatemia and a low TmP/GFR (tubular maximum phosphate reabsorption per glomerular filtration rate) were found only in a minority of the patients with absorptive hypercalciuria.²⁵⁷ Differences in urinary calcium or serum 1,25(OH)₂D levels between stone formers with and without a renal phosphate leak was detected in one study, but not in others.^{262–265} Moreover, the underlying mechanism(s) of the renal phosphate leak in this population has not been fully investigated. An association of fibroblast growth factor 23 (FGF23) with renal phosphate leak and potential pathogenic role in patients with calcium nephrolithiasis has been reported in two studies.^{265,266} It is still possible to have phosphate wasting and depletion without frank hypophosphatemia—serum phosphate is a practical but imperfect surrogate for phosphate stores.²⁶⁷

The conversion of circulating serum 25-hydroxyvitamin D (25[OH]D) occurs principally in the kidneys, and this metabolic conversion is enhanced with vitamin D deficiency.²⁶⁸ However, CYP24A1 is highly expressed in the kidney and converts 1,25(OH)₂D and 25(OH)D to the inactive metabolites,²⁶⁸

such as 1,24,25-trihydroxyvitamin D and 24,25-dihydroxyvitamin D, respectively. Thus, CYP24A1 hydroxylase acts as a potent inhibitor of vitamin D action by diminishing both 25(OH)D and 1,25(OH)₂D.

One study has described two patients with an elevated serum 1,25(OH)₂D level, nephrolithiasis, and nephrocalcinosis associated with hypercalciuria because of undetectable activity of 1,25(OH)₂D 24-hydroxylase (CYP24A1), an enzyme that inactivates 1,25(OH)₂D.²⁶⁹ This loss-of-function mendelian disease was described as being as a result of biallelic inactivating mutations of CYP24A1. The frequency of the CYP24A1 variant, based on a National Center for Biotechnology Information (NCBI) database, was estimated to be 4% to 20%. This report is proof of principle for the role of the 24-hydroxylase in D-dependent calcium stones, but it is difficult to determine whether variants in this gene predispose to urolithiasis in the general population.

An epidemiologic analysis of men in a Health Professionals Follow-Up Study over 12 years has found that the odds ratio of incident symptomatic kidney stones in the highest compared with the lowest quartile of serum 1,25(OH)₂D, within the normal range, was 1.73 after adjusting for body mass index, diet, plasma factors, and other covariates.²⁷⁰

1,25-Dihydroxyvitamin D-Independent Absorptive Hypercalciuria

In humans, two-thirds of absorptive hypercalciuria patients exhibit increased intestinal calcium absorption, with normal levels of 1,25(OH)₂D.^{234,242,250–252,271} Moreover, a triple-lumen intestinal perfusion study has supported 1,25(OH)₂D-independent selective jejunal hyperabsorption of calcium in this population, which differs from the less selective gastrointestinal (GI) effects of 1,25(OH)₂D in normal subjects.²⁷²

Pharmacologic probes were used to alter serum 1,25(OH)₂D concentrations and assess intestinal calcium absorption in hypercalciuric subjects.^{252,255–257} Ketoconazole reduces serum 1,25(OH)₂D in normal subjects and patients with primary hyperparathyroidism.^{273–275} Nineteen patients with absorptive hypercalciuria were treated with ketoconazole (600 mg/day $\times$ 2 weeks; Fig. 38.10)²⁵⁵ and, in 12 patients, ketoconazole lowered 1,25(OH)₂D concentration, intestinal calcium absorption, and 24-hour urinary calcium. Moreover, intestinal calcium absorption assessed directly by ⁴⁷Ca absorption correlated with serum 1,25(OH)₂D and 24-hour urinary calcium excretion. Importantly, in 7 patients, despite the significant reduction of 1,25(OH)₂D, there was no change in calcium absorption or urinary calcium excretion.²⁵⁵ In these patients, intestinal calcium absorption and urinary calcium excretion were not correlated with serum 1,25(OH)₂D levels. There have also been studies using thiazide,²⁵² glucocorticoids,²⁵⁶ and orthophosphates²⁵⁷ in hyperabsorptive patients. Results indicated that lowering 1,25(OH)₂D concentration is not associated with a reduction in intestinal calcium absorption, indicative of a calcitriol-independent mechanism in the development of hypercalciuria in this population.

Increased Abundance of Vitamin D Receptor

Expression and activation of the vitamin D receptor (VDR) are necessary for vitamin D action.²⁷⁶ A genetic association of intestinal hyperabsorption of calcium and calcium stone

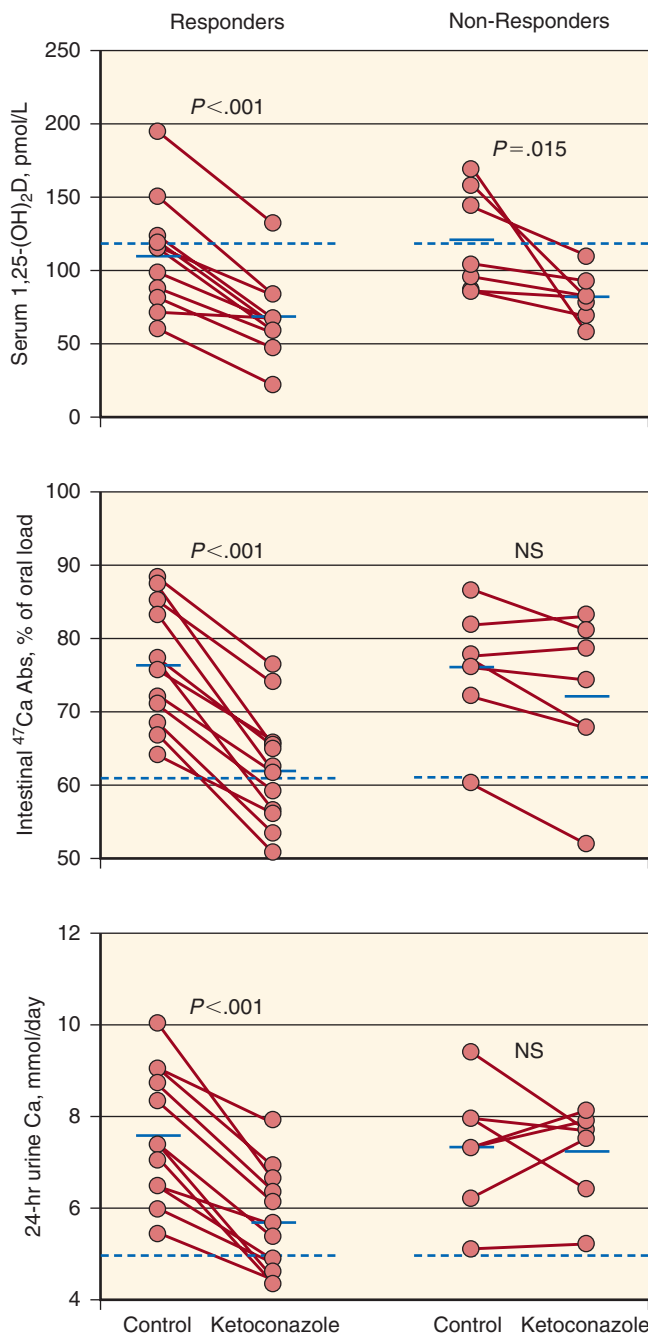


Fig. 38.10 Heterogeneity of absorptive hypercalciuria. Shown are the responses of patients with hypercalciuria to a 2-week course of ketoconazole (600 mg/day). Abs, Absorption; NS, nonsignificant. (From Breslau NA, Preminger GM, Adams BV, et al. Use of ketoconazole to probe the pathogenetic importance of $1,25\text{-dihydroxyvitamin D}$ in absorptive hypercalciuria. *J Clin Endocrinol Metab.* 1992;75[6]:1446–1452.)

formation with the VDR locus has been described.^{277,278} Two common single-nucleotide polymorphisms (SNPs) were linked to altered expression and/or function of VDR protein,^{279–282} but these results were not confirmed by others.^{283,284} Because intestinal calcium absorption in hyperabsorptive kidney stone formers mainly occurs with normal serum $1,25\text{(OH)}_2\text{D}$,^{234,250–252} there might be an alteration in VDR abundance or sensitivity to $1,25\text{(OH)}_2\text{D}$. VDRs are expressed in peripheral monocytes and T and B lymphocytes.^{285–289}

In 10 male hypercalciuric CaOx stone formers compared with age- and gender-matched controls without a history of stone disease, the abundance of VDR was twofold higher in peripheral blood monocytes (PBMC) in stone formers.²⁹⁰ In another study of absorptive hypercalciuric patients, a Scatchard analysis has shown an increase in VDR abundance in a subset of patients with normal serum $1,25\text{(OH)}_2\text{D}$ levels. These results suggest heterogeneous patterns with respect to VDR abundance.²⁸⁵

Genetic Hypercalciuric Stone-Forming Rat Model of Hypercalciuria

Genetic hypercalciuric stone-forming (GHS) rats are a polygenic model for hypercalciuria.^{291–294} Breeding of Sprague-Dawley rats with high-end physiologic calcium excretion yielded robust constant hypercalciuria after 60 generations.^{295,296} These animals have normal serum calcium and calcitriol levels, increased intestinal calcium absorption, formation of calcium stones, excessive bone resorption, and decreased renal tubular calcium reabsorption, making this a feasible model for studying the $1,25\text{(OH)}_2\text{D}$ –VDR axis.^{297–301} On a low-calcium diet, there was an increase in serum $1,25\text{(OH)}_2\text{D}$ levels in control and GHS rats. The increase was lower in GHS rats compared with controls, but there was greater net increase in intestinal calcium absorption in GHS rats, suggesting amplified bioactivity of $1,25\text{(OH)}_2\text{D}$.^{297,302} This is analogous to patients with absorptive hypercalciuria, in whom elevated intestinal calcium absorption and urinary calcium excretion are associated with normal $1,25\text{(OH)}_2\text{D}$ levels.^{234,250–252} The increased abundance of duodenal and kidney VDR^{300,303,304} may account for the heightened biologic effect of $1,25\text{(OH)}_2\text{D}$ in GHS rats.^{234,250–252} The high tissue VDR levels were as a result of an increase VDR gene expression, protein synthesis, and half-life of the VDR protein,^{304–306} with no change in VDR DNA sequence,³⁰⁵ but the transcription factor Snail, a negative regulator of VDR gene expression, seemed to be lower in GHS rats.³⁰⁶

Transcellular intestinal calcium transport involves initial calcium entry across the apical membrane, which is facilitated across the cells by the 9-kDa calbindin, followed by extrusion through the plasma basolateral membrane.³⁰⁷ The biologic action of VDR in GHS rats was supported by increased expression of vitamin D–dependent genes, such as increased expression of calbindin-9 in the duodenum.³⁰⁴ The 28-kDa calbindin is the principal calcium-binding protein in the renal distal convoluted tubule and plays a role in renal calcium transport.^{308–310} Baseline calbindin-28k levels were higher in GHS rat kidneys and, after a single dose of $1,25\text{(OH)}_2\text{D}$, calbindin-28k increased much more in GHS rats compared with control rats.³⁰⁴ One may conclude that abundance of tissue VDR in GHS rats, as well as humans with absorptive hypercalciuria, may amplify the biologic effect of normal serum $1,25\text{(OH)}_2\text{D}$ levels and consequently increase the expression of vitamin D–dependent genes and their protein products to regulate intestinal, renal, and bone calcium transport.

RENAL LEAK HYPERCALCIURIA

Renal leak of calcium is caused by diminished renal tubular calcium reabsorption and is associated with secondary increases in serum PTH and $1,25\text{(OH)}_2\text{D}$ levels and a compensatory increase in intestinal calcium absorption.³¹¹

The underlying mechanisms for impaired renal tubular calcium reabsorption have not yet been fully explored. However, several potential mechanisms have been purported to affect renal tubular calcium reabsorption, including a primary proximal defect, hyperinsulinemia, calcium-sensing receptor (CaSR) gene expression and its interaction with calcitriol, and claudin-14 expression. Seminal human metabolic studies performed by Worcester and associates have demonstrated an exaggerated fractional excretion of calcium (FE_{Ca}) in hypercalciuric patients in a postprandial state compared with normal subjects.³¹² Furthermore, increased FE_{Ca} was not a consequence of differences in filtered calcium load, urinary sodium excretion, or serum PTH levels (Fig. 38.11).

It was proposed that the postprandial inhibition of renal tubular calcium reabsorption may be as a result of differences in insulin levels in hypercalciuric stone formers.^{313,314} This was based on previous findings showing an exaggerated calciuric response to an oral carbohydrate load in hypercalciuric stone

formers.^{315–317} These studies did not segregate the effects of hyperinsulinemia from those of the serum glucose concentration. A study using euglycemic hyperinsulinemic clamp has demonstrated that the rise in insulin-induced calciuria is small and no different between hypercalciuric stone formers and non-stone-forming control subjects, suggesting that insulin is unlikely to play a significant pathogenic role in hypercalciuric stone formers.³¹⁸ Insulin resistance and hyperinsulinemia are features of obesity and are associated with hypercalciuria and the risk of kidney stone formation^{10,12,319,320} but FE_{Ca} was comparable in the hypercalciuric stone formers, overweight and obese non-stone formers, and lean controls, independent of peripheral insulin sensitivity measured as glucose disposal rates. This study is at variance with a previous report using semipurified insulin extracted from islets rather than recombinant insulin, which showed significant insulin-induced hypercalciuria.³²¹

The CaSR is expressed in multiple tissues, including the parathyroid glands, kidney, and GI tract.³²² In the parathyroid

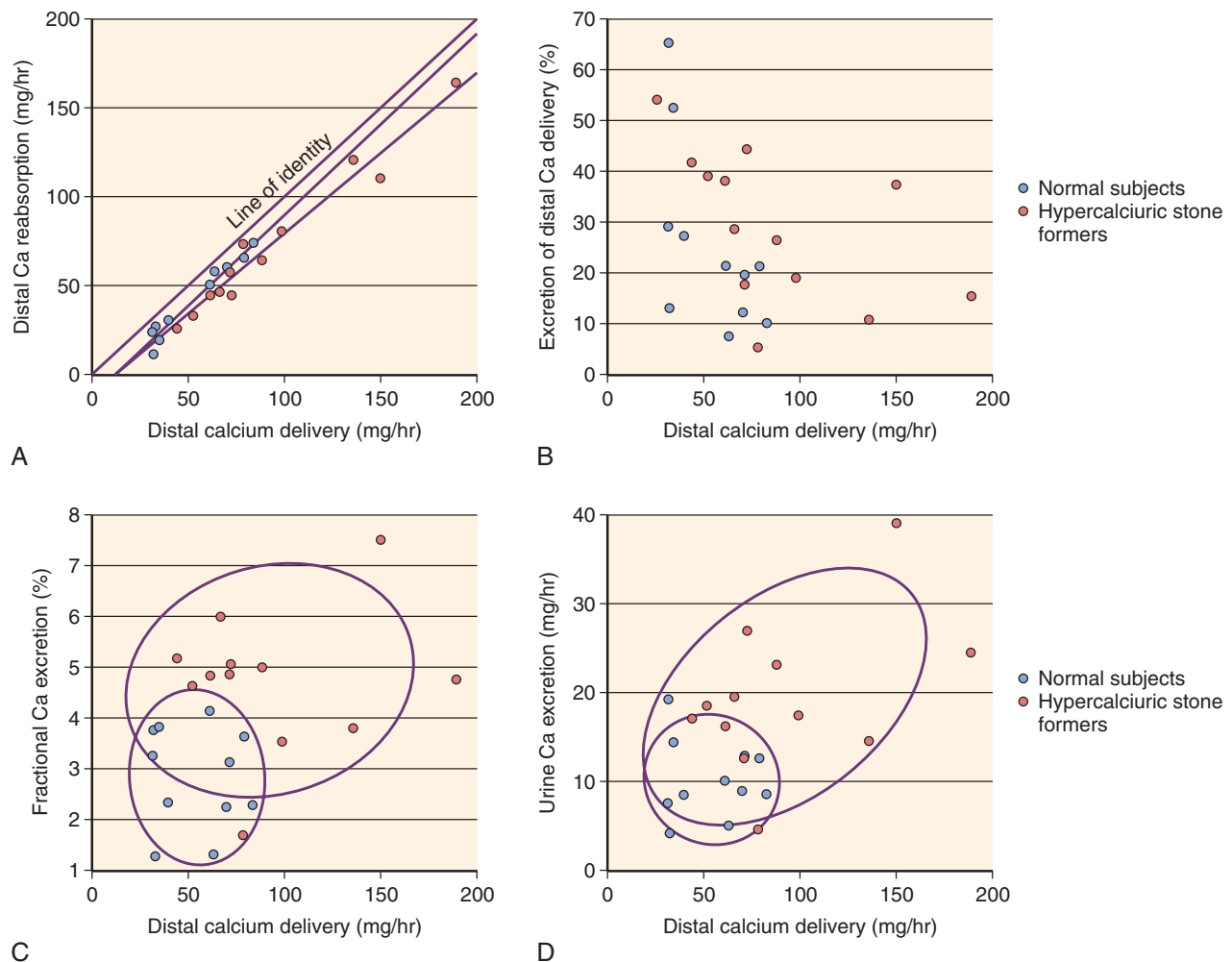


Fig. 38.11 Relationship between distal calcium reabsorption, renal calcium excretion, and distal calcium delivery. (A) Distal calcium reabsorption increases similarly with calcium delivery in normal subjects and hypercalciuric stone formers. Both lines are positioned slightly beneath the line of identity. This suggests normal distal calcium handling but higher distal calcium delivery. (B) The actual fraction (%) of distally delivered calcium that was excreted fell with increasing distal delivery, and stone formers lie above normal at comparable distal delivery. (C) Fractional and total (D) calcium excretions were high in many stone formers versus normal subjects at comparable deliveries. (From Worcester EM, Coe FL, Evan AP, et al. Evidence for increased postprandial distal nephron calcium delivery in hypercalciuric stone-forming patients. *Am J Physiol Renal Physiol.* 2008;295[5]:F1286–F1294.)

glands, the CaSR responds to calcium and suppresses PTH release. In the kidney, the CaSR responds to increased circulating ionized calcium and reduces renal tubular calcium reabsorption in the thick ascending limb (TAL) and distal convoluted tubal (DCT).³²³ An Italian cohort has linked hypercalciuria in stone formers to polymorphism in the CaSR gene. Serum calcitriol levels were not defined in these subjects. Vitamin D-responsive elements (VDREs) have been identified in the CaSR gene, which suggests that calcitriol can upregulate kidney CaSR expression and reduce renal tubular calcium reabsorption.³²⁴ Although the physiologic function of CaSR in the GI tract has not been defined, the CaSR is highly expressed in gastric parietal G cells, which in turn increases gastric acid and gastrin release. The underlying mechanism for the intestinal hyperabsorption of calcium in kidney stone formers was suggested to be related to the increased gastrin and gastric acid secretion to oral calcium and protein load in patients with absorptive hypercalciuria.^{325–328}

Tight junction proteins of the claudin family expressed in the TAL are critical for paracellular Ca^{2+} reabsorption in the kidney.³²⁹ A genome-wide association study conducted in Iceland and the Netherlands has found an association between sequence variations of the *CLDN14* gene and kidney stones and reduced bone mineral density at the hip.³³⁰ One study has suggested that CaSR regulates claudin-14 expression via two microRNAs, miR-9 and miR-374, and that dysregulation of the renal CaSR–claudin-14 pathway might be a possible cause of hypercalciuria.³³¹ This model proposes that CaSR increases claudin-14, which inhibits the complex of claudin-16 and claudin-19 that regulates paracellular Ca^{2+} reabsorption in the TAL.³²⁹

RESORPTIVE HYPERCALCIURIA

Resorptive hypercalciuria refers to hypercalciuria caused by enhanced calcium mobilization from bone. This can be PTH-dependent or PTH-independent.

Parathyroid Hormone-Dependent Resorptive Hypercalciuria

Primary hyperparathyroidism (PHPT) is the most common cause of resorptive hypercalciuria and is associated with calcium stones in 2% to 8% of patients.²²¹ Because of the much earlier diagnosis of PHPT, asymptomatic cases are common and renal complications have significantly decreased.³³² A retrospective study of 271 renal ultrasounds from asymptomatic cases with surgically proven PHPT has shown a 7% prevalence of stones compared with 1.6% in 500 age-matched subjects who underwent sonography examinations for other reasons.³³³ Although hypercalciuria has been perceived as a cause of kidney stones in this population, the exact relationship between hypercalciuria and the risk of nephrolithiasis in patients with primary hyperparathyroidism is debatable.^{334,335} The relative contribution from enhanced skeletal calcium mobilization versus enhanced intestinal calcium absorption is not clear.^{334,336,337} The serum $1,25(\text{OH})_2\text{D}$ level is higher, and there is increased calciuric response to oral calcium load in patients with PHPT and kidney stones.^{336–339} One study has shown that kidney stones are more frequently seen in younger patients because of their higher synthetic capacity of $1,25(\text{OH})_2\text{D}$ compared with older patients and hence higher intestinal calcium absorption.³³⁶

In a single-center prospective study over 5 years in 140 patients with PHPT consisting of 127 women, of whom 85% were postmenopausal women, approximately 50% had kidney stones by ultrasound, two-thirds had osteoporosis by dual-energy x-ray absorptiometry (DXA), and one-third had a history of vertebral fracture documented by x-ray. More kidney stones were detected in symptomatic than asymptomatic patients. The results of the study suggested that kidney stones and bone fractures are common in asymptomatic patients with PHPT.³⁴⁰ Urinary calcium excretion was not different between kidney-stone formers and those without kidney stones, suggesting that other factors might be responsible for the development of kidney stones. The result of this study did not agree with two previous reports showing that stone-forming patients with PHPT had higher hypercalciuria than non-stone formers and higher intestinal calcium absorption.^{337,341}

Parathyroid Hormone-Independent Resorptive Hypercalciuria

Elevated serum $1,25(\text{OH})_2\text{D}$ levels or amplified sensitivity to $1,25(\text{OH})_2\text{D}$ at target organs may lead to enhanced bone resorption and diminished bone collagen synthesis in hypercalciuric stone-forming subjects.^{290,342} In a retrospective multivariable analysis in 250 male and 182 female kidney stone formers, no significant relationship was found between urine calcium (either ad lib or restricted diet) with spine and femoral neck bone mineral density (BMD) in men and in postmenopausal women treated with estrogen. This suggests that calciuria may be a surrogate but not a causative factor in bone loss and higher fracture risk in kidney stone formers.³⁴³ A similar phenotype has been shown in GHS rats whose bones are more sensitive to exogenous $1,25(\text{OH})_2\text{D}$ compared with normal control rats.³⁰³ Other mechanisms include diminished bone expression of transforming growth factor- β , which stimulates bone formation and mineralization, and/or perturbation in the RANK/RANK-L/OPG system, which increases bone resorption. The latter, in association with other cytokines and growth factors, acts synergistically with high $1,25(\text{OH})_2\text{D}$ levels to affect bone remodeling.^{230,344,345}

HYPERURICOSURIA

Hyperuricosuria as a single abnormality is detected in much lower frequency than in combination with other metabolic abnormalities among calcium stone formers.^{222,346} There is a high frequency of CaOx stones in patients with gout.^{347,348} A structural similarity was described between crystals of uric acid (UA), sodium hydrogen urate, and CaOx , attributed to the growth of one crystal on another.³⁴⁹ This is described in more detail later.

Pathophysiologic Mechanism of Hyperuricosuria

UA production is from de novo synthesis, tissue catabolism, and dietary purine load. It is estimated that approximately 50% of the typical daily uric load is endogenous, with the remainder from dietary sources.³⁵⁰ Increased dietary purine intake has been implicated in most hyperuricosuric calcium urolithiasis (HUCU) patients. A metabolic study comparing HUCU stone formers with age- and weight-matched normal subjects has shown that stone-forming patients ingest a higher intake of purine-rich food such as meat, fish, and poultry.³⁵¹ With kinetic studies, endogenous overproduction of UA has

been suggested in approximately one-third HUCU stone formers consuming purine-free diets.³⁵¹ About one-third of synthesized UA is excreted into the intestinal tract, followed by intestinal uricolysis and the remainder by the kidney.³⁵² However, defective renal tubular reabsorption of UA has not been shown to be responsible for hyperuricosuria in this population.³⁵³ In most mammals, ingested purines are converted to UA and then further to allantoin enzymatically in the peroxisomes of hepatic cells.³⁵⁴ However, humans and higher primates lack functional uricase, the enzyme that converts UA to allantoin, thereby making UA the final product of purine metabolism in these species.^{355,356}

Physicochemical Mechanism of Hyperuricosuria-Induced Calcium Stones

The physicochemical basis for hyperuricosuria-induced calcium oxalate stones has been postulated but not yet been proven. There are three potential and nonmutually exclusive mechanisms.^{102,357,358} In two studies, the underlying mechanisms linking UA to CaOx crystallization was attributed to heterogeneous nucleation^{102,103} (Fig. 38.12). Another study showed that colloidal monosodium urate from supersaturated urine in these patients may attenuate inhibitor activity against CaOx crystallization^{359,360} but, when GAGs were specifically examined, UA did not affect its inhibitor activity.³⁶¹ Monosodium urate can diminish the solubility of CaOx in solution, a process referred to as *salting out*.^{357,362} Salting out is also known as *antisolvent crystallization*, which can typically be used to precipitate a nonelectrolyte multicharged macromolecule (usually protein) at a high electrolyte concentration (usually ammonium sulfate). Despite the uncertainty regarding the mechanism, clinical studies in hyperuricosuric patients have shown a drastic decline in the rate of recurrent kidney stone formation in those treated with xanthine oxidase inhibition.³⁶³

Increasing urinary supersaturation with monosodium urate in conjunction with the decreased limit of metastability of CaOx, because of the adsorption of a macromolecular inhibitor of CaOx crystallization by monosodium urate in hyperuricosuric patients, increases the risk of CaOx stone formation.³⁶⁴ Contrary to laboratory and clinical evidence, a retrospective population-based study in a large number of patients did not show a relationship between urinary UA and CaOx stone formation.³⁶⁵ The urate effect is likely imperceptible in a large, unselected, calcium stone-forming population.

HYPOCITRATURIA

Hypocitraturia as an isolated abnormality occurs in about one-third of patients with calcium nephrolithiasis.¹⁵⁶ The presence of citrate in human urine was described in 1917³⁶⁶ and, a decade later, it was shown that acid-base homeostasis plays a key role in determining urinary citrate excretion in that a higher urinary citrate excretion rate was observed in alkalotic patients and very low excretion rate in those with metabolic acidosis or an acid load.³⁶⁷ In 1954, a combination of oral sodium citrate, potassium citrate, and citric acid was used to alkalinize the urine in patients with uric acid and cystine stones.³⁶⁸ Citrate is one of the most abundant organic anions (in molar quantities) in human urine and is an important inhibitor of calcium stone formation via its effects on calcium chelation, prevention of crystallization, and aggregation, in addition to its effect to reduce ionized calcium and, some have postulated, its ability to promote crystal detachment from cells (Fig. 38.13).^{156,160,369,370} Tricarboxylate citrate has pKa values of 2.9, 4.3, and 5.6 and is found mostly as a trivalent anion (citrate³⁻) in blood. Citrate is freely filtered, and approximately 10% to 35% of filtered citrate is excreted in urine, varying with acid-base status. The reabsorption of

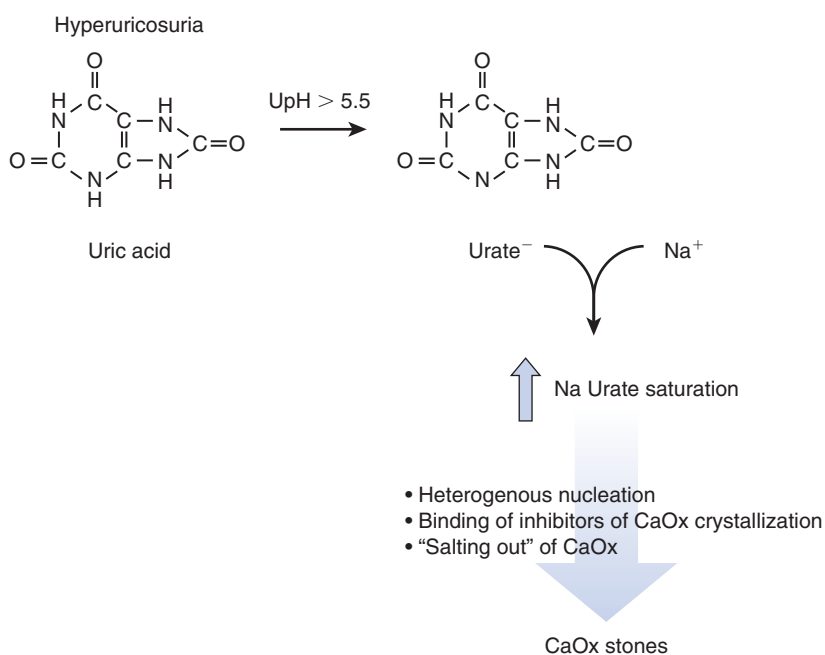


Fig. 38.12 Physicochemical scheme for urate-induced CaOx stones. Hyperuricosuria in the absence of acidic urine renders high free urate activity levels in the urine. Sodium activity is perpetually higher than urate by two orders of magnitude. The higher sodium urate promotes calcium oxalate crystallization by three nonexclusive mechanisms. UpH, Urinary pH.

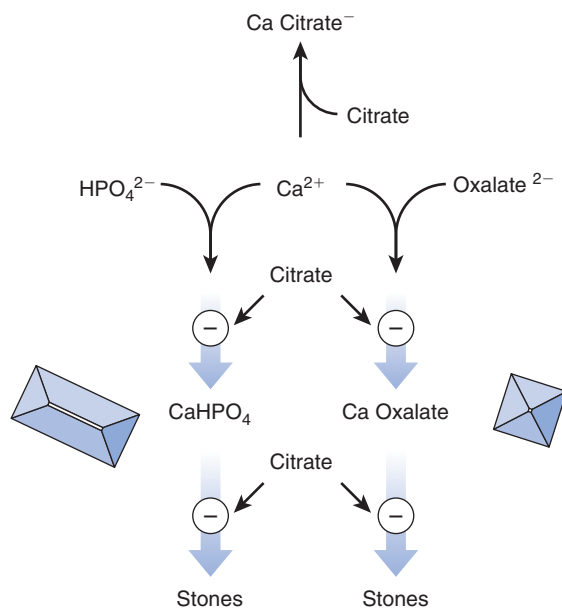


Fig. 38.13 Protective effects of citrate against urolithiasis. Citrate binds calcium with high affinity to form soluble calcium citrate complexes and lowers the ionized calcium and calcium activity. Citrate also directly inhibits calcium oxalate and calcium phosphate crystallization and aggregation.

citrate occurs in the proximal convoluted tubule and, to a lesser extent, in the proximal straight tubule.^{161,371,372} Apical membrane transport occurs through a coupled sodium-dicarboxylate cotransporter (NaDC-1).^{373,374} In contrast, basolateral citrate reabsorption occurs via a tricarboxylate cotransporter (NaDC-3).³⁷⁵

Role of Acid-Base Factors

It is well known that alterations of acid-base status plays a key role in renal tubular citrate reabsorption by multiple mechanisms (see Chapter 8). The highest pK_a of citric acid is 5.6, so the concentration of citrate²⁻, the transported ionic species, increases at lower luminal pH.³⁷⁶ There is also a direct gating effect of pH on NaDC-1 activity, independent of the substrate concentration.³⁷⁴ The abundance of NaDC-1 cotransporter increases in the apical membrane of the proximal renal tubule during acidosis. Finally, cytosolic ATP citrate lyase and mitochondrial aconitase activity increase in acidosis, which lowers the intracellular concentration of citrate and promotes its reabsorption.^{377–379}

Other Factors

The contribution of factors other than acid-base status to renal tubular citrate handling has not been extensively studied. However, it has been suggested that vitamin D, PTH, calcitonin, lithium, calcium, sodium, magnesium, and a variety of organic anions may alter urinary citrate excretion.^{380,381} The effect of organic acid has been attributed to its competition with renal citrate for reabsorption via NaDC-1 transporters.³⁸¹ The effect of increasing apical calcium and magnesium has been demonstrated in cultured proximal tubular cell, showing decreased citrate uptake that may involve a transporter other than NaDC-1.³⁸²

Table 38.2 Clinical Conditions Associated With Hypocitraturia

Low Extracellular Fluid pH	Normal or High Extracellular Fluid pH
Overproduction acidosis - Chronic diarrhea - Exercise-induced lactic acidosis	Potassium deficiency
Underexcretion acidosis - Congenital or acquired distal RTA - Acetazolamide, topiramate	Angiotensin II-related - ACE inhibitors - Salt excess - Excess dietary protein

ACE, Angiotensin-converting enzyme; RTA, renal tubular acidosis.

Clinical Conditions

Clinical conditions associated with hypocitraturia may be divided into those with systemic extracellular acidosis and those without (Table 38.2). dRTA is the most prominent cause of hypocitraturia and is frequently encountered in patients with recurrent nephrolithiasis.^{383–387} Other conditions associated with systemic acidosis include carbonic anhydrase inhibition,^{388,389} strenuous physical exercise,³⁹⁰ and chronic diarrheal states.^{391–393}

Hypocitraturia is also found in normobicarbonatemic states, including incomplete dRTA, chronic renal insufficiency, mild chronic metabolic acidosis, high protein consumption, thiazide treatment with hypokalemia, primary aldosteronism, excessive salt intake, and angiotensin-converting enzyme (ACE) inhibitor treatment.^{158,380,394–398} In experimental animals and, to a certain extent, in humans, ACE inhibitors lower the intracellular pH in the proximal tubule.³⁹⁸ Potassium depletion in rats upregulates renal citrate reabsorption in the proximal tubule apical membrane by altering the abundance of transporters.³⁹⁹

Actions of Citrate

The physicochemical basis for the inhibitory role of citrate involves the formation of soluble complexes from the reduction of the ionic calcium concentration in the urine.⁴⁰⁰ Direct inhibition of crystallization of CaOx and CaP is facilitated by the inhibition of spontaneous precipitation of CaOx and agglomeration of preformed CaOx crystals.^{116,135}

HYPEROXALURIA

Hyperoxaluria is present in isolation or in combination with other risk factors in 8% to 50% of kidney stone formers.^{223,401–403} In CaOx stone formers, both urinary oxalate and calcium are responsible for driving CaOx supersaturation, although the activity of calcium is one order of magnitude higher than oxalate (Fig. 38.14).⁴⁰⁴ Under normal circumstances, the physiologic concentration of the CaOx complex in the urine far exceeds its solubility constant.^{404,405} The mechanisms underlying hyperoxaluria can stem from multiple factors and are summarized in Fig. 38.15.^{406–410}

Increased Hepatic Production

Monogenic disorders of oxalate metabolism are rare but serious causes of kidney stones. Oxalate is a dicarboxylic acid

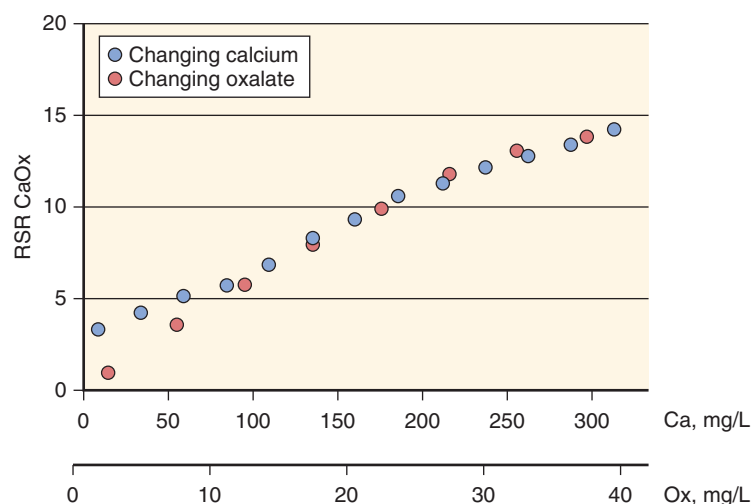


Fig. 38.14. Relationship between calcium oxalate relative supersaturation (*RSR*) and calcium or oxalate concentration. Calcium or oxalate was varied individually; the other was kept constant. (Modified from Pak CY, Adams-Huet B, Poindexter JR, et al. Rapid communication: relative effect of urinary calcium and oxalate on saturation of calcium oxalate. *Kidney Int.* 2004;66[5]:2032–2037.)

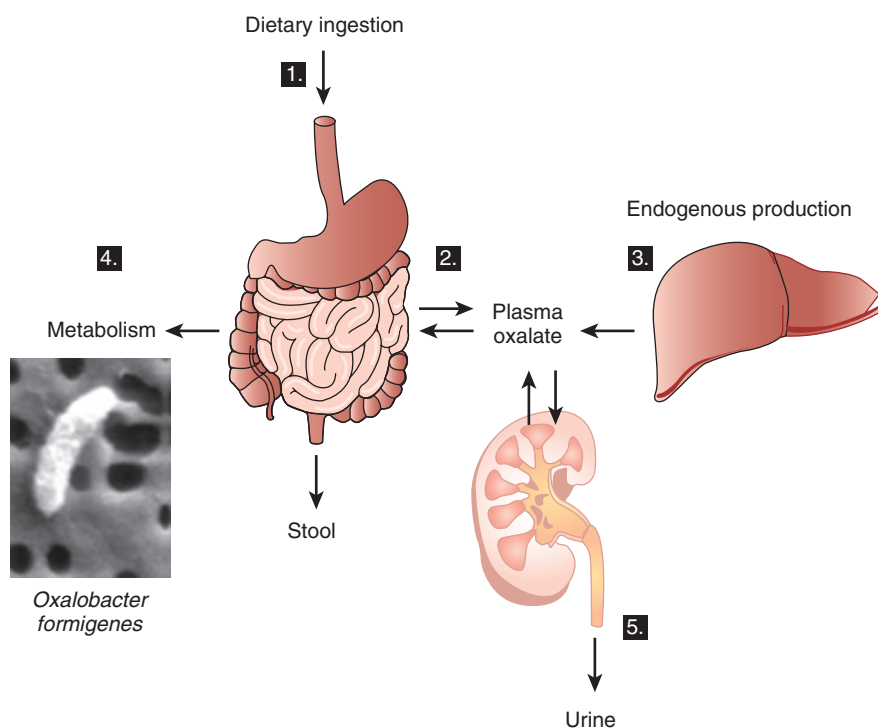


Fig. 38.15 Pathophysiologic mechanisms of hyperoxaluria. Oxalate balance is determined by ingestion and endogenous production versus intestinal and urinary excretion. Intestinal handling can be bidirectional, and luminal degradation is facilitated by microbial degradation. Urinary excretion is the net result of filtration minus tubular reabsorption. Hyperoxaluria can be caused by the following: (1) increased dietary ingestion; (2) increased gut absorption or decreased secretion; (3) increased endogenous hepatic production; (4) decreased intestinal bacterial metabolism; or (5) renal hyperexcretion.

and in mammals is either ingested or is an end product of hepatic metabolism.^{222,406,411} Primary hyperoxaluria (PH) is an autosomal recessive disorder of inborn enzymatic defects resulting in massive hepatic overproduction of oxalate. The three genetic forms converge onto the excess glyoxylate, which is the precursor of oxalate. These forms are classified into types I, II, and III. Type I PH (Online Mendelian

Inheritance in Man [OMIM] 259900) is the consequence of a deficiency or mistargeting of hepatic alanine glyoxylate transferase (AGT), which is a pyridoxal 5'-phosphate-dependent enzyme that transaminates glyoxylate to glycine.⁴¹² It accounts for about 80% of PH cases. Type II (OMIM 260000) is caused by a deficiency in the cytosolic enzyme glyoxylate reductase-hydroxypyruvate reductase (GRHPR), which reduces glyoxylate

to glycolate,⁴¹³ and accounts for about 10% of cases. The least common variety, type III (OMIM 613616), is as a result of activating mutations in the mitochondrial 4-hydroxy-2-oxoglutarate aldolase enzyme (HOGA), which converts hydroxyproline to glyoxalate.^{414–417} Since this is a gain-of-function mutation, type III heterozygotes can have calcium oxalate stones.^{416,417} PH type 3 accounts for about 5% cases; another 5% of PH have no known genetic mutation.

Multiple other metabolic precursors of oxalate metabolism, including by-products of the breakdown of ascorbic acid—fructose, xylose and hydroxyproline—potentially contribute to oxalate production. However, their contribution under normal physiologic circumstances has not yet been fully elucidated.^{418,419}

Dietary Intake and Bioavailability

Dietary oxalate contributes significantly to urinary oxalate excretion. There is a wide variation in estimated oral intake of oxalate, ranging from 50 to 1000 mg/day.^{223,403} Dietary oxalate and its bioavailability contribute approximately 45% of the urinary oxalate excretion.⁴²⁰ The main sources of most bioavailable oxalate-rich food are seeds, chocolate derived from tropical cocoa trees, leafy vegetables, including spinach and rhubarb, and tea. The relationship between oxalate absorption and dietary oxalate intake has been demonstrated to be nonlinear.⁴²⁰

Intestinal Absorption

Despite seminal advances in rodent intestinal physiology of oxalate handling,⁴²¹ the specific intestinal segments in the

human intestine involved in oxalate absorption and secretion is not fully known. It was proposed that the main site of oxalate absorption is in the small intestine because most oxalate absorption occurs in the first 4 to 8 hours after ingestion.^{401,409,422,423} Nevertheless, it has been suggested that the colon may also participate, to a lesser extent, in oxalate absorption.⁴²³

Oxalate is both absorbed and secreted in the GI tract through both paracellular and transcellular pathways. Paracellular oxalate absorption has not yet been confirmed in intact organisms. However, it has been proposed that at a low gastric pH, part of the dietary oxalate is converted into small hydrophobic molecules, which could possibly diffuse through the lipid bilayer^{424–427} and thereby increase urinary oxalate excretion.

Role of Anion Exchanger Slc26a6

The anion exchanger Slc26a6 is involved in intestinal oxalate transport in rodents,^{428,429} where it is expressed in the apical membrane of the duodenum, jejunum, and ileum and, to a lesser extent, the large intestine.⁴³⁰ There is defective net oxalate secretion in mice with targeted inactivation of *Slc26a6*⁴²⁸ (Fig. 38.16). *Slc26a6* null mice on a control oxalate diet showed increased plasma oxalate concentration, decreased fecal oxalate excretion concentration, and high urinary oxalate excretion.⁴²⁸ The abnormalities were attenuated following stabilization on an oxalate-free diet, suggesting that diminished net gut oxalate secretion is responsible for the rise in plasma oxalate concentration and elevated urinary oxalate excretion. However, the influence of this putative anion exchanger on

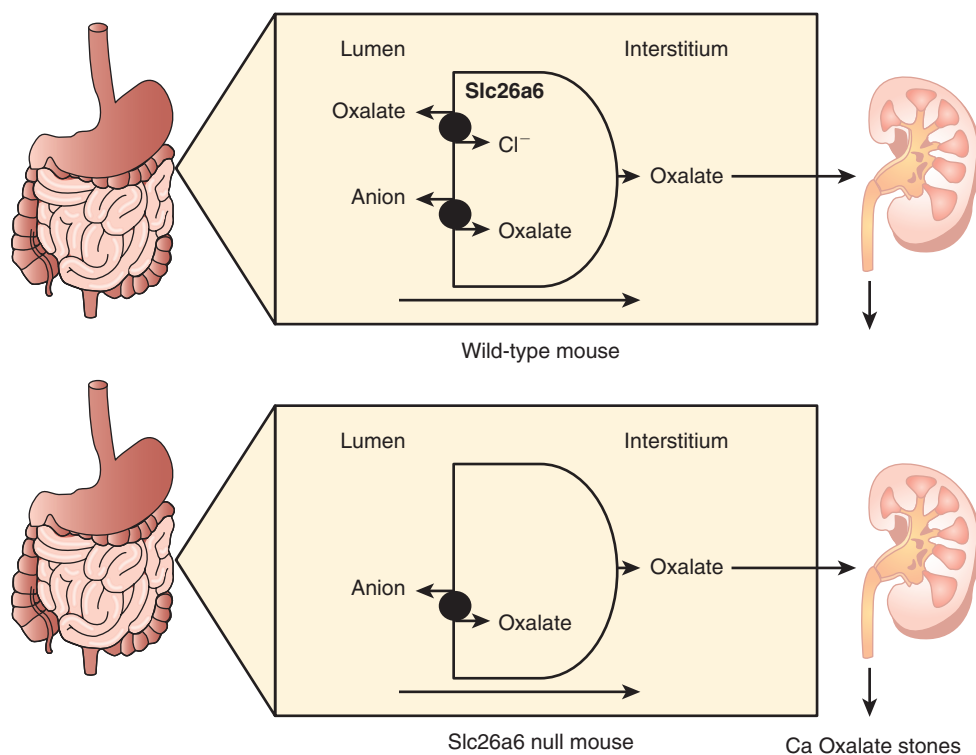


Fig. 38.16 Comparison of the wild-type versus the *Slc26a6* mouse. *Slc26a6* mediates intestinal oxalate secretion in wild-type mice, so that only 10% of ingested oxalate is normally absorbed and excreted into the urine. Targeted deletion of the *Slc26a6* gene in mice unmasks a large intestinal absorptive flux of oxalate and leads to increased plasma oxalate levels, hyperoxaluria, and calcium oxalate stones.

intestinal oxalate absorption, and specifically its role in kidney stone formation in humans, has not yet been demonstrated. Studies with exogenous radiolabeled oxalate in normal humans have shown that renal excretion accounts for most of the disposal of oxalate.⁴³¹ In contrast, report of a case of enteric hyperoxaluria showed drastically diminished expression of *Slc26A6* protein in the intestine.⁴³²

The potential underlying mechanisms of hyperoxaluria in obesity have been investigated in *ob/ob* mice.^{433,434} These *ob/ob* mice had significantly higher urinary oxalate excretion than pair-fed controls, demonstrating that hyperoxaluria is not because of increased oxalate intake. When jejunal tissue was mounted on an Ussing chamber, defective oxalate secretion was shown in *ob/ob* mice. There was less *Slc26A6* mRNA and protein expression in *ob/ob* mice compared with controls. The contribution of regional and systemic inflammation in this model was supported by high plasma and jejunal tumor necrosis factor interferon and interleukin-6 levels in *ob/ob* mice.

Role of *Oxalobacter Formigenes*

Many gut bacteria, including *Oxalobacter formigenes* (OF), degrade oxalate in the intestinal lumen via oxalate decarboxylase.⁴³⁵ OF was initially found in ruminants, but is also present in other species, including humans.^{436,437} Colonization with this bacteria begins approximately during childhood and may be found in feces of 60% to 80% of adults.⁴³⁸ Dietary oxalate intake influences the colonization of in the GI tract. In animal studies, a significant decline in urinary oxalate follows administration of or increase in OF colonization.^{439,440} In a cross-sectional study, patients with recurrent calcium oxalate stones had lower OF colonization than normal subjects matched for age and gender.⁴⁴¹ However, urinary oxalate excretion was no different between the two populations because the study was performed under an uncontrolled diet. In another study, Seiner and colleagues⁴⁴² have shown that OF-negative calcium oxalate stone formers have higher urinary oxalate excretion than OF-positive patients on a controlled diet⁴⁴² (Fig. 38.17). Intestinal oxalate absorption using ¹³C₂-labeled oxalate was similar in OF-positive and

OF-negative patients, but plasma oxalate concentrations were significantly higher in the OF-negative population. The results supported the role of reduced oxalate secretion in OF-negative calcium oxalate kidney stone formers. An ex vivo Ussing chamber study has demonstrated that OF not only degrades intestinal luminal oxalate, but has the capacity to stimulate net intestinal oxalate secretion.⁴⁴³ This pathophysiologic scheme from animal experiments was confirmed in patients with type I PH, in subjects with normal renal function, and in patients with CKD, showing a transient reduction of urinary oxalate following the oral administration of OF.⁴⁴⁴

Renal Excretion

The kidney plays a major role in oxalate homeostasis. Oxalate is not significantly protein-bound and is freely filtered at the glomerulus. With impaired kidney function, plasma oxalate concentration steadily increases and exceeds its saturation in the blood, therefore enhancing the risk of systemic tissue oxalate deposition. Renal oxalate clearance studies in human subjects are more controversial. Radiolabeled oxalate studies^{445,446} have demonstrated net oxalate secretion, whereas endogenous renal oxalate clearance assessments^{447,448} using direct measurements of serum and urine oxalate levels have demonstrated net reabsorption. However, renal secretion of oxalate with high fractional excretion has been reported in patients with primary or enteric hyperoxaluria.⁴⁴⁶⁻⁴⁴⁸

In the renal proximal tubule, both reabsorption and secretion of oxalate have been shown.^{449,450} *Slc26a6* is expressed in the apical membrane of the proximal renal tubule and modulates the activity of multiple other apical anion exchangers.^{451,452} However, the physiologic role of this anion exchanger in renal oxalate handling has not yet been fully demonstrated. In the *Slc26a6* null mice, the hyperoxaluria appears to be driven mostly by hyperoxalemia rather than by a renal leak of oxalate.⁴²⁸

Clinical Hyperoxaluria

PH typically presents during early childhood with calcium oxalate stones and nephrocalcinosis. This disease is severe and is associated with frequent stone recurrence and impaired

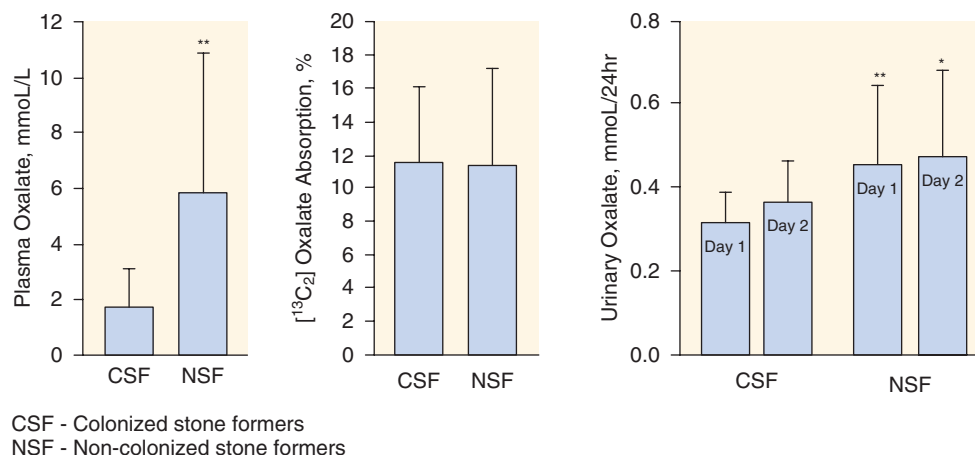


Fig. 38.17 Impact of *Oxalobacter formigenes* on oxalate balance in stone formers. Shown is the plasma oxalate concentration, urine oxalate excretion rate, and intestinal oxalate absorption using ¹³C₂-labeled oxalate in *O. formigenes*-colonized stone formers (CSF) and noncolonized patients (NSF). (From Siener R, Bangen U, Sidhu H, et al. The role of *Oxalobacter formigenes* colonization in calcium oxalate stone disease. *Kidney Int.* 2013;83[6]:1144–1149.)

kidney function.⁴⁵³ From 1985 to 1992, the estimated prevalence of PH was reported to be 1 to 3 cases/million population, a calculated incidence rate of 1/120,000/year.^{454,455} Furthermore, end-stage kidney disease (ESKD) as a result of PH has been demonstrated to occur in up to 10% of children.^{456,457} Occasionally, milder forms of PH are found in adult subjects.

Enteric hyperoxaluria because of inflammatory bowel disease, jejunioileal bypass, and modern bariatric surgeries for morbid obesity are the most common cause of hyperoxaluria in the clinical practice.^{458–463} Roux-en-Y gastric bypass (RYGB) is a weight reduction procedure theoretically combining restrictive and malabsorptive mechanisms for the treatment of obesity. Urolithiasis is a well-known complication of subjects who underwent RYGB. A comparison in 4690 patients following RYGB and an obese control group has shown 7.5% of kidney stones in RYGB patients compared with 4.6% in obese controls. The result of this study was confirmed by another cross-sectional study of 762 patients following bariatric surgery (mostly RYGB) matched with obese

controls who did not undergo surgery.⁴⁶⁴ Even though the prevalence of kidney stones was similar between the two populations at baseline, following RYGB, the incidence of nephrolithiasis increased to 11% compared with 4% at follow-up in obese non-stone-formers. Another retrospective cohort with 972 RYGB patients has shown an 8.8% stone prevalence before surgery, whereas 3.2% developed new stones postoperatively.⁴⁶⁵ The pathophysiologic mechanisms for lithogenesis are complex and variable and may be because of hyperoxaluria, hypocitraturia, aciduria, and low urinary volume.^{392,393,466–474} (Table 38.3). A national insurance claim database from 2002 to 2006 supported the notion that gastric banding is not associated with increased kidney stone risk. After gastric banding, 1.49% of subjects formed stones compared with 5.97% in obese controls.⁴⁷⁵

The underlying mechanisms for hyperoxaluria following inflammatory bowel disease and bariatric procedures is not yet fully defined. Purported mechanisms have linked hyperoxaluria to intestinal fat malabsorption.^{471,476} In this model, unabsorbed fatty acids sequester calcium, which would

Table 38.3 Kidney Stone Risk Profiles Following Roux-en-Y Gastric Bypass Surgery

Parameter	Before RYGB	Following RYGB	Reference
Urinary oxalate (mg/day)	N/A	79	Nelson ⁴⁶⁶
	31 ± 16	65 ± 39 ^a	Sinha ³⁹²
	N/A	85 ± 44 ^a	Asplin ³⁹³
	31 ± 10	41 ± 18 ^a	Duffey ⁴⁶⁸
	N/A	48 ± 4	Penniston ⁴⁷²
	32 (median)	40 (median) ^a	Park ⁴⁶⁷
	N/A	45 ± 21 ^a	Maalouf ⁴⁰⁸
	N/A	61 ± 4 ^a	Patel ⁴⁷⁰
	26 ± 13	32 ± 11 (NS)	Kumar ⁴⁷¹
	N/A	26 (median; NS)	Froeder ⁴⁷³
Urinary citrate (mg/day)	660 ± 297	444 ± 376 (NS)	Sinha ³⁹²
	N/A	477 ± 330 ^a	Asplin ³⁹³
	N/A	441 ± 71 ^a	Penniston ⁴⁷²
	675 (median)	456 (median) ^a	Park ⁴⁶⁷
	N/A	358 ± 357 ^a	Maalouf ⁴⁰⁸
	N/A	621 ± 40 ^a	Patel ⁴⁷⁰
Urinary pH	N/A	472 (median; NS)	Froeder ⁴⁷³
	N/A	5.72 ± 0.31 ^a	Asplin ³⁹³
	5.96 ± 0.38	5.78 ± 0.59 (NS)	Sinha ³⁹²
	5.82 ± 0.54	5.66 ± 0.43 (NS)	Duffey ⁴⁶⁸
	6.03 (median)	5.75 (median; NS)	Park ⁴⁶⁷
	N/A	5.78 (median; NS)	Froeder ⁴⁷³
Urinary volume (mL/day)	1380 ± 400	900 ± 430 ^a	Duffey ⁴⁶⁸
	1800 (median)	1440 ^a	Park ⁴⁶⁷
	N/A	1900 ± 900 (NS)	Maalouf ⁴⁰⁸
	2091 ± 768	1316 ± 540 ^a	Kumar ⁴⁷¹
	N/A	1140 ^a	Froeder ⁴⁷³
Urinary calcium (mg/day)	206 ± 111	112 ± 92 ^a	Sinha ³⁹²
	N/A	141 ± 61 ^a	Asplin ³⁹³
	206 ± 111	112 ± 92 ^a	Duffey ⁴⁶⁸
	161 ± 22	92 ± 15 ^a	Fleischer ⁴⁷⁴
	N/A	100 ± 12 ^a	Penniston ⁴⁷²
	176 (median)	135 ^a (median)	Park ⁴⁶⁷
	N/A	115 ± 93 ^a	Maalouf ⁴⁰⁸
	N/A	89 ^a (median)	Froeder ⁴⁷³

^aSignificant compared with control.

N/A, Not available; NS, statistically nonsignificant; RYGB, Roux-en-Y gastric bypass.

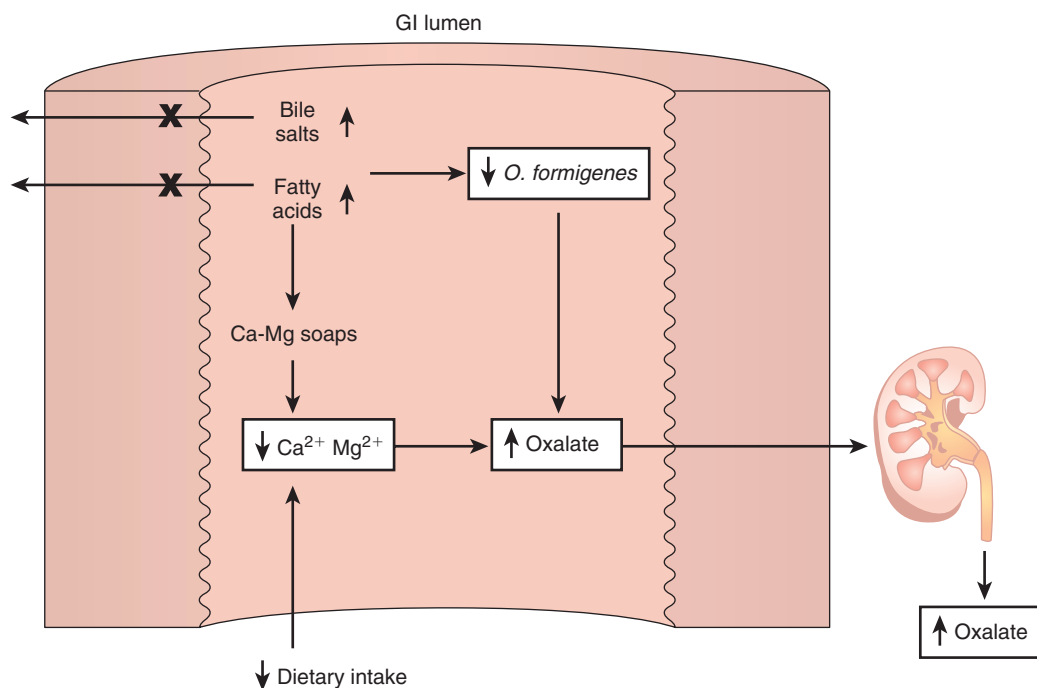


Fig. 38.18 Pathophysiologic mechanisms of hyperoxaluria in inflammatory bowel disease or following bariatric surgery. Fatty acids and bile salts precipitate luminal calcium, which can be compounded by low dietary intake. Low luminal calcium frees up and increases unbound oxalate. Lack of *Oxalobacter formigenes* because of excess bile salts also contributes to high luminal oxalate levels. *GI*, Gastrointestinal.

otherwise bind oxalate in the intestinal lumen, and thereby increase free luminal concentrations of oxalate, enhancing its availability for absorption. An additional mechanism may involve increased permeability of the colon from exposure to unconjugated bile acids and long-chain fatty acids in both inflammatory bowel disease and bariatric procedures^{410,477} (Fig. 38.18). Finally, it was proposed that changes in intestinal microbiota may occur in patients with recurrent CaOx nephrolithiasis,^{439,478,479} enteric hyperoxaluria,^{480,481} and cystic fibrosis,^{441,482} all of which are conditions that can potentially modify the colonization of lower intestinal flora with respect to OF.^{441,482}

Hyperoxaluria can develop early or late after bariatric surgery. The timing varies depending on the type of surgical procedure and nutritional and health status of the subjects.^{468,469} In one study following RYGB,⁴⁶⁴ urinary oxalate levels increased over time, with a more significant rise in those who developed kidney stones. CaOx supersaturation was found to be highest in those patients who developed kidney stones. As a result, the incidence of CaOx stones following RYGB increased but the incidence of stones with hydroxyapatite, struvite, and uric acid did not change.

Physicochemical Effects of Hyperoxaluria

Oxalate and calcium are both important in raising calcium oxalate supersaturation in urine.⁴⁰⁴ In human serum, the oxalate concentration is 1 to 5 μM but its concentration in the urine is 100 times higher than that of serum.⁴⁸³ At physiologic pH, oxalate forms an insoluble salt with calcium. Normal urine is usually supersaturated with calcium oxalate; the solubility of CaOx in an aqueous solution is approximately 57 μM at a pH of 7. Considering that normal urine volume

is between 1 and 2 L/day, normal urinary excretion is $<45 \mu\text{mol/day}$. In most cases, normal urine is supersaturated with CaOx salt but blood is undersaturated. Nevertheless, in patients with PH and renal insufficiency, when the serum oxalate concentration rises above 30 μM , the blood also becomes supersaturated with CaOx.⁴⁸⁴

ALTERATIONS OF URINARY PH

Both highly acidic (≤ 5.5) and highly alkaline (≥ 6.7) urine increase the propensity for calcium kidney stone formation. With unduly acidic urinary pH, urine becomes supersaturated with undissociated uric acid that can contribute to CaOx crystallization.^{102,485} Highly alkaline urine increases the abundance of monohydrogen phosphate (dissociation constant $\text{pK}_a \approx 6.7$), which, in combination with calcium, transforms to thermodynamically unstable brushite ($\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$) and, finally to hydroxyapatite, $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$. Over the past 4 decades, the average CaP content of stones has progressively increased.⁷⁶

The rise in the prevalence of CaP stones has been attributed to but not proven to be as a result of extracorporeal shock wave lithotripsy (ESWL), increasing use of medications including topiramate, and alkali therapy.^{20,224,486,487} The three main risk factors for the development of brushite stone are alkaline urine, hypercalciuria, and hypocitraturia (Fig. 38.19). It has been suggested that urinary pH elevation plays the most important role in the transformation of CaOx to CaP stones. A retrospective study of 62 patients has found that high urinary pH is the primary physiologic abnormality in those who evolved from CaOx to CaP.⁷⁶ Defective renal acidification was proposed to be related to renal tissue damage from ESWL.²⁰ Patients with CaP stones usually carry a greater stone burden and are less likely to be stone-free after urologic

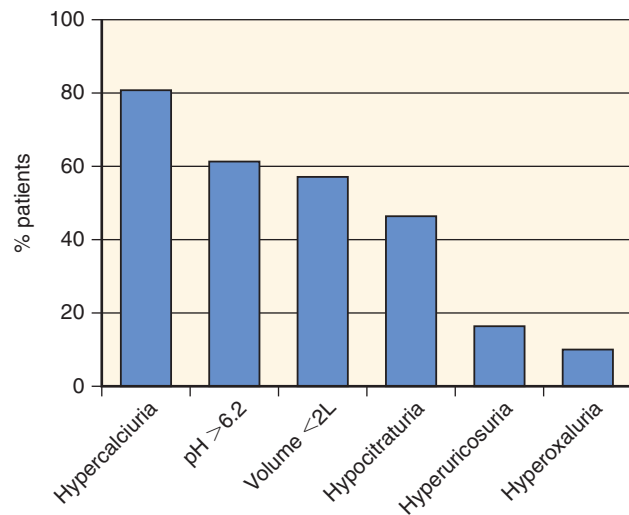


Fig. 38.19 Urinary stone risk profiles in 82 brushite kidney stone formers. (Modified from Krambeck AE, Handa SE, Evan AP, Lingeman JE. Brushite stone disease as a consequence of lithotripsy? *Urol Res*. 2010;38[4]:293–299.)

procedures; they also experience resistance to ESWL and ultrasonic lithotripsy.^{488–490}

URIC ACID STONES

Higher primates have higher serum and urinary UA levels because of lack of the enzyme uricase, which transforms UA into more soluble allantoin.^{355,356,491} Uricase deficiency is partially compensated by repression of xanthine oxidase⁴⁹² but the high serum UA level is also because of low renal fractional excretion of UA.³⁵⁶ The principal sources of UA production are de novo synthesis, tissue catabolism, and dietary purine load.³⁵⁰ Generally, 50% of the typical daily urate load is provided by de novo synthesis and tissue catabolism, with the remainder originating from dietary purines.³⁵⁰ About 30% of synthesized UA is excreted through the intestinal tract, where it undergoes uricolysis, and the remainder by the kidney.³⁵²

The cause of uric acid nephrolithiasis can be genetic^{355,493,494} or acquired,^{390,395,495} with the metabolic syndrome being the principal cause (Table 38.4). The primary pathophysiologic mechanisms accounting for UA nephrolithiasis include hyperuricosuria, and low urinary volume; the most important contributor is an unduly low urinary pH.

PHYSICOCHEMISTRY OF URIC ACID

Given that UA solubility in a urinary environment is limited to about 96 mg/L, and UA excretion in humans typically exceed 600 mg/day, the risk of UA precipitation is constant.^{222,401} Uric acid has a pK_a of 5.35 at 37°C,^{496,497} so UA solubility is determined primarily by pH in that acidic urine (pH ≤5.5) titrates urate to UA, which is sparingly soluble and therefore precipitates.^{401,498,499} This may also indirectly induce mixed UA-CaOx nephrolithiasis, which can be mediated through heterogeneous nucleation and epitaxial crystal growth.^{104,500,501} Generally, urine is metastably supersaturated with respect to UA. This suggests that the absence of an inhibitor may influence the propensity for UA nephrolithiasis;

Table 38.4 Causes and Mechanisms for Uric Acid Nephrolithiasis

Causative Factors	Low Urine Volume	Low Urinary pH	Hyperuricosuria
Acquired			
• Diarrhea	+	+	+
• Myeloproliferative			+
• High animal protein		+	+
• Uricosuric drugs			+
• Primary gout		+	+
• Metabolic syndrome		+	
Congenital			
• Enzyme disorders of purine metabolism			+
• Mutations in uric acid transporters			+

this has been supported by experimental evidence demonstrating the presence of macromolecules that attenuate UA crystal adherence to renal tubular epithelium.⁵⁰²

PATHOPHYSIOLOGY OF URIC ACID STONES

Hyperuricosuria

Hyperuricosuria may be caused by genetic, metabolic, or dietary factors.⁴³ Hyperuricosuria is seen in rare hereditary disorders with mutations in enzymatic pathways responsible for UA production. These conditions include X-linked phosphoribosyl synthetase overactivity, autosomal recessive glucose-6-phosphatase deficiency, and glycogenosis types III, V, and VII.^{43,355,494} The clinical presentations include gout, kidney stones, and even ESKD. Biochemical abnormalities include a highly elevated serum UA concentration (≥10 mg/dL) and urinary UA excretion of 1000 mg/day or higher. The disease manifestations commonly present during childhood; however, they may remain silent until puberty.

Renal handling of UA is complex and involves secretion and reabsorption.³⁵⁶ Inactivating mutation of the urate transporter, URAT1, causes renal uric acid wasting, resulting in hyperuricosuria, hypouricemia, UA kidney stones, and exercise-induced acute renal failure.^{503,504} In addition, in the Sardinian population, a putative gene locus localized on chromosome 10q122 has been linked to UA nephrolithiasis. The putative culprit gene codes for a hypothetical protein with zinc finger motifs, ZNF365, of unknown function.⁵⁰⁵

Hyperuricosuria may also occur with excessive tissue breakdown in malignancy, especially during chemotherapy.^{506,507} Certain uricosuric drugs such as high-dose salicylates, probenecid, radiocontrast agents, and losartan have been shown to increase UA excretion, which may increase the risk of UA stone formation.^{497,508} In the large majority of cases of UA stones, there are no identifiable congenital or acquired causes of hyperuricosuria.

Low Urinary Volume

Low urinary volume increases the urinary supersaturation of all stone-forming constituents.⁵⁰⁹ Volume depletion from chronic diarrhea in inflammatory bowel disease has an impact on UA stone formation. In addition, UA stones comprise up to two-thirds of all stones in patients following ileostomy.⁵¹⁰ In this population, low urinary volume, in addition to the aciduria from intestinal alkali loss, is common.^{511–513}

Low Urinary pH

In the large majority of cases, hyperuricosuria or low urine volume are not the culprits. Rather, the lithogenesis is driven by a low urine pH. This is collectively called, by default, idiopathic uric acid nephrolithiasis (IUAN). The necessary requirement is unduly acidic urine, which is the single invariant feature in all patients with IUAN^{47,401,514} (Fig. 38.20).

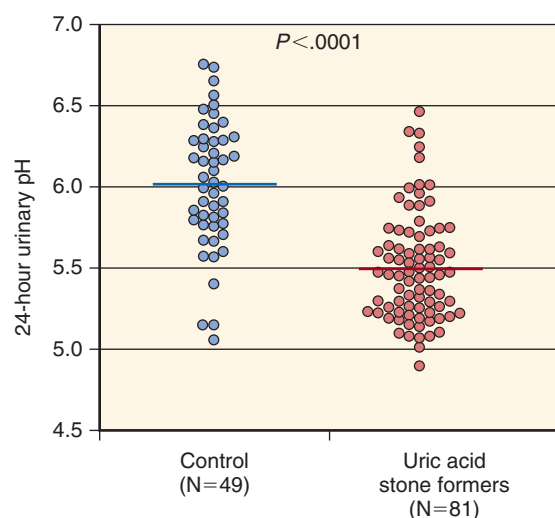


Fig. 38.20 Shown is 24-hour urine pH in idiopathic uric acid nephrolithiasis versus control subject. Urine was collected under a fixed metabolic diet. Comparison was by an unpaired t-test.

ORIGIN OF LOW URINARY PH

IUAN is the stone type that shares numerous common characteristics with the metabolic syndrome.^{49,493,515,516} Several cross-sectional studies have supported the association between UA stones, diabetes, and obesity^{48,50,51,517–520} (Fig. 38.21). Over the past 1.5 decades, two major causative factors, increased acid load to the kidneys and impaired ammonium (NH_4^+) excretion, have been shown to result in unduly acidic urine.^{47,401}

Increased Acid Load to Kidneys

In a steady state, net acid excretion (NAE) equals net acid production. Higher NAE can occur as a consequence of increased endogenous production, increased dietary ingestion (high dietary acid consumption or low alkali intake), or alkali loss.⁵²¹ Several metabolic studies of a constant controlled metabolic diet have shown that NAE is higher in IUAN patients and type 2 diabetics without stones compared with control subjects,^{47,515,516,522–524} suggesting that endogenous acid production is elevated (Fig. 38.22). The nature and source of these putative organic acids is unknown at present but likely has an enterohepatic origin.⁵²⁵ Adults with type 2 diabetes without kidney stones have similar features to those with UA stones and similar differences in gut microbiota compared with nondiabetic adults.⁵²⁶

Impaired NH_4^+ Excretion

Under normal physiologic circumstances, NH_4^+ excretion plays a key role in the regulation of acid-base balance as a result of its high pK_a , of 9.3 ($\text{NH}_3/\text{NH}_4^+$ system). NH_4^+ can effectively buffer a major portion of secreted H^+ as a result of its high capacity and high pK_a .^{527,528} The alternative source of buffering is by many H^+ acceptors, which are collectively referred to as titratable acid (TA).⁵²⁷ In IUAN, there is defective NH_4^+ production and excretion (Fig. 38.23). More H^+ is buffered by TA (including urate) to maintain acid-base homeostasis.^{47,401,516} The tradeoff is propensity for UA precipitation.^{47,514} The defective NH_4^+ excretion in IUAN occurs at a steady state while the patient is on a fixed metabolic diet, as

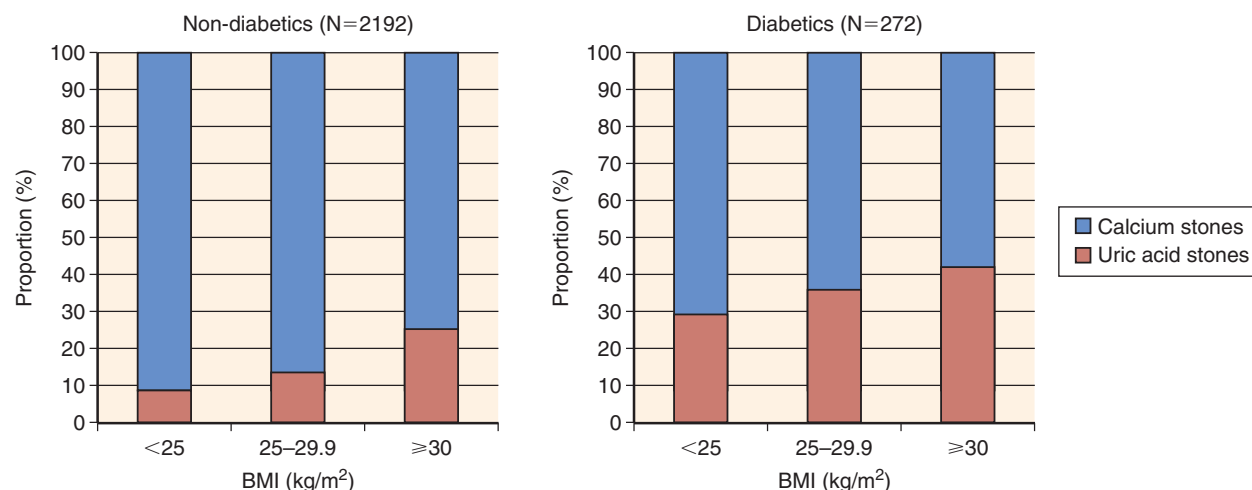


Fig. 38.21 Distribution of stone type with respect to body mass index (BMI) and diabetes status. (Based on data from Daudon M, Traxer O, Conort P, et al. Type 2 diabetes increases the risk for uric acid stones. *J Am Soc Nephrol.* 2006;17[7]:2026–2033.)

well as after acute acid loads.⁴⁷ Defective NH_4^+ excretion is not unique to UA stone formers but remains a shared feature between metabolic syndrome and type 2 diabetic non-stone formers (Fig. 38.23).^{49,493,516,520}

Role of Renal Lipotoxicity

Most urinary NH_4^+ is produced and secreted by the renal proximal tubular cell.^{529,530} NH_4^+ is directly transported by a sodium hydrogen exchanger (NHE3) as $\text{Na}^+/\text{NH}_4^+$ into the proximal tubular lumen or by nonionic diffusion of NH_3 into the renal proximal tubular lumen, trapped as NH_4^+ by luminal H^+ secretion by NHE3.^{529–535} When energy intake exceeds utilization, as occurs in patients with IUAN, obesity,

diabetes, and metabolic syndrome, fat accumulates, and this can occur in nonadipose tissues⁵³⁶ (Fig. 38.24). Ectopic fat is termed *steatosis*, and the ill effect of steatosis is termed *lipotoxicity*,^{536,540–544} which is the consequence of the accumulation of toxic metabolites such as acyl coenzyme A, diacylglycerol, and ceramide.^{545–547} In a rodent model of the metabolic syndrome, the Zucker diabetic fatty (ZDF) rat,⁵⁴⁸ as well as in cultured proximal tubular cells, steatosis led to deranged ammoniogenesis and transport.⁵³⁷ ZDF rats have higher renal cortical triglyceride content commensurate with lower urinary NH_4^+ and pH ⁵³⁷ (see Fig. 38.24). Similar findings were also seen in humans with a high BMI.⁵³⁸ Kidney cell lines incubated with a mixture of long-chain fatty acids have a decrease in NH_4^+ production.⁵³⁷ Treatment of ZDF rats with thiazolidinediones or removal of fatty acids from cultured cells reduced renal steatosis and restored acid-base parameters toward normal, demonstrating causality (see Fig. 38.24).⁵³⁹ Another potential mechanism in UA stone formers and/or in ZDF rats is defective NH_4^+ synthesis because of substrate competition—namely, fatty acid instead of glutamine as a source of energy—thus removing the source of nitrogen for ammoniogenesis.⁵⁴⁹

CYSTINE STONES

OVERVIEW

Cystine stones are exclusively seen in cystinuria, which is a hereditary mendelian disorder of a heterogenous nature caused by inactivating mutations of the subunits of a dibasic amino acid transporter in the proximal tubule's rBAT or b⁰AT (SLC3A1 or SLC7A9—heavy and light chain, respectively); this is equivalent to the system b⁰ of amino transport (see Chapter 8; Fig. 38.25). Cystinuria is the most common primary inherited aminoaciduria (OMIM 220100). There are no systemic symptoms of amino acid deficiency in this disorder, but it causes 1% to 2% of renal stones in adults and 6% to 8% in pediatric patients.⁵⁵⁰ Normally, reabsorption of amino acids from the urine is almost complete. Inactivation

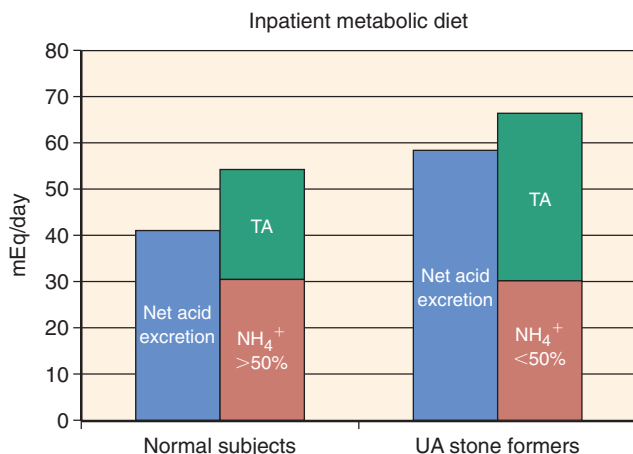


Fig. 38.22 Comparison of net acid excretion between normal subjects and uric acid (UA) stone formers. Studies were performed with a controlled metabolic diet. Despite equivalent amounts of exogenous acid intake (same urinary sulfate excretion, not shown), uric acid stone formers were found to have higher net acid excretion, and a lower fraction of the net acid was carried by ammonium. TA, titratable acid. (Adapted from data from references 516, 523, and 524.)

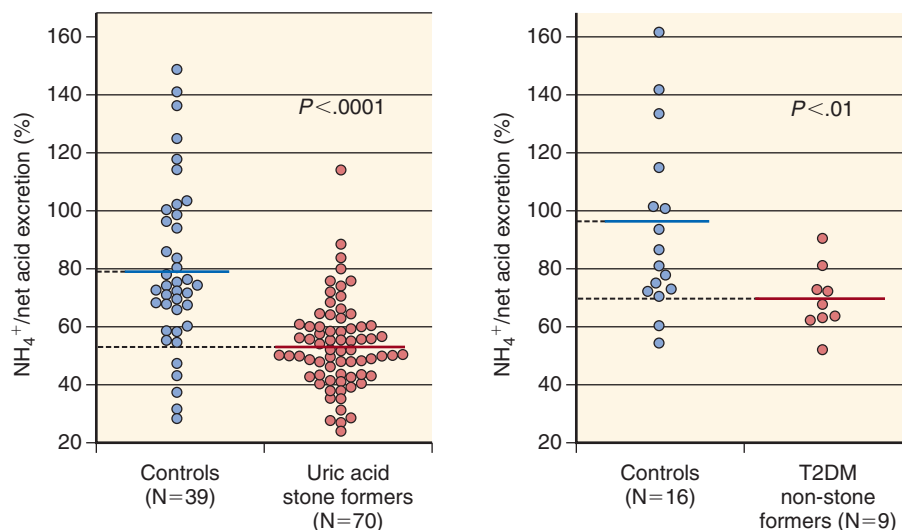


Fig. 38.23 Decreased ammonium excretion as a fraction of net acid excretion in idiopathic uric acid stone formers and in subjects with type 2 diabetes (T2DM) without stones. Individuals were studied under controlled metabolic diet so all differences are intrinsic to the subjects. (From Maalouf NM, Cameron MA, Moe OW, Sakhaee K. Metabolic basis for low urine pH in type 2 diabetes. *Clin J Am Soc Nephrol*. 2010;5[7]:1277–1281.)

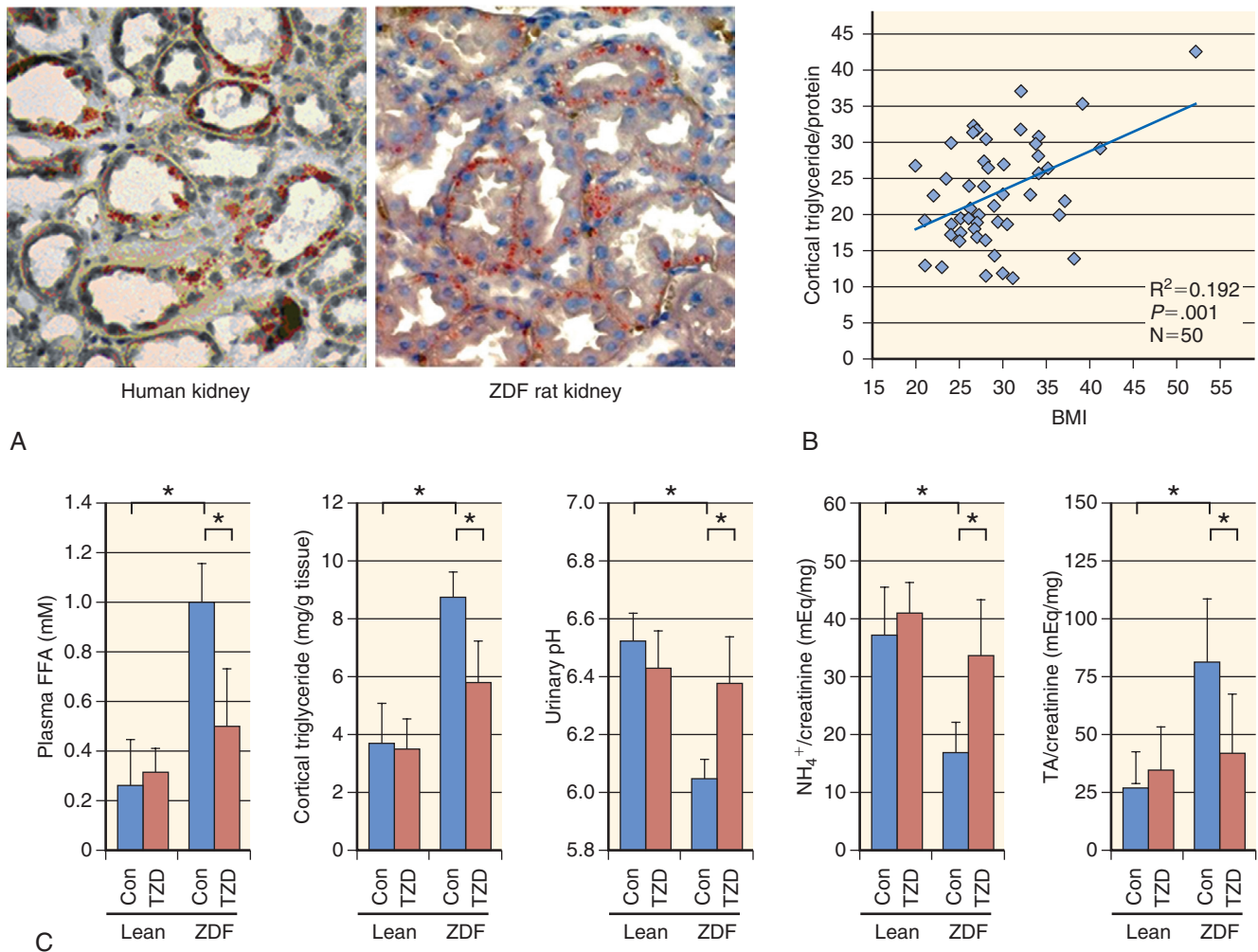


Fig. 38.24 Renal steatosis and lipotoxicity in humans and rats. (A) Oil red O stain of triglyceride in human kidney from an obese individual and the Zucker diabetic fatty (ZDF) rat. (B) Cortical triglyceride content in human biopsy samples showing the relationship between triglyceride and body mass index (BMI). (C) Lean rats or ZDF rats were treated with vehicle control, (Con), or rosiglitazone thiazolidinedione (TZD). A number of parameters were compared among the four groups. The amelioration of plasma free fatty acids (FFAs) and renal cortical triglycerides was accompanied by an increase in urine pH and ammonium excretion, whereas titratable acid (TA) excretion was decreased. (Adapted from references 537 to 539.)

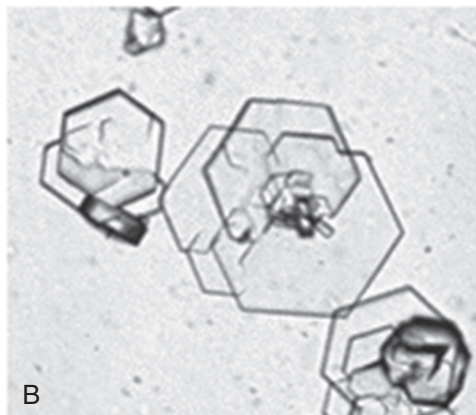
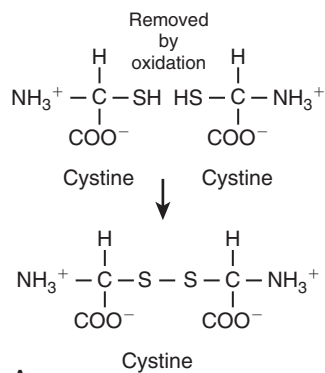


Fig. 38.25 (A) Cystine is a dimer of cysteine formed by disulfide bonding under oxidizing conditions. (B) Appearance of cystine crystals in the urine of a patient with cystinuria.

of rBAT/b⁰⁺AT leads to urinary wasting of many cationic amino acids but only cystine has a low enough solubility to precipitate. Because rBAT/b⁰⁺AT is also present in the gut, there is some malabsorption of cationic amino acids but there is no clinically discernible intestinal phenotype.

MOLECULAR BIOLOGY AND GENETICS

The rBAT and b⁰⁺AT subunits are linked by a disulfide bridge in the heterodimer, which is characteristic of the heteromeric amino acid transporters⁵⁵¹ and mediates obligatory exchange of cationic amino acids, (e.g., cystine = two cysteines bound by a disulfide bridge) and neutral amino acids with 1:1 stoichiometry (see Chapter 8). In addition to human disease, the causal link between rBAT/b⁰⁺AT and cationic aminoaciduria has been corroborated by mouse experimental models with defective rBAT (D140G mutation)⁵⁵² or b⁰⁺AT (knock-out),⁵⁵³ and by Newfoundland dogs with defective rBAT (naturally occurring nonsense mutation),⁵⁵⁴ all of which have cystinuria similar to that in humans.

Over 130 human mutations in rBAT (*SLC3A1*; cystinuria type A) and 100 mutations in b⁰⁺AT (*SLC7A9*; cystinuria type B) have been identified, including missense, nonsense, splice site, frame shift, and large chromosomal rearrangements⁵⁵⁰ in around 90% of the subjects studied. In around 3% of patients with clinical cystinuria, there are no mutations in the two candidate loci. These may be as a result of mutations in promoter, regulatory, or intronic regions. It is believed that all cases of classic and isolated cystinuria are as a result of mutations in system b⁰⁺.

Interestingly, haplotypes with b⁰⁺AT polymorphisms have been reported in cystinuria and some b⁰⁺AT heterozygotes who are cystine stone formers.^{555,556} Isolated cystinuria (cystine wasting without cationic aminoaciduria) can be caused by a heterozygous b⁰⁺AT mutation.⁵⁵⁶⁻⁵⁵⁸ The molecular biology is in accordance with metabolic data. System b⁰⁺ (rBAT/b⁰⁺AT) is the principal transport system for cystine reabsorption in the proximal tubule. Cystine clearance approximates the GFR (glomerular filtration rate; i.e., no reabsorption) in classic cystinuria.⁵⁵⁹ In contrast, clearance of cationic amino acids is only partly affected (GFR = 40–60 mL/min × 1.73 m²) in cystinuria,⁵⁶⁰ suggesting that other apical transport systems participate in the renal reabsorption of these amino acids.

CLINICAL PRESENTATION

Cystine was discovered in the urine of a patient with bladder stones by Wollaston in 1810.⁵⁶¹ The prevalence is reported to be highest in individuals of Libyan-Jewish descent (1:2500). A history of consanguinity may or may not be present because compound heterozygotes can come from seemingly unrelated partners. Patients present with hematuria, renal colic, and urinary obstruction, although their symptoms tend to be more severe and more likely to have staghorn, surgery, and chronic kidney disease (CKD).⁵⁶² The age of onset varies widely but probably 50% of patients present to pediatricians.⁵⁶³ The historical type I versus type II classification is now replaced by type A (mutations in both alleles of *SLC3A1*; genotype AA) and type B (mutations in both alleles of *SLC7A9*; genotype BB).⁵⁶⁴ Uncommon digenic (*SLC3A1* and *ALC7A9*) inheritance (type AB) has also been described, but individuals with this pattern are less likely to suffer from kidney stones because they have a 25% chance of having a completely wild-type b⁰⁺ system.

Urinary cystine quantification is not routinely performed in all stone formers. The features that raise suspicions are family history of cystinuria, staghorn calculi, positive non-quantitative screening test with sodium nitroprusside (>75 mg/L; 0.325 mM), and the presence of pathognomonic hexagonal cystine crystals on urinalysis (Fig. 38.25).

Quantitation of urinary cystine excretion (normal, <30 mg/day; 0.13 mmol/day) is mandatory to diagnose cystinuria. Patients with cystinuria often excrete more than 400 mg/day (1.7 mmol/day). Poor cystine solubility and precipitation can lead to misleadingly low results, so alkalinization of collected urine is often required.⁵⁶⁵ One caveat is the inability of current assays to distinguish cystine from soluble adducted drug-cystine complexes. Ex vivo disintegrated thiol cysteine can self-recombine to form dimeric cystine. These problems have led to the development of a solid-phase assay, which is reliable even in the presence of thiol drugs.⁵⁶⁶ One form of physicochemical assay similar to that for calcium stones is potentially useful to empirically measure the propensity for cystine stone formation.⁵⁶⁶

MANAGEMENT

Fluid, alkali, and dietary modification (salt and protein) are first-line therapies, with thiol therapy as second line. To reduce urine cystine supersaturation to less than 1.0, or concentration less than 243 mg/L (1 mmol/L) requires drinking around 4 L/day of fluids, including nocturnal intake.⁵⁶⁷ Off-label use of a vasopressin receptor antagonist has been proposed to induce pharmacologic diabetes insipidus,⁵⁶⁸ but the long-term effects of this drug has not been evaluated, and the cost is prohibitive. In conjunction with the safe and easily available water, it is unclear whether there is an advantage to this therapy.

Cystine excretion has been shown to be reduced by decreasing dietary sodium, although no clinical outcome studies have been performed.⁵⁶⁹ The physiology of this effect is not known but the use of the b⁰⁺ system for L-DOPA (levodopa) uptake to synthesize nephrogenic dopamine is a possibility,⁵⁷⁰ so in a high-salt diet, the b⁰⁺ system may be occupied with dopamine synthesis. In the absence of ill effects of salt restriction, it is reasonable to prescribe restricting salt intake to about 2 g/day. Reduction of animal protein has also been proposed because of the reduction of dietary cystine, and its precursor methionine may reduce cystine excretion,⁵⁷¹ and the associated increase in urinary pH (UpH).

Pharmacologic Therapy

Cystine solubility increases with increasing pH, which is achieved with oral potassium citrate. (This is discussed in the section on alkali therapy of kidney stones.) One can start with 20 mEq bid, which may or may not be sufficient based on periodic UpH testing. In over 50% of patients, the previously mentioned measures are unlikely to suffice, and treatment with a thiol agent is indicated. The two available agents are D-penicillamine and alpha-mercaptopyrionylglycine, or tiopronin. These drugs work by reducing the disulfide bond of cystine, producing mixed compounds with cysteine, which are all more soluble than cystine.⁵⁷² The incidence of adverse effects is lower with tiopronin, thus rendering it as a first-line agent. D-Penicillamine is often used for patients with adverse reactions to tiopronin. The prescription of these drugs is discussed in more detail in a subsequent section.

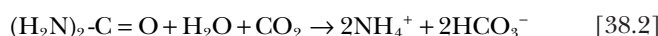
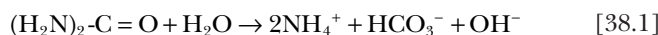
INFECTION STONES

PATHOPHYSIOLOGY

This category of stones does not form because of intrinsic host defects but because of an infected urinary environment. Struvite ($\text{MgNH}_4\text{PO}_4 \cdot 6\text{H}_2\text{O}$) stones account for a low percentage of all kidney stones and often also contain carbonate apatite, $\text{Ca}_{10}(\text{PO}_4)_6 \cdot \text{CO}_3$. These stones are rapidly growing, branch, and enlarge and fill the renal collecting system to form staghorns. The key culprits are urea-splitting bacteria. These stones are difficult to treat medically. Even with surgical removal, any remaining fragments containing the infecting bacteria furnish a nidus for further rapid stone growth. The nature of these stones to grow rapidly, recur, and cause morbidity and mortality has led to the appellation *stone cancer*.

Struvite stones occur more frequently in women than in men, largely because of the higher incidence of urinary tract infections. Chronic urinary stasis or infections predispose to struvite stones so that old age, neurogenic bladder, indwelling urinary catheters, and urinary tract anatomic abnormalities are all predisposing factors. The presence of large stones in infected alkaline urine should alert the clinician to the potential presence of struvite. Given their potential for rapid growth and substantial morbidity, early detection and eradication are essential.^{573,574}

The urease of urea-splitting organisms hydrolyzes the following reactions:



Urea contains two nitrogens and one carbon. On the product (right-hand) side of the equation, 2NH_4^+ represents two acid equivalents, and either HCO_3^- plus OH^- or 2HCO_3^- represents two base equivalents. The phosphate and magnesium combines with the NH_4^+ to form struvite, and the calcium and phosphate combine with the carbonate to form carbonate apatite. Ammonium can bind to the sulfates on the GAGs on the urothelium,⁵⁷⁵ impair the hydrophilic activity of the GAGs, and increase crystal adhesion. During infections with urease-producing organisms, there is a simultaneous elevation in urine NH_4^+ , pH, and carbonate concentration. With successful antimicrobial treatment of the underlying infection, the struvite can actually dissolve because the urine is generally undersaturated with respect to struvite.^{576,577} However, urine is not undersaturated with respect to carbonate apatite,⁵⁷⁷ so successful antimicrobial therapy is not expected to dissolve this component of the stone. Whether a stone will dissolve with prolonged antibiotic is dependent on the amount of carbonate apatite.

Although many bacteria (gram-negative and gram-positive), *Mycoplasma*, and yeast species can produce urease, most urease-producing infections are caused by *Proteus mirabilis*. In addition, *Haemophilus*, *Corynebacterium*, and *Ureaplasma* have been identified to cause struvite stones. All these bacteria use urease to split urea and supply their need for nitrogen (in the form of NH_3). Colony counts may be low, so the laboratory should be instructed to identify any bacteria and determine sensitivities, no matter how low the number of colony-forming units. If routine urinary cultures are negative but a urease producer is suspected,

the laboratory should be specifically instructed to culture for *Mycobacterium* and *Ureaplasma urealyticum*, which are also urease-positive.⁵⁷⁵

UNCOMMON STONES

Rare kidney stone disease can be genetic or acquired. Among the genetic diseases, the most common types are xanthine stones and 2,8-dihydroxyadenine stones. Acquired causes may be iatrogenic with the use of medications or caused by toxins or other diseases.

GENETIC CAUSES

2, 8-Dihydroxyadenine (DHA) stone is characterized by excessive production and urinary excretion of DHA as a consequence of adenine phosphoribosyl transferase deficiency (APRT), which is inherited as an autosomal recessive disorder (OMIM, 102600). In subjects with impaired APRT, adenine is converted to 8-hydroxyadenine, which is further metabolized to DHA by xanthine dehydrogenase.⁵⁷⁸ Consequently, APRT deficiency results in high urinary levels of DHA, which is insoluble and forms crystals that aggregate, grow, and form kidney stones.^{579,580} This disorder can present at any age but, in approximately 50% of subjects, symptoms do not occur until adulthood.⁵⁸¹ The diagnosis is made by the microscopic appearance of DHA crystals in the urine, which are pathognomonic for this disease. Typically, DHA crystals by polarized microscopy are round and reddish-brown, with a characteristic central Maltese cross pattern.⁵⁷⁸ However, the diagnosis is established by the absence of APRT enzyme activity in red cell lysates or the identification of functionally significant mutations in *APRT*.

Xanthine stones are present in about one-third of subjects with classic xanthinuria, which is an inborn error in metabolism inherited as an autosomal recessive trait.^{582,583} Hereditary xanthinuria is caused by mutations in xanthine dehydrogenase (XDH),⁵⁷⁸ leading to the overproduction of xanthine and minimal production of uric acid. Patients have very low serum urate levels but suffer from elevated levels of xanthine in the urine, leading to xanthine stones, hematuria, and sometimes occult kidney failure.⁵⁸⁴ Urinary excretion of xanthine and hypoxanthine are significantly increased. Xanthine stones occur more often than hypoxanthine because of the lower solubility of xanthine in urine. Xanthinuria should be suspected if a patient has significant hypouricemia and hypouricosuria in the presence of radiolucent stones.

Xanthine stones may also be acquired following allopurinol treatment in patients with significant hyperuricemia such as Lesch-Nyhan syndrome and in those undergoing chemotherapy for myeloproliferative disorders.^{585,586}

ACQUIRED CAUSES

Rare acquired kidney stones may arise from disease and also after ingesting toxins or using certain drugs with poor solubility in urinary environments.⁵⁸⁷ Ammonium urate stones are radiolucent stones that occur in patients with chronic diarrhea, inflammatory bowel disease, ileostomy, and laxative abuse; all are associated with intestinal alkali loss and compensatory renal hyperexcretion of ammonium.⁵⁸⁸⁻⁵⁹² This occurs because of the relative abundance of urinary ammonium accompanied with decreased urinary sodium and potassium levels, thereby

making the urine supersaturated with respect to poorly soluble ammonium urate. The catabolic state of these patients may also cause hyperuricosuria.

Consumption of over-the-counter (OTC) drugs, such as guaifenesin as an expectorant and ephedrine as a stimulant and for weight reduction, accounts for approximately one-third of the drug-induced U.S. kidney stone incidence.^{593,594} After the introduction of protease inhibitors for the treatment of human immunodeficiency virus (HIV) patients, indinavir-treated patients began to show a high incidence of indinavir-associated stones.^{595,596} In later years, it was shown that other antiproteases, such as nelfinavir, tenofovir, atazanavir, and antinucleosidic drugs, including efavirenz, also cause kidney stone formation.^{597–600}

Other drugs such as triamterene, various antimicrobial agents, including sulfonamides, penicillin, cephalosporin, quinolones, and nitrofurantoin, as well as agents such as magnesium trisilicate, have been shown to be associated with drug-induced stones.⁵⁸⁷

Other environmental factors play a role in kidney stone formation. Melamine is an organic nitrogenous compound used in the industrial productions of plastics, dyes, fertilizers, and fabrics.⁶⁰¹ In 2008, kidney stone cases were reported in infants and children in China consuming melamine-contaminated milk.⁶⁰² A study in Taiwan screened 1129 children with potential exposure to contaminated milk formula. The results showed that those with a high exposure had an increased incidence of nephrolithiasis. The age of the group of children with kidney stones was reported to be significantly younger than those without. Metabolic workups did not disclose any evidence of hypercalciuria, and the stones were radiolucent, indicating that these stones were related to melamine ingestion.^{603,604}

GENETICS

HUMAN GENETICS

There are a number of factors involved.

FAMILIAL CLUSTERING

It is well accepted that a significant portion of the risk of kidney stones in humans, including its major risk factor, hypercalciuria, is hereditary in origin.^{605–608} The complex polygenic trait with phenocopying, multiple loci, loci heterogeneity, and lack of intermediate phenotypes in a clinical database all render the deciphering of the genetics of hypercalciuria and kidney stones a formidable task. The familial clustering of kidney stones and of hypercalciuria has been documented for decades.²⁹¹ Despite the fact that family members share similar lifestyles, a number of studies have corrected for confounding factors and still show an increased risk of kidney stones in relatives of stone formers. This is also the anecdotal experience of most clinicians. In an epidemiologic study of more than 300,000 patient-years, adjusting for dietary factors, age, and BMI, there was a 2.6-fold higher risk of kidney stones when there was a positive family history.⁶⁰⁹ Case-control studies with combined number of subjects in cases and controls exceeding 1000 have shown the relative risk to be two- to fourfold higher and the hereditary contribution (defined as the ratio of genetic variance

to the total phenotypic variance) to be 40% to 60%.^{610–613} In a segregation analysis of more than 200 stone-forming families, hypercalciuria was found to have a heritability of 60% and to be transmitted in a polygenic fashion.⁶¹⁴

ETHNICITY

African-Americans have a lower prevalence of kidney stones than U.S. whites that cannot be accounted for by diet.³ A longitudinal study has also shown persistently lower stone incidence in African-Americans in the 1970s through the 1990s, suggesting that the difference is intrinsic.¹ A similar difference has been noted in urinary calcium excretion, with much lower rates in blacks.⁶¹⁵ Values for U.S. Hispanics and Asians are intermediate between U.S. whites and blacks. In a cross-sectional, population-based study, immigrants from various ethnic backgrounds maintained the relative risk of stones of their native country, despite considerable assumption and homogenization of Western lifestyle.⁶¹⁶

HUMAN GENETIC STUDIES

Twin Studies

Comparison of monozygotic to dizygotic twins and/or plain siblings reared in virtually identical environments have been informative.^{617,618} Goldfarb and coworkers sampled dizygotic and monozygotic twins from the Vietnam Era Twin Registry and found a concordance rate of 32% in monozygotic twins compared with 17% in dizygotic twins, an effect that cannot be explained by the documentable dietary information.⁶¹⁹ Other twin studies obtained urinary chemistries and found the heritability of urinary calcium excretion rate to be around 50%.^{617,618}

Candidate Genes

This approach is based on educated guesses of suspected loci. Studies have demonstrated an association of polymorphisms of genes along the vitamin D axis with one form of phenotypic parameter or another,^{620–623} but corresponding phenotype studies have been negative.^{283,284} Sib-pairs studies in French-Canadians have yielded positive results on candidate genes, including the vitamin D receptor, 1 α -hydroxylase, the CaSR, and crystallization modifiers such as osteopontin, THP, and osteocalcin-related gene, but thus far no firm conclusions have been drawn.^{278,624–626} Association studies in an Italian cohort have suggested that a CaSR functional polymorphism (R990G) mutation^{627–629} was a possible locus. In a Swiss cohort, three nonsynonymous polymorphisms in the intestinal calcium channel, TRPPV6, were found with higher frequency in calcium stone formers. Interestingly, the contemporaneous presence of all three polymorphisms led to increased channel activity.⁶³⁰

Halbritter and associates have demonstrated the power of the candidate gene approach in a complex polygenic disease⁶³¹ by determining the percentage of cases that could be accounted for by mutations in any one of 30 known kidney stone genes. They used a high-throughput mutation analysis from multiple kidney stone clinics of 272 genetically unresolved individuals (106 children and 166 adults) from 268 families with nephrolithiasis or isolated nephrocalcinosis. In their study, 50 putative gene variants (deemed likely but not proven to be mutations) in 14 of 30 analyzed genes were

detected, yielding a genetic diagnosis in 15% of all cases, with 40% previously undocumented base changes. The frequency of monogenic cases was remarkably high in both the adult (11%) and pediatric cohorts (21%). Recessive causes were more frequent among children and dominant disease more in adults. Their report showed that monogenic causes of urolithiasis are much more frequent than expected and illustrate the principle of “one who does not look will not find.” One caution is that causality is still not established in these individuals.

Genome-Wide Association Studies

To date, success using genome-wide association studies (GWAS) has been modest in regard to the identification of loci of kidney stones. A small-scale, whole-genome linkage analysis of absorptive hypercalciuria and low bone mineral density found the soluble adenylyl cyclase (sAC) as a possible locus.^{632,633} Polymorphisms in this gene are also associated with BMD variation in healthy premenopausal women and men. There is a whole host of functions for the sAC protein, but one appears to be as a mediator of low bicarbonate-induced bone resorption.⁶³⁴ A GWAS of >3700 cases and >42,500 controls from Iceland and the Netherlands found synonymous variants in the claudin-14 gene that were associated with kidney stones,³³⁰ with carriers estimated to have a 1.64 times greater risk. Claudin-14 is a paracellular protein that regulates calcium transport in the renal thick ascending limb. The same variants are also associated with reduced bone mineral density.³³⁰ From a similar Icelandic and Dutch database, another GWAS found a variant positioned next to the UMOD gene that encodes uromodulin (THP), and this variant seems to protect against kidney stones.⁶³⁵ Uromodulin confers protection from kidney stones through yet unknown mechanisms.^{186,636}

There is no doubt that there is a genetic component to hypercalciuria and kidney stones. However, the genes remain largely elusive after decades of study, a situation similar to that for hypertension, dyslipidemia, and diabetes mellitus. Unraveling the origins of this complex polygenic trait is a formidable challenge. Correlation of the genotype with gene product function and whole-organism physiology and pathophysiology is a critical part of this venture. The knowledge gained from monogenic diseases in humans and animals, and from polygenic animal models, is valuable in unraveling this mystery because the function and dysfunction of the gene products can be linked to physiology and pathophysiology, respectively.

MONOGENIC CAUSES OF UROLITHIASIS

Using intermediate or endophenotypes, one can appreciate clear mendelian conditions in humans that cause hypercalciuria and a predisposition to stones. A large number of candidate loci have been identified from animal gene deletions that result in hypercalciuria. These conditions are summarized in Table 38.5. One powerful feature of mendelian diseases is that they have lesions in one gene product that allows one to connect a point of origin to a discrete phenotypic endpoint. The pathophysiologic mechanisms of hypercalciuria and/or kidney stones in these conditions are extremely diverse. This underscores the fact that defects in many organs can all converge on hypercalciuria as a phenotype. The question is whether some of these loci have alleles that

contribute to the risk in the general hypercalciuric stone-forming population. It is possible that a minimal or mild form of these mutant proteins are polymorphic alleles that are more prevalent in the generation population and affect their functions only slightly but yet affect urinary stone risk. A prime example is the fact that heterozygous hypomorphic alleles of the B1 subunit of the H⁺-ATPase can contribute to an unduly high urine pH and CaP stone risk.^{652–654} Multiple loci that by themselves are each weak can then each contribute to small but yet additive effects on calciuria. This monogenic database will be most valuable in tackling the polygenic complex trait of calciuria.

POLYGENIC ANIMAL MODEL

There are numerous animal models of monogenic stone formation.⁶⁵⁵ However, human nephrolithiasis is clearly a polygenic disorder, and most people with calcium-containing kidney stones are hypercalciuric.^{618,656} A powerful model of polygenic hypercalciuria is the GHS rat developed by Bushinsky and coworkers and other.* After more than 90 generations of successive inbreeding of the most hypercalciuric Sprague-Dawley (SD) rats, a strain emerged with 10 times as much urinary calcium as control SD rats. GHS rats have a triple defect of increased gut absorption, renal leak, and bone resorption. The defects persisted *ex vivo*, with cultured bone releasing more calcium. There is an increased number of VDRs and CaSRs and an exaggerated response of VDR gene expression to 1,25(OH)₂D₃ in the intestine and kidney in GHS rats. The elevated levels of VDR are regulated by a decreased level of the transcription factor Snail.

Regions of five chromosomes—1, 4, 7, 10, 14—were linked to hypercalciuria,⁶⁶⁰ but no specific genes have yet been identified. When normocalciuric Wistar-Kyoto rats were bred with the GHS rats to yield congenic rats with the chromosome 1 locus on the Wistar-Kyoto background, the congenic rats were also hypercalciuric but to a lesser extent than the parenteral GHS rats,⁶⁶⁴ supporting both the importance of this locus and the polygenic nature of the hypercalciuria in the GHS rats.

NEPHROLITHIASIS AS A SYSTEMIC DISORDER

Traditionally, kidney stone disease has been recognized as an isolated, benign, painful local condition of the urinary tract. However, the association of urolithiasis with gout and degenerative vascular disease in postmortem examinations was noted by Morgagni back in the 1760s.⁶⁶⁵ In recent years, the prevalence of kidney stones has increased, along with the ever-expanding epidemic of obesity, T2DM, and metabolic syndrome.^{10,11,666–670} Concern with metabolic syndrome and the risk for nephrolithiasis has not been limited to adults because this link was also reported in obese adolescents and the pediatric kidney stone population.^{65,671} It is presently unknown whether the link between kidney stone disease and metabolic syndrome reflects the same underlying pathophysiologic mechanisms in both disorders or is simply an association.

*References 179, 298, 299, 301, 303, 305, and 657–663.

Table 38.5 Rodent Models of Monogenic Hypercalciuria

Gene, Gene Product	Phenotype	Reference(s)
<i>CLC5</i> , chloride channel	- Hypercalciuria - Hyperphosphaturia - Proteinuria - Increased gut calcium absorption - Spinal deformities	637–639
<i>NPT2</i> , renal-I specific Na-coupled phosphate cotransporter	- Hypercalciuria - Hyperphosphaturia - Renal calcifications - Retarded secondary ossification	640
<i>NHERF-1</i> , Na/H exchanger regulatory factor; docking protein	- Hypercalciuria - Hyperphosphaturia - Hypermagnesuria - Female—reduced BMD and fractures	641
<i>TRPV5</i> , epithelial calcium channel	- Hypercalciuria - Hyperphosphaturia - Increased intestinal calcium absorption	642
<i>VDR</i> , vitamin D receptor	- Reduced trabecular and cortical thickness of bones - Hypercalciuria on high-calcium and lactose diet - Rickets	643–645
<i>CalB</i> , calbindin-D28k; intracellular calcium buffer	- Normocalcemia - Hypercalciuria - Normocalciuria	645–648
<i>NKCC2</i> , Na-K-Cl cotransporter	- Hypercalciuria - Polyuria; hydronephrosis - Proteinuria	649
<i>CAV1</i> , caveolin-1; scaffolding protein	- Hypercalciuria in males - Bladder stone	650
<i>AKR1B1</i> , aldoketoreductase	- Hypercalciuria - Hypercalcemia - Hypermagnesemia	651

BMD, Bone mineral density.

OBESITY, WEIGHT GAIN, DIABETES MELLITUS, AND RISK FOR NEPHROLITHIASIS

A prospective epidemiologic study of over 200,000 subjects has demonstrated that obesity and weight gain increase the risk of kidney stones¹⁰ (Fig. 38.26). The relative risk for stone formation in men with a body weight of 100 kg or more was significantly higher than men with a body weight 68 kg or less.¹⁰

Similarly, the relationship between T2DM and the risk of kidney stone formation was seen in three large cohorts from the Nurses' Health Study I comprising older women, the Nurses' Health Study II consisting of younger women, and the Health Professionals Follow-up Study in men. The relative risk of prevalent kidney stones in subjects with T2DM compared with those without is 1.38 in older women, 1.60 in younger women, and 1.31 in men.¹¹

ASSOCIATION BETWEEN METABOLIC SYNDROME AND NEPHROLITHIASIS

Metabolic syndrome is characterized by a cluster of features, including dyslipidemia, hyperglycemia, hypertension, obesity, and insulin resistance.^{672,673} In addition to its relationship to T2DM and cardiovascular risks, metabolic syndrome is associated with nephrolithiasis and chronic renal disease.^{10,11,65,666–677}

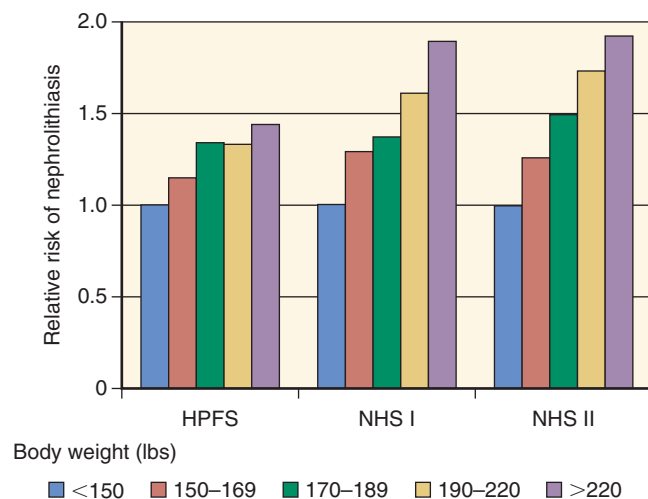


Fig. 38.26 Obesity and the risk of nephrolithiasis. The data were pooled from three databases. *HPFS*, Health Professionals Follow-Up Study; *NHS I*, Nurses' Health Study I; *NHS II*, Nurses' Health Study II. (Modified from Sakhaee K, Maalouf NM, Sinnott B. Clinical review. Kidney stones 2012: pathogenesis, diagnosis, and management. *J Clin Endocrinol Metab.* 2012;97[6]:1847–1860.)

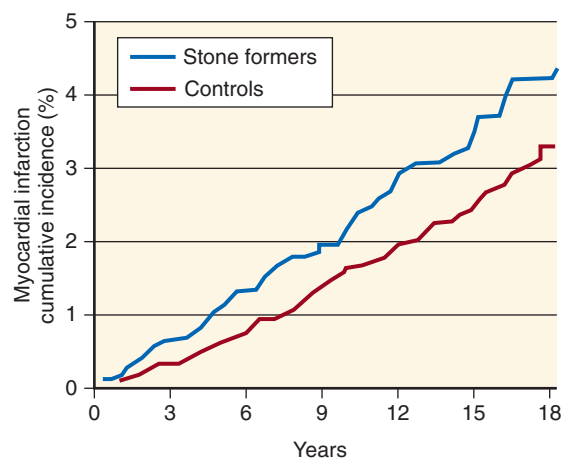
Despite this strong association between metabolic syndrome and kidney stone disease, one consistent limitation has been the lack of documentation of stone composition in most studies. The link between obesity and documented CaOx nephrolithiasis has been in part explained by dietary factors, such as higher consumption of salt and animal proteins.^{319,320} The contribution of the metabolic syndrome to urinary calcium and urinary oxalate excretion in calcium stone formers has not been fully elucidated.⁶⁷⁸ In one large Japanese cohort, hypertension and hyperlipidemia were associated with hypercalciuria, but hyperoxaluria was only associated with hyperlipidemia.⁶⁷⁷ The result of this study was not supported by two other studies of kidney stone formers in the United States and Switzerland.³¹⁹ To this end, other investigators showed that being overweight and diabetic, but not other metabolic syndrome components, were significantly associated with increased urinary oxalate excretion in both kidney stone formers and non-kidney stone formers in the United States.⁶⁷⁹

Uric acid stones is more prevalent in patients with T2DM than in nondiabetic stone formers and more in obese than in nonobese stone formers.^{47,48,50,51,517,519} Higher BMI and T2DM are shown to be independent risk factors for uric acid nephrolithiasis.⁵¹⁸ Cross-sectional studies in healthy non-stone-forming subjects and in kidney stone formers have shown an inverse relationship among urinary pH, body weight, and increasing features of metabolic syndrome.^{493,520} The relationship between low urinary pH, supersaturation index of UA, and adiposity has been specifically shown to be related to fat distribution, with a significant relationship between total body fat and trunk fat associated with increased risk factors for uric acid stone formation.⁶⁸⁰ Further evidence in ZDF rats has demonstrated the causative role of renal steatosis in the pathogenesis of urinary buffer defects in this animal model.⁵³⁷ Furthermore, treatment with thiazolidinediones (TZDs) to improve insulin resistance by redistributing fat to adipocytes has been shown to restore urinary biochemical profiles in this animal model compared with control animals. These changes were associated with reduced renal triglyceride accumulation.⁵³⁷

NEPHROLITHIASIS, CARDIOVASCULAR DISEASE, AND HYPERTENSION

Traditional Framingham risk factors for coronary artery disease such as atherosclerosis, hypertension, diabetes, and metabolic syndrome are also seen frequently in patients with kidney stones.^{8,11,666,681–684} Cross-sectional studies linking coronary artery disease to kidney stones have been inconsistent,^{685–687} with one study showing a positive association⁶⁸⁵ and others demonstrating lack of such an association.^{686,687} Another cross-sectional study in a large number of Portuguese subjects has shown a significant correlation between self-reported history of kidney stones and myocardial infarction solely in females, following multivariable adjustments.⁶⁸⁸

One longitudinal study in a U.S. population over 9 years has shown a multivariable adjusted hazard ratio (HR) for developing myocardial infarction to be higher in patients with kidney stones compared with non-stone-forming control participants⁶⁸⁹ (Fig. 38.27). Although the study was adjusted for multiple variables, risk factors such as dietary calcium and use of thiazide diuretics were not accounted for. In a



Incidence (no. at risk)

Control	0 (10,860)	0.8 (6,689)	2 (3,184)	4.2 (1,010)
Stone formers	0 (4,564)	1.3 (2,686)	3 (1,276)	5.2 (404)

Fig. 38.27 Increased risk for myocardial infarction in stone formers. Data collected from Olmsted County, Minnesota, residents. (Modified from Rule AD, Roger VL, Melton LJ 3rd, et al. Kidney stones associate with increased risk for myocardial infarction. *J Am Soc Nephrol.* 2010; 21[10]:1641–1644.)

large number of patients among the two female cohorts in Nurses' Health Study I (NHS I) and Nurses' Health Study II (NHS II), a positive history of kidney stones was associated with elevated risk of coronary artery disease but such a link was not detected in a separate cohort in men in the Health Professionals Follow-Up Study (HPFS).⁶⁹⁰ The causal relationship among kidney stones, coronary artery disease, and gender specificity has not been proven.

Multiple cross-sectional studies have demonstrated an association between nephrolithiasis and blood pressure.^{691–695} In 895 Swedish men whose blood pressure was measured once, the prevalence of nephrolithiasis increasing from 1% in those with the lowest blood pressure compared with 13% in those with the highest blood pressure.⁶⁹¹ Another study in Italian adults detected a higher prevalence of nephrolithiasis in subjects in the highest quintile of diastolic blood pressure compared with those in the lowest quintile (5.2% vs. 3.4%).⁶⁹³ In study of U.S. men and women, the prevalence of nephrolithiasis was higher in hypertensive subjects compared with normotensive participants adjusted for age and race.⁶⁹⁵ The results of these cross-sectional studies were supported by a longitudinal study conducted over an 8-year period; this study demonstrated that 17.4% of subjects with a history of nephrolithiasis versus 13.1% without a history of kidney stones received a new diagnosis of hypertension when adjusted for age, BMI, and dietary sodium, potassium, magnesium, and alcohol consumption.⁶⁹⁶ In a Canadian study of 25,000 subjects compared with people without kidney stones, and after adjustment for potential confounders, subjects with at least one kidney stone had a higher risk of subsequent myocardial infarction, revascularization, and stroke. The magnitude of the excess risk associated with a kidney stone appeared more pronounced for younger people than for older people ($P < .001$) and for women than men ($P = .01$). The causal relationship has not yet been fully

demonstrated but some have suggested that alterations in calcium metabolism may potentially link the development of kidney stone disease and hypertension.^{697,698} Hypercalciuria, which is prevalent in subjects with calcium nephrolithiasis and primary hypertension, has been proposed by some as a major underlying mechanism, but the role of calcium in human hypertension has never been established.^{697–701} A model of hypercalciuria and blood pressure elevation was proposed in spontaneously hypertensive rats.^{702,703} The NHS showed an association among alterations in acid-base balance, hypocitraturia, and hypertension.⁷⁰⁴ Similar associations were found in spontaneously hypertensive rats, demonstrating evidence of metabolic acidosis.^{705,706}

KIDNEY STONE DISEASE AND CHRONIC KIDNEY DISEASE

Kidney stone disease and CKD can potentially be causally related because of recurrent obstruction and infection, repeated shock wave therapy, or to the same comorbid conditions. A study conducted in Olmsted County, Minnesota, has shown that symptomatic kidney stone formers compared with matched controls followed over 9 years were at increased risk of developing ESKD, independent of diabetes, hypertension, dyslipidemia, and gout.⁷⁰⁷ The association between kidney stones and risk of ESKD was found to be increased in those with urologic abnormalities, including a history of hydronephrosis, recurrent urinary tract infections, single kidney, neurologic bladder, and ileal conduit.⁷⁰⁷ However, other studies have shown that the risk factors for CKD in patients with nephrolithiasis are the same as in the general population—namely, hypertension and diabetes.^{11,51,708,709} Moreover, the apparent association of kidney stones to CKD has been attributed to infectious stone-causing staghorn calculi and to patients with cystinuria.^{710–713} In one epidemiologic study, the prevalence of ESKD deemed to be as a result of nephrolithiasis in the general population was estimated to be approximately 3.1 cases/million per year.⁷¹⁴ This result is comparable to data from the US Renal Data System involving over 200,000 subjects who started dialysis between 1993 and 1997, which showed that approximately 1.2% with ESKD listed nephrolithiasis as the cause.⁷¹⁵ In NHANES III, the estimated glomerular filtration rate (eGFR) of 876 subjects with a history of kidney stone disease was compared with 14,000 subjects without a history of stones.⁷¹⁶ After adjustment for confounders, the eGFR in stone formers with a BMI of 27 kg/m² or higher was significantly lower than in non-stone formers. However, no difference in eGFR between stone formers and non-stone formers was found in subjects with a BMI of 27 kg/m² or lower. The association between kidney stones and CKD is gender-specific, with a significantly higher risk for ESKD development, doubling of serum creatinine level, and CKD (stages 3b–5) in women with a history of kidney stones than in men.⁷¹⁷

URINARY TRACT CANCERS AND KIDNEY STONES

The association of renal pelvis cancer with kidney stones has been reported in several case-control studies.^{718–720} However, the association of renal parenchymal cancer has only been reported in case reports.⁷²¹ In one large population-based cohort study from a Swedish national inpatient registry and

a cancer registry in 61,144 patients who were hospitalized for kidney or ureter stones from 1965 to 1983, there was an increased risk of renal pelvis, ureter, or bladder cancer beyond 10 years of follow-up. However, it was proposed that chronic infection or irritation may play a pathogenic role in the development of cancer.⁷²² In this large population-based cohort study, no association was found between kidney or ureteral stones and renal cell cancer. This result is contrary to the high risk reported in the case-control studies.^{723–726} The possibility of selection bias may exist.

CALCIUM STONES AND BONE DISEASE

EPIDEMIOLOGY

Bone disease is an underemphasized condition in nephrolithiasis. Several epidemiologic studies have established an association between a history of kidney stones and a higher prevalence of fractures.^{727–729} In a population-based study in Rochester, Minnesota, the incidence of first vertebral fracture in patients with symptomatic kidney stone disease followed for 19 years was fourfold higher than in the comparable general population⁷²⁷ (Fig. 38.28).²³⁰ The NHANES III study, including 14,000 men and women, demonstrated an association among a history of kidney stone, low BMD, and higher incidence of fracture.⁷²⁸ This risk of fracture was found to be higher in males than females. In the most comprehensive population-based study, the Osteoporotic Fractures in Men Study (“Mr. OS”), of almost 6000 men has shown an association of kidney stones with lowered BMD at the spine and hip.⁷²⁹ However, in recent studies of 9856 postmenopausal women, kidney stone formation was not shown to be associated with BMD or bone fractures after being adjusted for covariates that are linked with osteoporosis and kidney stones.⁷¹⁰ Major controversies have arisen from population-based cohort studies relying on self-report of kidney stone incidence, dietary composition, calcium and vitamin D supplementation, medications known to affect the risk of kidney stone and bone disease, ethnicity, and general health status of the studied

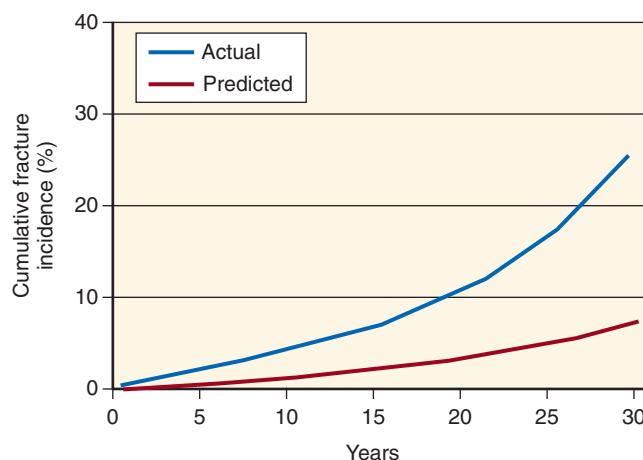


Fig. 38.28 Cumulative incidence of vertebral fractures in stone formers; data from Rochester, Minnesota, residents following an initial episode of symptomatic nephrolithiasis. The elevated fracture risk was vertebral and was present in both genders. (Modified from Melton LJ 3rd, Crowson CS, Khosla S, et al. Fracture risk among patients with urolithiasis: a population-based cohort study. *Kidney Int.* 1998;53[2]:459–464.)

population.^{27,727,730–733} A recent systematic review and comparative meta-analysis in 24 case-control studies in subjects with nephrolithiasis has shown that a lower BMD involves all skeletal sites and that the risk of osteoporosis in patients with nephrolithiasis is four times more than in healthy controls.⁷³⁴

PATHOPHYSIOLOGIC MECHANISMS LINKING OSTEOPOROSIS AND KIDNEY STONES

Osteoporosis is a heterogeneous disorder characterized by disordered bone remodeling resulting in reduced BMD, impairment of microarchitectural integrity, reduced bone strength, and increased fracture risk.⁷³⁵ Numerous studies have found reduced BMD in calcium stone-forming subjects^{259,736–764} (Table 38.6).²³⁰ The loss of BMD is generalized at all skeletal sites, with 40% of patients showing diminished BMD at the vertebral spine, 30% at the proximal hip, and 65% at the radius.²³⁰ Although low BMD is present in hypercalciuric and normocalciuric stone formers,^{738,741,745} it is most prominent in those with hypercalciuria.^{738,741,745,750,756} Diminished BMD was not universally detected in normocalciuric kidney stone formers.^{744,747,750,752} Given that BMD may be one of the important surrogates of bone strength, it is reasonable to suggest that hypercalciuric stone-forming subjects may carry the highest risk of bone fracture. Altered bone remodeling is a significant finding in hypercalciuric stone formers.*

Most histomorphometric studies have agreed that defective bone formation rather than excessive bone resorption plays a key role in the development of bone disease in this population^{259,757–760,762,766} (Table 38.7).²³⁰

The pathophysiologic mechanism(s) of bone disease in stone formers is not known, although it may be related to interplay among environmental, genetic, and hormonal influences and perturbations in the formation of local cytokines.²³⁰ A major emphasis has been placed on the link of idiopathic hypercalciuria and the development of bone disease. It has been assumed that hypercalciuria and negative calcium balance is pathogenic for bone loss in kidney stone formers in several cross-sectional studies of a small number of kidney stone formers.^{241,744,767,768} However, the presence of negative calcium balance,⁷⁶³ osteopenia, and osteoporosis has been demonstrated primarily in patients with dRTA, as well as medullary sponge kidney,⁷⁶⁹ in whom there is a high prevalence of acidosis,^{770,771} which can be the main cause of bone loss.

Dietary Factors

Salt and protein intake is associated with an increased risk of kidney stones²³⁰ and bone disease.^{744,745} Multiple pathophysiologic mechanisms contribute to the high salt- and protein-induced hypercalciuria, including diminished renal tubular calcium reabsorption, hyperfiltration, reposition of acid load, and increased urinary prostaglandin excretion.^{396,772–777} The presence of subclinical metabolic acidosis is common in protein-induced hypercalciuria.⁷⁷⁵ Metabolic acidosis inhibits osteoblastic matrix protein synthesis and alkaline phosphatase activity by stimulating prostaglandin E2 production.^{778–780} Increased prostaglandin E2 production may enhance the

osteoblastic expression of receptor activator for nuclear factor kappa B ligand (RANKL), a major downstream cytokine, which stimulates osteoblastic production by binding to its receptor RANK.⁷⁸¹ Low bicarbonate in vitro alters osteoblastic extracellular matrix proteins, including type I collagen^{778,780} osteopontin, matrix Gla protein,^{779,782} and expression of cyclooxygenase-2⁷⁸³ and RANKL.^{781,784} The effect of protein is only partially mediated by its acid content⁷⁸⁵ because, in humans, the total neutralization of the acid of dietary protein does not reverse the hypercalciuria.⁷⁸⁶

Genetic Factors

The genetic link of vertebral bone loss in kidney stone patients with absorptive hypercalciuria was identified in a genome-wide linkage approach study showing polymorphism in the soluble adenylyl cyclase gene (*ADCY10*) on chromosome 1q23–4q24.^{633,787} Another genome-wide screening study associated sequence variants in the claudin-14 gene (*CLDN14*) with kidney stone and reduced BMD at the hip in a large population of kidney stone formers from Iceland and the Netherlands.³³⁰ Although soluble *ADCY10* has been shown to regulate osteoclastic function by ambient bicarbonate concentration,⁶³⁴ and claudin-14 is recognized as a paracellular protein mediating renal tubular calcium reabsorption, it is unclear in both cases how these putative genes influence bone remodeling. Others have shown an association between CaSR gene polymorphism and increased urinary calcium excretion in hypercalciuric patients⁶²⁷ and reduced forearm BMD in healthy subjects and postmenopausal women.^{788,789} An initial report claimed that mutation in the sodium phosphate transporter, NaPi2a, is linked to kidney stones and defective bone mineralization,⁷⁹⁰ but additional studies did not confirm this association.^{791,792}

Numerous genetically heterogeneous disorders that present in childhood or early adulthood are associated with hypercalciuria, nephrocalcinosis, nephrolithiasis, and rachitic bone disease, including Dent disease (mutation of the chloride-proton transporter, CLC-5) and Lowe syndrome (mutation of the phosphatidylinositol-4,5 bisphosphonate 5-phosphatase, OCRL1).^{793–795} However, the role of these genes in bone disease in the general kidney stone-forming population has not yet been illustrated.

Hormones and Local Cytokines

Elevated serum 1,25(OH)₂D levels occurs in one-third of hypercalciuric stone formers.^{241,246,251,253} Peripheral blood monocyte VDR levels are increased, which may amplify the target organ action of circulating 1,25(OH)₂D on bone, kidney, and intestine.²⁹⁰ High doses of 1,25(OH)₂D in vitro enhance bone resorption and decrease bone collagen synthesis.³⁴² Low expression of transforming growth factor-β (TGFβ), which is known to enhance bone formation and bone mineralization,^{342,345} was demonstrated in bones of hypercalciuric stone formers,³⁴⁴ as was increased production of bone resorptive lymphokines.^{741,747,748} A study using undecalcified bone in idiopathic hypercalciuric patients has shown an elevated bone expression of RANKL.³⁴⁴ Therefore it is plausible that the interplay of cytokines and RANK/RANKL/OPG (osteoprotegerin) may stimulate bone resorption, whereas lowered TGF-β, high 1,25(OH)₂D, and/or amplified vitamin D activity may impair osteoblastic bone formation in this population.

*References 259, 752, 757, 759, 760, 762, and 765.

Table 38.6 Changes in Bone Mineral Density in Calcium Kidney Stone Formers

Reference	Study Subjects	No. of Subjects			BMD Site			Measurement Technique
		Male (M)	Female (F)	Control	Spine	Hip	Radius	
Alhava ⁷³⁶	Single or recurrent urolithiasis	54	21	21			↓	Americium-241 gamma ray attenuation
Lawoyin ⁷³⁷	Absorptive hypercalciuria	94	23	Not reported			↔	Single-photon absorptiometry
	Renal hypercalciuria	28	16				↓	
	Primary hyperparathyroidism	22	31				↓	
Fuss ⁷³⁸	Osteoporosis	14	55	Not reported			↓	Single-photon absorptiometry
	Absorptive hypercalciuria	24	19				↓	
	Renal or resorptive hypercalciuria	7	18				↓	
Barkin ⁷³⁹ Fuss ⁷⁴⁰	Normocalciuria	35	6				↓	Neutron-activation analysis Single-photon absorptiometry
	Idiopathic hypercalciuria	86	23	84	↓ Calcium-binding index			
	Free diet	63	0	16			↔	
Pacifi ⁷⁴¹	Low calcium diet	60	0				↓	Quantitative computed tomography (CT)
	Absorptive hypercalciuria	29	18	28 (24 M, 4 F)	↓			
Bataille ⁷⁴²	Fasting hypercalciuria	16	7		↓			Quantitative CT
	Dietary hypercalciuria	12	6	61 (41 M, 20 F)	↔			
Borghi ⁷⁴³	Dietary independent hypercalciuria	17	7		↓			Dual-photon absorptiometry
	Dietary hypercalciuria	13	7	0	↔			
Pietschmann ⁷⁴⁴	Dietary independent hypercalciuria	14	7		↓			Dual-energy x-ray absorptiometry; dual-photon absorptiometry; single-photon absorptiometry
	Absorptive hypercalciuria	42	20	0	↓			
	Fasting hypercalciuria	24	3		↓			
Jaeger ⁷⁴⁵	Normocalciuric	25	6		↔			Dual-energy x-ray absorptiometry
	Hypercalciuric	49	0	234	↓	↓		
Zanchetta ⁷⁴⁶	Normocalciuric	61	0		↓	↓		Dual-photon absorptiometry
	Fasting hypercalciuria	15	23	50 (20 M, 30 F)	↓			
Weisinger ⁷⁴⁷	Absorptive hypercalciuria	5	7		↓ F, ↔ M			Dual-energy x-ray absorptiometry
	Hypercalciuric	4	13	12 (4 M, 8 F)	↓			
Ghazali ⁷⁴⁸	Normocalciuric	4	8		↔			Quantitative CT
	Idiopathic hypercalciuria	15	1	10 (8 M, 2 F)	↓			
Giannini ⁷⁴⁹	Dietary hypercalciuria	9	1		↔			Dual-energy x-ray absorptiometry
	Fasting hypercalciuria	8	23	13 (10 M, 3 F)	↓	↓		
Trinchieri ⁷⁵⁰	Absorptive hypercalciuria	13	5		↓	↔		Dual-energy x-ray absorptiometry
	Hypercalciuria	10	0		↔	↔		
Tasca ⁷⁵¹	Normocalciuria	34			↔	↔		Dual-energy x-ray absorptiometry
	Fasting hypercalciuria	27	12	15	↓			
Misael da Silva ⁷⁵²	Absorptive hypercalciuria	20	11		Not reported			Dual-energy x-ray absorptiometry
	Idiopathic hypercalciuria	11	11	10 (5 M, 5 F)	↓	↓		
Asplin ⁷⁵³	Normocalciuric	8	10		↔	↔		Dual-energy x-ray absorptiometry
	Stone formers	15	7	37 (14 M, 23 F)	↓	↓		
Vezzoli ⁷⁵⁴	Stone formers	29	35	0	↓	↓		Dual-energy x-ray absorptiometry
	Hypercalciuria	15	27		↓	↓		
Caudarella ⁷⁵⁵	Normocalciuria	102	94	196			↓	Dual-energy x-ray absorptiometry; quantitative ultrasonography
	Stone formers (27% with hypercalciuria)			(102 M, 94 F)				

↓, Lower bone mineral density (BMD); ↔, no change in BMD.

Modified from Sakhaee K, Maalouf NM, Kumar R, et al. Nephrolithiasis-associated bone disease: pathogenesis and treatment options. *Kidney Int.* 2011;79(4):393–403.

Bone histomorphometric analyses performed in kidney stone formers have shown impaired bone formation as a unanimous finding.^{259,762} These findings would not be expected if only hypercalciuria played a pathogenetic role in the development of bone disease in this population.^{259,762} Idio-

pathic hypercalciuria is a heterogeneous and complex disorder that could result from increased production or sensitivity to calcitriol and affect the target organs differently. Thus, whereas there is evidence of increased intestinal calcium absorption in this population, the high levels of calcitriol attained in

Table 38.7 Bone Histomorphometric Characteristics in Kidney Stone Formers

Reference	Study Subjects	No. of Subjects	Bone Histomorphometric Profiles
Bordier ²⁵⁹	Dietary hypercalciuria	20	None
	Renal hypercalciuria	19	Increased osteoclast and osteoblast surfaces
	Hypophosphatemia	21	Increased osteoclastic and eroded surface (within normal range), decreased osteoblast surfaces, decreased osteoid parameters
Malluche ⁷⁵⁷	Controls	12	Low-normal osteoclastic bone resorption, low osteoblastic activity, decreased fraction of mineralizing osteoid seams, decreased mineralization apposition rate
	Absorptive hypercalciuria	15	
de Vernejoul ⁷⁵⁸	Controls	22	Decreased trabecular volume, decreased active osteoblastic surface, decreased active bone resorption surface
	Idiopathic hypercalciuria	30 (20 M, 10 F)	
Steiniche ⁷⁵⁹	Controls	187	Increased bone resorption surfaces (decreased refilling of lacunae with low bone formation), decreased bone formation rate, increased mineralization lag times
	Idiopathic hypercalciuria	33 (22 M, 11 F)	
Heilberg ⁷⁶⁰	Controls	30 (19 M, 11 F)	Increased eroded surface, decreased osteoid surface, decreased bone formation rate with a complete lack of tetracycline double labeling
	Fasting hypercalciuria	6 male	
Bataille ⁷⁶¹	Controls	No information	Low eroded surface, low bone volume, low osteoid surface, thickness, mineral apposition rate, adjusted apposition rate and bone formation rate
	Idiopathic hypercalciuria	24 (20 M, 4 F)	
Misael da Silva ⁷⁵²	Control	18 (9 M, 9 F)	High eroded surface, increased osteoblastic bone surface, no change in trabecular thickness
	Idiopathic hypercalciuria	22	
Heller ⁷⁶²	Control	94	Relatively high bone resorption (osteoclast surface, bone surface—mean value within the normal limit), lower indices of bone formation (osteoblast surface/bone surface), decreased wall thickness
	Absorptive hypercalciuria	9 (6 M, 3 F)	
	Control	9 (6 M, 3 F)	

hypercalciuric stone formers may result in diminished osteoblastic bone formation.^{255,256,290,796} The inhibitory actions of calcitriol on bone collagen synthesis, as well as in osteoblastic cell proliferation, has been shown to be exerted at 10^{-10} M, which is no different from the level of calcitriol attained in hypercalciuric stone formers.^{253,306,342}

EVALUATION

CLINICAL PRESENTATION

SYMPTOMS AND SIGNS

Kidney stones may be asymptomatic and incidentally present in patients during an imaging procedure. Renal colic, pain localized to the back and flank during passage, is a common clinical manifestation of kidney stones. The pain occurs as the kidney stone is propelled through the ureter and is a consequence of increased intraluminal pressure, causing stimulation of nerve endings in the ureteral mucosa. Pain is usually intense and intermittent, originating in the back or flank, radiating around the torso to the groin, and ending up in the testicles or labia for male or female subjects, respectively. Stones in the midportion of the ureter may imitate appendicitis on the right side or diverticulitis on the left side. Renal colic can be associated with symptoms such

as nausea and vomiting because the GI tract shares common innervation with the genitourinary system. When the kidney stone approaches the urinary bladder, it frequently causes bladder symptoms such as urinary frequency, dysuria, suprapubic pain, and incontinence.⁷⁹⁷

The most important physical finding is costovertebral angle tenderness, although its presence and severity are not as consistent and intense as in pyelonephritis. The abdominal examination is usually negative. Hypertension and tachycardia, if present, are most likely as a result of severe relentless pain. Fever and muscle spasm are infrequent and may represent underlying diseases or complications. A toxic-appearing patient during stone passage may indicate obstruction, infection, and urosepsis.

ENVIRONMENTAL, LIFESTYLE, AND MEDICAL HISTORY

It is important to elicit a history of any systemic disorders. These include disorders of calcium homeostasis such as primary hyperparathyroidism, conditions accompanied by extrarenal calcitriol production such as granulomatous disease, and disorders such as obesity, type 2 diabetes, gout, recurrent urinary tract infections, inflammatory bowel disease, bowel resection, pancreatic disease, bariatric surgery, distal renal tubular acidosis, and medullary sponge kidney.^{222,798,799}

Patients should be asked about climate exposure, work environment, and exercise frequency and intensity because these affect the risk of development of kidney stones. Higher temperatures and prolonged summer seasons in the equatorial part of the globe increase the risk of kidney stones. Daily exposure to hot conditions, military cohorts training in a high-temperature environment, and individuals who engage in physical exercise during the summer months are also factors in stone development.^{15,799,800}

Dietary history should include daily fluid intake (from water and various beverages), intake of calcium, sodium, amount and type of protein intake, food containing oxalate, and alkali-rich foods, including fruits and vegetables. The history may also detail the use of OTC medications and/or supplements, including calcium, vitamin D, multivitamins, and ascorbic acid.

A careful drug history must be taken because prescription and over-the-counter (OTC) drugs have been demonstrated to increase the risk of kidney stones.⁷⁹⁸ In some cases, the increased risk is caused by metabolic alterations in the urinary environment—for example, calcium, vitamin D and vitamin C supplements, carbonic anhydrase inhibitors (acetazolamide, topiramate, and zonisamide), laxatives,^{588,589} probenecid, ascorbic acid, lipase inhibitors,^{801,802} excessive alkali treatment, and chemotherapeutic agents.⁷⁹⁸ However, in other cases, there is a risk of crystallization of the drug in urine because of its poor solubility. This class of drugs includes triamterene, protease inhibitors (indinavir, atazanir, and nelfinavir), guaifenesin, ephedrine, antacids (magnesium trisilicate), and antimicrobials (sulfonamides and quinolones).^{389,487,798,799,803}

FAMILY HISTORY

It is very common to obtain a positive family history in a stone former. The pattern may or may not be mendelian in nature but nonetheless needs to be documented. The interpretation of whether a certain pedigree has a mendelian trait is often difficult because of phenocopy, incomplete penetrance, and loci heterogeneity. Certain monogenic diseases present with kidney stones that may have specific phenotypic characteristics. Hypercalciuric nephrolithiasis among male subjects associated with renal impairment and low-molecular-weight proteinuria that is X-linked recessive is found in patients with Dent disease.⁸⁰⁴ The familial occurrence of kidney stones and nephrocalcinosis in male subjects with early cataracts or glaucoma is suggestive of Lowe syndrome.⁸⁰⁵ Aggressive nephrolithiasis, nephrocalcinosis, retarded growth, and deafness can be seen in patients with dRTA, presenting with both autosomal dominance and recessive inheritance.^{805,806} A presentation with nephrocalcinosis and impaired kidney function with hyperoxaluria should suggest the diagnosis of primary hyperoxaluria.⁸⁰⁷

LABORATORY EVALUATION

Laboratory diagnoses include blood and urinary metabolic profiles, stone analysis, and imaging (Tables 38.8 to 38.10).^{222,797} There are limited prospective studies regarding the appropriate extent of the metabolic evaluation in the kidney stone population. Therefore, the proposed evaluation varies, depending on the severity of the kidney stone disease and structure of different practices. A few studies have suggested that single stone formers share the same

metabolic abnormalities with those with recurrent kidney stone disease.^{808–810} Furthermore, there are no surrogate metabolic markers that identify those at high risk for recurrent nephrolithiasis, except for low urine volume.⁸¹⁰ However, in a recent retrospective study of a large number of first-time adult kidney stone formers followed for 15 years (Recurrence of Kidney Stones [ROKS]), a nomogram was developed to predict recurrence.⁸¹¹ Among the risk factors for recurrence were younger age, male gender, white race, presence on imaging of both asymptomatic or nonobstructing stones and symptomatic renal pelvic or lower pole stones, gross hematuria, and uric acid stones. Based on this nomogram, the 10-year recurrence rate ranged from 12% to 56% in the lowest to the highest risk quintile. However, because of the lack of presentation of 24-hour urinary metabolic risk factors, no distinction was made between metabolically active and metabolically inactive stone formers.^{812,813}

SERUM CHEMISTRY

All kidney stone formers require the determination of full fasting serum chemistries (e.g., electrolytes, including calcium and phosphorus, renal function, uric acid) and PTH. Fasting glucose and a full lipid panel are also justified considering the high prevalence of diabetes and metabolic syndrome in stone formers. Increased serum total calcium concentration and low serum phosphorus levels, in association with a high PTH concentration, are indicative of primary hyperparathyroidism, although this full triad is often not present. The finding of a low serum phosphorus level with a normal serum PTH level is suggestive of a renal phosphorus leak. In the latter condition, hypercalciuria may ensue as a result of increased intestinal calcium absorption from an elevated serum calcitriol level because of hypophosphatemia. The measurement of serum 1,25(OH)₂D may be considered for specific situations or for research purposes. A serum 25(OH)D measurement is helpful in patients with high or high-normal PTH levels and mild hypercalcemia to exclude vitamin D deficiency as a cause of the high PTH level. The finding of a low serum potassium level and low serum total CO₂ content is suggestive of dRTA or a chronic diarrheal state. Hyperuricemia may suggest the diagnosis of UA nephrolithiasis.^{222,656,797,799}

URINE CHEMISTRY

Spot Urinalysis

A spot urine collection is of some value because of its simplicity. The UpH obtained from the colorimetric reaction of the dipstick is notoriously inaccurate, and only extreme values are informative. A very low dipstick pH (<5.5) suggests UA stones, whereas a high pH (>6.5) suggests dRTA, and a very high pH (>7.4) should raise suspicion for infection. Urine culture must be obtained in patients suspected of urinary tract infection (UTI) and in patients with established struvite stones. Crystalluria per se is not abnormal because it is found in normal urine, except that cystine crystals are never present in normal urine (see Fig. 38.22B).

To date, the available clinical studies have not shown that first-time kidney stone formers are metabolically and clinically separable from those with recurrent disease. Thus, all patients, irrespective of the severity of kidney stone disease, should have a metabolic evaluation. It has been suggested that a simplified metabolic evaluation may be considered for single

Table 38.8 Simplified Ambulatory Metabolic Evaluation and Interpretation of Urinary Parameters^a

Random 24-Hour Urinary Profile	Expected Values (per day)	Interpretation
Total volume	≥2.5 L	Indicative of daily fluid intake (minus insensible losses); diminishes with low fluid intake, sweating, and diarrhea
pH	5.9–6.2	<5.5—increases risk of uric acid precipitation; commonly found in idiopathic uric acid stone patients, subjects with intestinal disease and diarrhea, and in those with intestinal bypass surgery >6.7— increases risk of calcium phosphate precipitation; commonly found in patients with dRTA, primary hyperparathyroidism, alkali, and carbonic anhydrase treatment >7.0–7.5— indicates urinary tract infection from urease-producing bacteria.
Creatinine	15–25 mg/kg body weight (0.13–0.22 mmol/kg body weight)	Assessment of completeness of collection: 15–20 mg/kg body weight (0.13–0.15 mmol/kg body weight) in females, 20–25 mg/kg body weight (0.15–0.22 mmol/kg body weight) in males; valid only in steady state of constant serum creatinine concentration with time
Sodium	100 mEq (100 mmol)	Reflects dietary sodium intake (minus extrarenal loss); much lower than dietary intake in diarrhea and with excessive sweating; high sodium intake is major cause of hypercalciuria
Potassium	40–60 mEq (100 mmol)	Reflects dietary potassium intake (minus extrarenal loss); much lower than dietary intake in diarrhea states; gauge of dietary alkali intake because most dietary potassium accompanied by organic anions
Calcium	≤250–300 mg (≤6.24–7.49 mmol)	A higher value expected in males; in states of zero balance, urinary calcium excretion is net gut absorption minus net bone deposition; secondary causes should be ruled out before making the diagnosis of idiopathic hypercalciuria
Magnesium	30–120 mg (1.23–4.94 mmol)	Low urinary magnesium detected with low magnesium intake, intestinal malabsorption (small bowel disease), and following bariatric surgery; low magnesium may increase risk of calcium stones.
Oxalate	≤45 mg (≤0.51 mmol)	Commonly encountered with intestinal disease with fat malabsorption, such as inflammatory bowel disease and following bariatric surgery; values >100 mg/day (1.14 mmol/day) suggest primary hyperoxaluria (PH); the diagnosis of PH I and PH II is further established with high urinary glycolate and L-glycerate levels.
Phosphorus	≤1100 mg (35.5 mmol)	Indicative of dietary organic and inorganic phosphorus intake and absorption; a higher excretion may increase the risk of calcium phosphate stone formation.
Uric acid	600–800 mg (3.57–4.76 mmol)	Hyperuricosuria is encountered with overproduction of endogenous uric acid or overindulgence of purine-rich foods such as red meat, poultry, and fish; mainly a risk factor for calcium oxalate stones when UpH is >5.5 but is a risk factor for uric acid stone when UpH < 5.5.
Sulfate	≤20 mmol	Sulfate is a marker of dietary acid intake (oxidation of sulfur-containing amino acids).
Citrate	≥320 mg (≥1.67 mmol)	Inhibitor of calcium stone formation; hypocitraturia is commonly encountered in metabolic acidosis, dRTA, chronic diarrhea, excessive protein ingestion, strenuous physical exercise, hypokalemia, intracellular acidosis, with carbonic anhydrase inhibitor drugs (e.g., acetazolamide, topiramate, zonisamide), but rarely with ACE inhibitors
Ammonium	30–40 mEq (30–40 mmol)	Ammonium is a major carrier of H ⁺ in the urine; its excretion corresponds with urinary sulfate (acid load); a higher ammonium-to-sulfate ratio indicates GI alkali loss.
Chloride	100 mEq (100 mmol)	Chloride varies with sodium intake.
Cystine	<30–60 mg (<0.12–0.25 mmol)	Cystine has a limited urinary solubility, at 250 mg/L.

^aExpected values should be cross-checked with reference laboratory recommendations because these values may differ. These limits are mean +2 standard deviations (for calcium, oxalate, uric acid, pH, sodium, sulfate, and phosphorus) or mean –2 standard deviations (for citrate, pH, and magnesium) from normal.

ACE, Angiotensin-converting enzyme; dRTA, distal renal tubular acidosis; GI, gastrointestinal; UpH, urinary pH.

stone formers, and an extensive evaluation is needed for recurrent kidney stone formers and individuals at high risk for recurrent stone formation, such as those with primary hyperparathyroidism, dRTA, chronic diarrhea, UA stones, obese subjects, patients with CKD, or bone disease and in pediatric populations with kidney stones. Kidney stone formers

of a certain profession, such as pilots, and frequent travelers should also be considered for metabolic evaluation.

Simplified Metabolic Evaluation

A simplified workup is appropriate for single stone formers with a low risk of recurrence. The evaluation should include

Table 38.9 Simplified Ambulatory Metabolic Evaluation and Interpretation of Other Parameters

Other Tests	Values	Interpretation
Complete metabolic panel	Variable ^a	Low serum potassium, high serum chloride, and low serum total CO ₂ content are suggestive of a diarrheal state or dRTA.
PTH	10–65 pg/mL (1.06–6.90 pmol/L)	High PTH, when accompanied by high serum calcium and low serum phosphorus, is suggestive of primary hyperparathyroidism.
1,25-dihydroxyvitamin D	Variable ^a	Normal serum calcium, normal PTH, and elevated 1,25 dihydroxyvitamin D is suggestive of absorptive hypercalciuria. Normal serum calcium, normal PTH, low serum phosphorus and elevated 1,25 dihydroxyvitamin D is suggestive of renal phosphorus leak.
Bone mineral density measurements (DXA)	Z-score >–2 T-score >–2.5	Z-score <–2 or T-score <–2.5 indicates bone loss. This finding may be more prevalent in hypercalciuric kidney stone formers.

^aExpected values should be cross-checked with reference laboratory recommendations since these values may differ.
dRTA, distal renal tubular acidosis; DXA, dual-energy x-ray absorptiometry; PTH, parathyroid hormone.

Table 38.10 Extensive Ambulatory Metabolic Evaluation and Interpretation^a

Restricted 24-hour Urinary Profile	Expected Values (per day) ^b	Interpretation
Calcium	<200 mg (<4.99 mmol)	A restricted diet urinary Ca <200 mg/day (4.99 mmol/d), accompanied by a random diet urinary Ca >200–250 mg/day (6.24–7.49 mmol/day) is reflective of dietary Ca indiscretion. Persistent high urinary Ca on a restricted diet is reflective of intestinal hyperabsorption of Ca.
Sodium	100 mEq (100 mmol)	Urinary Na > 100 mEq/day (100 mmol/day) may reflect high dietary salt intake. An increment of 100 mEq/day (100 mmol/day) of Na increases urinary calcium by approximately 40 mg/day (1 mmol/day).
Oxalate	40 mg (<0.45 mmol)	Urinary oxalate >40 mg/day (0.45 mmol/day), in the absence of a chronic diarrheal state, is indicative of high dietary oxalate intake.
Sulfate	<20 mmol	Urinary sulfate >20 mmol/day indicates overindulgence of high-acid-ash diet found in animal proteins.
2-hour fasting calcium-to-creatinine ratio (Ca:Cr)	<0.11 mg/100 mL GF (<2.7 μmol/100 mg GF)	Elevated fasting Ca:Cr, high serum calcium, and elevated PTH suggestive of primary hyperparathyroidism; elevated fasting Ca:Cr, normal serum calcium, and normal or suppressed PTH suggestive of resorptive hypercalciuria; Elevated fasting Ca:Cr, normal serum calcium, and elevated PTH suggestive of renal hypercalciuria.
4-hour Ca:Cr following a 1-g oral calcium load	≤ 0.20 mg/mg Cr (≤ 0.56 mmol/mmol Cr)	Elevated Ca:Cr following a 1-g oral calcium load is suggestive of absorptive hypercalciuria.

^aAfter 1 week of dietary restrictions.

^bExpected values should be cross-checked with reference laboratory recommendations since these values may differ. These limits are mean + 2 standard deviations (for calcium, oxalate, uric acid, pH, sodium, sulfate, and phosphorus) or mean – 2 standard deviations (for citrate, pH, and magnesium) from normal.

GF, Glomerular filtrate; PTH, parathyroid hormone.

a comprehensive history and physical examination, dietary and record of fluid consumption, and review of medications.

Although it is not perfect, studies have shown a correlation between kidney stone composition and urinary biochemical profiles.^{90,91} A simple evaluation includes a single, random, 24-hour urinary profile on an ad libitum diet, including total volume, pH, and creatinine, CaOx, citrate, UA, sulfate, chloride, and ammonium levels (Table 38.8). There is some controversy regarding the recommended number (one or two) of random 24-hour urine collections required to determine kidney stone risk.^{814–816} Regardless of the number of samples collected, physicians should take into account the dietary intake of known risk factors such as salt and protein because these are outpatient collections from those on a

random diet. Patients should be instructed to adhere to their usual daily activities and diet before and during the collection. If two collections are contemplated, restriction of dietary sodium, oxalate, and protein may be considered prior to a second collection to evaluate the contribution from dietary aberrations without embarking on a fully executed dietary control program (see later). Imaging to determine the extent of the kidney stone disease and type of stones also may be of some value. A spot urine test for the evaluation of UTI and urine culture may be useful as part of this evaluation.

Extensive Metabolic Evaluation

An extensive evaluation is infrequently carried out in the evaluation of kidney stones.⁴⁷⁰ One study has shown that

low-risk patients with kidney stone recurrence are more interested in a complete metabolic evaluation than the practicing physician.⁸¹⁷ The justification for a complete metabolic evaluation is based on the economic burden of years of medical treatment in contrast to surgical intervention.⁸¹⁸ See Table 38.9 for a listing of other tests.

An extensive evaluation includes a 24-hour urinary sample following a 1-week instructed, fixed metabolic diet consisting of 400 mg calcium/day, 100 mEq sodium/day, and avoidance of oxalate-rich food⁸¹⁹ (Table 38.10). Generally, high-risk and recurrent stone formers may benefit from an extensive metabolic evaluation. These include those with a systemic illness, such as primary hyperparathyroidism, dRTA, gout, type 2 diabetes, patients with a family history of stone disease, and those with a solitary kidney. The 24-hour urine profiles may be complemented by two additional tests, which can be performed at the completion of the restricted 24-hour urine collection—2-hour urinary fasting calcium-to-creatinine ratio to assess renal leak and/or excessive skeletal calcium mobilization, urinary fractional phosphorus excretion to monitor renal phosphorus leak (TmP/GFR), and a 4-hour urinary calcium-to-creatinine ratio following a 1-g oral calcium load to determine intestinal calcium absorption. However, these tests are usually performed only as a research tool or in a specialized stone clinic.

A comprehensive metabolic workup may also include the analysis of the BMD given that both cross-sectional and epidemiologic studies have shown low BMD, abnormal bone histology, and increased fracture risks among kidney stone formers compared with the general population.²³⁰ One major hurdle is the insurance constraints justifying BMD determination in these subjects.

In those with established cystine stones, a positive family history of cystine stone, or those suspected with the condition, urinary cystine measurement must be done. In those with urinary oxalate exceeding 100 mg/day, the diagnosis of PH may be suspected; 24-hour urine must be tested for glycolate and L-glycerate to confirm the diagnosis of PH types I and II, respectively. Increased urinary glycolate levels are detected in approximately two-thirds of PH type I subjects, so a normal urinary glycolate level does not exclude this diagnosis.^{820,821} With renal impairment, there is a fall in the urinary oxalate level, so its urinary measurements may not be accurate. In these circumstances, the plasma oxalate level should be assessed to confirm the diagnosis of PH. It is typically elevated at 80 $\mu\text{mol/L}$ compared with non-PH hyperoxaluric patients,

in whom plasma oxalate levels vary from 30 to 80 $\mu\text{mol/L}$.^{822–824} The most definitive diagnosis is by genetic analysis.^{825–828} In cases in which clinical suspicion of PH is considered to be high but DNA screening is nondiagnostic, a liver biopsy can establish the diagnosis.⁸²¹

Spot and Fasting Urine Specimen

Some patients may be reluctant or unable to collect 24-hour urine, and one may be concerned with the accuracy of the 24-hour urine collection. Several cross-sectional studies have suggested that spot urine could be substituted for the 24-hour urine in the assessment of urinary kidney stone risk profiles.^{829–835} In one study in a large population of stone and non-stone-forming subjects, fasting urinary pH correlated with a 24-hour urinary pH.⁸³⁶ However, significant variability was demonstrated between these two profiles in individual subjects, suggesting that the fasting urine pH could not substitute for the 24-hour urinary pH in this population.

Urinary Supersaturation Estimation

Urinary supersaturation measurement has not been widely used clinically (Table 38.11). However, it can be used in clinical and research settings to estimate risks and monitor response to treatment.^{90,91,837} Urinary supersaturation was used in stone research using the EQUIL2 software program (now offered commercially).^{83,838,839} Other programs, including JESS, have also been used in the calculation of urinary supersaturation.⁹² Furthermore, urinary supersaturation ratios are reported by commercial clinical laboratories and are available to practitioners. The RSR is defined as the ratio of calculated activity product in a given urine sample and respective thermodynamic solubility product.⁸³ Values higher than 1 indicate supersaturation, and values less than 1 indicate undersaturation. Another method, urinary RS, is calculated as the ratio of activity product (for UA, it is the concentration of undissociated UA) in a particular urine specimen and corresponding mean activity product (for UA, it is the undissociated UA) from normal subjects. The upper limit of normal for RS of CaOx, brushite, monosodium urate, and UA is defined as 2.⁵⁰⁹

Despite the lack of uniform hard evidence correlating urinary biochemical abnormalities and urinary supersaturation of stone-forming salt following dietary and pharmacologic treatment with clinical stone occurrence, several observational and case-control studies have shown a correlation between

Table 38.11 Urinary Supersaturation Estimation^a

Term	Definition	Readout
Relative supersaturation ratio (RSR)	Ion concentration product in a given solution (activity product); ion concentration product at equilibrium (solubility product)	Values <1 represent undersaturation; values >1 represent supersaturation.
Relative supersaturation (RS)	Urinary activity product in urine specimen; activity product from normal subjects	Upper normal limits = 2.

^aThe EQUIL2 software program is used for the assessment of urinary supersaturation in stone research as well as in a clinical setting. The JESS program, which is now mostly used in research programs also represents the supersaturation ratio but is defined as saturation index (SI) in that software program.

the calculated parameter and clinical outcome.^{813,840,841} For example, one study at a specialized kidney stone center has shown that a reduction of CaOx supersaturation is accompanied by a reduction in the number of formed kidney stones.⁸⁴²

There is no hard rule regarding structured longitudinal monitoring of urinary parameters following the initiation of a pharmacologic and/or dietary regimen. However, it is advisable to obtain a 24-hour urine annually, depending on the activity of stone disease.^{843,844}

A recent study has concluded that urinary supersaturation does not offer any diagnostic advantage in separating kidney stone formers from healthy non-kidney stone formers at presentation.⁸⁴⁵ However, like previous studies, this study has suggested that urinary supersaturation should be analyzed in clinical and research settings during the assessment of the effectiveness of medical treatment.

STONE ANALYSIS

Stone analysis provides valuable information that clarifies the differential diagnosis and assists in directing management.^{846,847} The stone crystallographic finding also helps identify cystine- and infection-induced stones, which may completely change management. Stone analysis also assists in the diagnosis of extremely rare stones, such as 2, 8-hydroxyadenine or drug-induced stones.^{798,799} Commonly, the presence of ammonium urate stones is encountered in patients with HIV-acquired immune deficiency syndrome as a consequence of laxative abuse⁵⁹² or because of protracted diarrhea.⁷⁹⁷ In addition, the presence of CaP stones may suggest conditions such as dRTA, primary hyperparathyroidism, medullary sponge kidney, and carbonic anhydrase inhibitor treatment.⁸⁴⁸ One may follow and reassess stone analysis in the management of recurrent stone formers who are unresponsive to specific medical treatment.

Although stone analysis plays an important role in the understanding of the underlying pathophysiology and selection of the treatment, one major limitation has been the lack of a standard method among different laboratory techniques to identify urinary stone composition.^{845,849} At present, there is a general agreement that chemical analysis of urinary stone composition must be substituted with more accurate spectroscopy techniques, including Fourier-transform infrared spectroscopy (FTIR), x-ray diffraction (XRD), and possibly scanning electron microscopy. Another drawback is that all current methods involve grinding and homogenization of the specimen, thus destroying any spatial information about the distribution of different components of mixed stones.

IMAGING STUDIES

Imaging studies could be considered in patients suspected of kidney stones and in the follow-up of treated stone formers to monitor stone activity. Various recognized imaging methods include plain abdominal radiography for the kidneys, ureters, and bladder (KUB), ultrasound examination, and noncontrast computed tomography (NCCT).

KIDNEYS, URETERS, AND BLADDER X-RAY

KUB is a plain x-ray of the abdomen that will detect opaque calcareous or cystine stones. It has wide availability, minimal

radiation exposure, and low cost. The drawback is limited sensitivity (45%–58%) and specificity (60%–77%)⁸⁵⁰ because of body habitus, overlying bowel gas, and extragenitourinary calcifications. It does provide adequate information in following the stone load during therapy in patients with radiopaque stones. For stable patients, annual imaging is adequate; however, additional imaging must be considered according to the clinical activity of the patient.⁸⁵¹

ULTRASOUND

Although ultrasound is a reliable, noninvasive, and fast technique and does not involve ionizing radiation, its major limitation is its low sensitivity. The lack of radiation and contrast renders it safe, particularly for children and pregnant women.^{852,853} It is also inexpensive and widely available. It is excellent in detecting hydronephrosis and hydroureter.⁸⁵⁴ Renal ultrasound is suitable for most patients with radiolucent stones (e.g., UA). However, ultrasound may miss a significant fraction of ureteral stones⁸⁵⁴ and may give a false-positive diagnosis of obstruction in patients with pyelonephritis, vesicoureteric reflux, and residual dilation following relief of obstruction. Finally, sonography tends to overestimate the size of a stone as a result of the inaccurate determination of the stone and tissue boundary.

COMPUTED TOMOGRAPHY

NCCT, synonymous with unenhanced helical CT screening, has the highest sensitivity (94%–100%) and specificity (92%–99%)^{855,856} and can be considered the current gold standard for the diagnosis of kidney stones.⁸⁵⁷ Stones as small as 1 mm can be diagnosed by NCCT. The disadvantages of this technique include radiation exposure, limited ability to evaluate degree of obstruction, and high cost. Newer techniques have been adapted that offer lower exposure to radiation (from traditional 8 to 16 millisieverts [mSv] down to 0.5–2 mSv).⁸⁵⁸ If NCCT imaging can be performed effectively with lower dose radiation, it will remove the only concern with this test, which is increased risk of malignancy with long-term usage in repeated stone formers.^{859–861}

Another advantage of the NCCT scan is the ability to determine stone density using Hounsfield units (HU), which may have prognostic value in terms of success in shock wave therapy.⁸⁶² Similarly as a preoperative test (e.g., before percutaneous nephrolithotomy), NCCT can detect possible altered anatomy and accurately assess stone size and location, both of which have an impact on the selection of optimal surgical intervention.

INTRAVENOUS PYELOGRAPHY

Intravenous pyelography (IVP) has been the historic gold standard imaging approach to diagnose nephrolithiasis. It provides excellent anatomic detail of the minor and major calyces, infundibula, renal pelvis, and ureters, and a calculus is visualized as a discrete filling defect. The disadvantage is the use of nephrotoxic contrast agents. The sensitivity and specificity of KUB for detecting renal calculi have been reported as 59% and 71%, respectively.⁸⁶³ The use of NCCT has largely supplanted IVP.^{864–866} IVP is still occasionally used following NCCT in some patients to guide in percutaneous or endoureteral surgical procedures. It is also used in diabetic patients with renal colic mimicking kidney stones caused by papillary necrosis.⁸⁵⁰

MAGNETIC RESONANCE IMAGING

Magnetic resonance imaging (MRI) is a potential alternative to NCCT for the diagnosis of nephrolithiasis and urinary tract obstruction. The obvious advantage is that it does not deliver ionizing radiation. Akin to sonography, it is useful in pregnant women and in children and adolescents. A related technique, magnetic resonance urography, has been reported as a method for the diagnosis of urinary tract obstruction, specifically in pregnant women.⁸⁶⁷ The disadvantage of MRI is its high cost.

In general, patients with a known history of previous renal colic because of stone disease may be initially evaluated by KUB or ultrasound. NCCT can be used in patients who have not been previously diagnosed and in those with atypical clinical presentation. Ultrasound is considered the method of choice in pregnant patients and children with kidney stone disease. IVP should be considered following NCCT if no stone was diagnosed or when planning an endoscopic evaluation or open intervention to assist the urologist in mapping the urinary tract. In situations in which NCCT is not available, one may start with KUB and ultrasound evaluations and only consider IVP if these two techniques fail to diagnose kidney stones. In the follow-up of patients, because of the high risk of cumulative radiation, it is advisable to select KUB and ultrasound until newer, low-radiation NCCT has been developed.

DIGITAL TOMOSYNTHESIS

Digital tomosynthesis is a modern technique that may accurately detect radiopaque kidney stones with limited radiation (0.54 mSv).^{868,869} This technique is comparable with NCCT for the detection of renal stones, but is not affected by stone size and BMI of the patient and has superb reproducibility. It has an advantage over CT scans because it offers minimal radiation with a decreased cost.

MANAGEMENT OF STONES

ACUTE MANAGEMENT

As an acute illness, urolithiasis is associated with significant pain, disability, and loss of productivity.⁸⁷⁰ Approximately 50% of patients experiencing acute upper urinary tract stones require surgical intervention.^{871,872} Medical treatment has been shown to facilitate spontaneous passage of ureteral stones. Stone size and location in the urinary tract are the major determinants of the likelihood of spontaneous stone passage. Spontaneous passage rates are higher for distal ureteral calculi compared with proximal and middle locations. Rates are also higher for smaller (<4 mm in diameter) compared with larger stones that are between 4 to 6 mm and those larger than 6 mm in diameter.^{873,874} Overall, spontaneous passage rates are 12%, 22%, and 45% for proximal, middle, and distal ureteral calculi, respectively, and 55%, 35%, and 8% for stones smaller than 4 mm, 4 to 6 mm, and larger than 6 mm, respectively.⁸⁷³ Because acute renal colic is associated with severe pain over an unpredictable time interval, adjunctive treatment to promote spontaneous stone passage and reduce symptoms is crucially important.

MANAGEMENT OF RENAL COLIC

The management of renal colic is necessary until spontaneous passage of a stone, which usually occurs in patients with smaller (<4 mm) and more distal ureteral stones within 48 hours after the onset of acute renal colic.⁸⁷⁵ During that period, patient will require supportive treatment, comprised of pain medications, including nonsteroidal antiinflammatory drugs (NSAIDs), opioids, hydration, and antiemetics.⁸⁷⁶ Patients with resolving colic can be discharged from the emergency room but hospitalization is indicated in cases of persistent, relentless pain and intractable vomiting, established infection, and obstruction. Nephrostomy must be considered urgently to relieve the obstruction, which can be followed by ESWL, percutaneous lithotripsy, ureteroscopy with laser lithotripsy, retrograde basket extraction of the stone or, in rare events, open surgical removal.^{877,878}

Medical Expulsive Therapy

Medical expulsive therapy (MET) has been used as conservative management in the treatment of ureteral stones.^{879,880} It can potentially improve the patient's quality of life by reducing episodes of pain,^{881,882} minimizing analgesic use,⁸⁸³ and diminishing the stone transit time^{883,884} and might decrease the costs and complications of ESWL and ureteroscopy.^{870,885} Therapies that have been tested include glucocorticoids, hormones, NSAIDs, phosphodiesterase-5 inhibitors, calcium channel blockers, and α -adrenergic blockers.⁸⁸⁶⁻⁸⁸⁹ Most of the available studies are of α_1 -adrenergic receptor antagonists, which decrease ureteral smooth muscle tone and frequency and force of peristalsis.^{4,5} There are limited data on the efficacy of calcium channel blockers, and tamsulosin has now been shown to be superior to nifedipine.⁸⁹⁰ A recent Cochrane systematic review that included 67 studies and a total of 10,509 participants has found that alpha blockers increase stone clearance and shorten the stone expulsion time. In a subgroup analysis, alpha blockers were shown to be less effective for smaller stones (≤ 5 mm) than larger stones (> 5 mm), perhaps because smaller stones already have a high rate of spontaneous passage.⁸⁹¹ In this study, only stone size, but not stone location or type of alpha blocker, was shown to affect the results.

Pradere and associates have reviewed the most recent 2016 guidelines for the management of urolithiasis.⁸⁹² Both the American Urological Association (AUA) and European Association of Urology (EAU) recommend offering alpha blockers as expulsive therapy for distal ureteral stones. The EUA recommends MET for stones larger than 5 mm, whereas the AUA recommends it for stones smaller than 10 mm. The AUA suggests 4 to 6 weeks of observation with conservative treatment from the initial clinical presentation.

CHRONIC MANAGEMENT

LIFESTYLE AND DIETARY MANAGEMENT

One major limitation in the field has been the scarcity of RCTs to compare the effects of specific dietary and fluid measures in recurrent kidney stone formers.

Fluid Intake

Fluid management should be considered in all patients with kidney stones, regardless of stone composition. Dilution of lithogenic elements in the urine should be the panacea of

kidney stone therapy. The effect of fluid intake was first examined in 108 idiopathic stone formers over approximately 5 years, with significant declines in new stone formation. One limitation of this study was a lack of a control group.⁸⁹³ However, in an RCT in men with recurrent CaOx stones over 5 years, high fluid intake to ensure urinary volume of approximately 2.5 L/day reduced stone recurrence by approximately 44%, compared with a control group with no dietary restrictions.⁸⁹⁴ However, this study targeted first-time stone formers, and the results of this study are not necessarily applicable to the kidney stone-forming population with residual stones at a high-risk of kidney stone recurrence. In a 3-year RCT, 1009 kidney stone formers were randomized to a control group that continued drinking soft drinks and an intervention group that avoided consumption of soft drinks.⁸⁹⁵ This study showed a marginal benefit in avoiding soft drinks. Another study suggested that fluid intake of fruit juices, specifically orange juice, is also effective in reducing urinary CaOx saturation.⁸⁹⁶ The same effectiveness is not seen with apple juice, grapefruit juice, cola, or some sport drinks because of their elevated oxalate and fructose concentrations.⁸⁹⁶⁻⁸⁹⁹

Fluid therapy is also recommended for cystinuria. Cystine has a limited solubility of 243 mg/L in urine.⁹⁰⁰ Reduction of urine cystine supersaturation to less than 1.0, or concentration to less than 243 mg/L (1 mmol/L), would require drinking around 4 L/day, including nocturnal intake.⁵⁶⁷ Off-label use of vasopressin receptor antagonists has been proposed to induce pharmacologic diabetes insipidus,⁵⁶⁸ but the long-term effects of this drug has not been evaluated. In conjunction with the exorbitant cost and safe and easily available water, it is unclear whether there is an advantage to this therapy.

Dietary Adjustment

Calcium Intake. In general, high dietary calcium intake appears to be protective against CaOx stones. In one study of Italian men with recurrent CaOx stones, a “traditional prescription” of low calcium diet was compared against liberal calcium (1200 mg/day) but low dietary sodium (<100 mEq/day) and animal protein consumption (50–60 g/day).²⁷ Despite only partial adherence, the low-sodium and protein diet resulted in less recurrence of stones.

Three large epidemiologic studies using food frequency questionnaires and history of kidney stone passage have suggested that low calcium and low fluid intake,^{22,24–26} sugar-containing beverages,⁸⁹⁷ and high animal protein intake are risk factors for the development of first-time kidney stones^{22,24–26} among women and younger men.

Unlike dietary calcium, supplemental calcium may increase the risk of stone formation. This association was shown in an observational study of older women treated with calcium supplements compared with those who did not take supplements.²⁶ However, this association was not seen in younger men and women.²⁴ In a large epidemiologic study (Women’s Health Initiative), calcium supplementation increased the risk of CaOx stone formation; one should exercise some caution because the total calcium intake was very high in those using supplements.³²

Dietary Oxalate. High urinary oxalate is associated with the risk of nephrolithiasis,⁹⁰¹ and dietary oxalate restriction has

generally been recommended. In epidemiologic studies, however, it is difficult to demonstrate the relationship between dietary oxalate and urinary oxalate.⁶⁷⁹ Although there is no doubt that dietary oxalate is absorbed, the huge variation in intestinal absorption, bacterial degradation, and superimposed endogenous synthesis renders it difficult to discern a clear relationship between dietary and urinary oxalate. The Dietary Approaches to Stop Hypertension (DASH) diet, which was not restricted in oxalate content, significantly reduced the risk of kidney stones.⁹⁰² However, the protective effect of the DASH diet may be caused by the ingestion of a high fruit and vegetable content (alkali) in conjunction with low animal protein intake. In general, urinary oxalate excretion is affected by calcium intake, which may determine intestinal bioavailability of oxalate and consequently its absorption. With the recommended daily dietary calcium, CaOx stone risk has been shown not to be significantly influenced, even with relatively high dietary oxalate.³⁸ Therefore, it is prudent to recommend patients with hyperoxaluria and CaOx stone to consume a total calcium of 1000 to 1200 mg/day ingested with meals to complex intestinal luminal oxalate. The restriction of dietary oxalate in subjects with idiopathic hyperoxaluria may not be efficacious but likely will not increase stone risk. In contrast, in patients with inflammatory bowel disease or who have undergone bariatric surgery, dietary oxalate restriction should be imposed in addition to higher total calcium intake from the diet and supplements at mealtime.^{130,903,904}

Animal Protein. In a multicomponent dietary intervention study, it was shown that a combination of low protein (50–60 g/day) and low sodium consumption (≤100 mEq/day) associated with normal calcium intake (1200 mg/day) significantly lowered the risk of kidney stone recurrence in Italian males with hypercalciuria.²⁷ However, it was argued that the protective effects may not be related to low protein consumption but was because of the consumption of low-salt and normal calcium intake. The result of this study was not supported in two other studies using a low-protein and high-fiber diet³⁹; another study showed no difference in stone occurrence between low animal protein intake (<three servings of fish or meat/week and dairy products <100 g/day) compared with a controlled diet.⁹⁰⁵

Ascorbic Acid. Vitamin C above the physiologic dose may increase urinary oxalate. In a metabolic study, 2 g of ascorbic acid elevated the 24-hour urinary oxalate (29–35 mg in non-stone formers; 31–41 mg in stone formers). Therefore, patients with hyperoxaluria should be cautious about OTC vitamin C supplements.^{906,907} Fortunately, most preparations are less than 1 g/day.

Dietary Intervention in Cystinuria. Cystine excretion has been shown to be reduced by decreasing dietary sodium, although no clinical outcome studies have been performed.^{567,569,908–910} The mechanism for this is not known but salt regulation of the b^{0,+} system for the purpose of L-DOPA uptake and synthesis of nephrogenic dopamine is a possibility.⁵⁷⁰ On a high-salt diet, we suggest that the b^{0,+} system may be occupied with dopamine synthesis and hence unavailable for cystine reabsorption. In the absence of ill effects of salt restriction, it is reasonable therefore to recommend restricting salt intake to about 2 g/day.

Reduction of animal protein intake has also been proposed to be beneficial because it would reduce dietary intake of cystine and its precursor methionine, which might reduce cystine excretion,⁵⁷¹ and would also increase the solubility of urinary cystine by increasing urinary pH. One study has shown that urinary cystine excretion decreases in cystinuric patients on a low-protein diet, compared with a total of 9% of caloric intake from protein compared with higher protein intake.⁵⁷¹ Restriction of protein must be used cautiously in children because the restriction of essential amino acids may compromise growth in children.

Other Dietary Interventions. It has been suggested that lowering urinary phosphorus, magnesium, and ammonium levels via dietary manipulation will lower the risk of struvite stones. There are limited data in humans exploring the efficacy of dietary modification in this population. Low phosphorus and calcium, combined with aluminum hydroxide gel, has been proposed,⁹¹¹ supposedly to lower urinary phosphorus excretion by limiting the intake and absorption of phosphorus. Treatment with estrogen over 3.5 years was proposed to diminish urinary calcium excretion mediated by decreasing bone resorption.⁹¹² However, these interventions were also associated with side effects, including GI distress, lethargy, bone pain, and hypercalciuria.

PHARMACOLOGIC TREATMENT

Pharmacologic treatment is essential in almost all recurrent kidney stone formers and in the management of patients with UA, cystine, recurrent calcium, and infection-induced stones, principally as a result of the lack of availability and/or consensus regarding the effectiveness of nonpharmacologic interventions.⁹¹³ One major limitation with respect to pharmacologic treatment is the paucity of randomized controlled data with stone episodes as hard evidence of treatment results.⁸⁷⁴ This is largely because stone events are rare, and such studies must be carried out in a large number of patients over a long period of time. There is also no consensus as to whether pharmacologic treatment should be targeted at specific metabolic abnormalities or should be given empirically, irrespective of underlying biochemical abnormalities.^{147,914-917} Although the benefit of directed medical treatment has not been decisively proven, some observational studies have suggested the beneficial effects of directed management of kidney stones.^{918,919}

PHARMACOLOGIC AGENTS USED

Thiazide Diuretics

Thiazide diuretics and their analogues are used commonly for lowering calcium excretion in hypercalciuric, recurrent, calcium stone formers.⁸⁷⁴ To date, six RCTs have studied thiazides in recurrent calcium stone formers. One study has shown a similar decrease in stone formation among thiazide-treated versus untreated patients.⁹²⁰ Another study has shown diminished hypercalciuria without any change in stone events in patients treated with hydrochlorothiazide compared with placebo.⁹²¹ The remaining four RCTs, which evaluated a total of 408 patients over durations of 26 to 36 months, have demonstrated significant reductions in recurrent kidney stones with thiazides and the thiazide analogue indap-

amide^{916,917,922-924} (Table 38.12). The results of these RCTs are consistent with uncontrolled studies totaling over 6600 patient-years of thiazide treatment for calcium nephrolithiasis patients.^{363,500,924-928,939} Two critical but often ignored facts of thiazide therapy deserve emphasis. First, the optimal effect of thiazide diuretics is attained with a low-salt diet. Although there is likely some direct effect of thiazides on the DCT, most of its effect is to induce slight extracellular volume contraction and increase proximal calcium reabsorption; thus, simultaneous ingestion of a high-salt diet will nullify the efficacy of thiazides.^{940,941} Second, the successful reduction in urinary calcium can be offset by hypocitraturia because of potassium depletion and proximal tubule intracellular acidosis. One must be vigilant in detecting potassium deficiency, and potassium supplements should be prescribed to avoid hypocitraturia if dietary potassium intake is not sufficient.³⁹⁷ Potassium citrate may provide an advantage over potassium chloride in these cases.⁹⁴²

Thiazide treatment, along with dietary sodium restriction to maximize the hypocalciuric effect of thiazide, is the treatment of choice in hypercalciuric calcium stone-forming subjects. The incidence of side effects on thiazide diuretic may approach 30%⁹²⁴ but side effects requiring discontinuation of the drug are rare. To date, long-term side effects of thiazide treatment in the kidney stone population have not yet been documented. A meta-analysis of clinical trials in hypertensive subjects has demonstrated a relationship between changes in serum glucose and potassium concentrations.⁹⁴³ This was also confirmed in a large cohort, the Antihypertensive and Lipid Lowering Treatment to Prevent Heart Attack Trial (ALLHAT), in which the incidence of diabetes mellitus was higher with chlorthalidone compared with amlodipine or lisinopril over 4 years. The exact pathophysiologic mechanism has not yet been fully uncovered.⁹⁴⁴⁻⁹⁴⁶ One potential mechanism is the relationship between potassium deficiency and defective insulin secretion and action.^{947,948} The list of commonly used drugs and recommended dosages in the treatment of hypercalciuric calcium nephrolithiasis with potential side effects is summarized in Table 38.13.

Alkali Treatment

Alkali treatment can be used alone or in combination with thiazides in recurrent calcium or uric acid stone formers. There have been four RCTs^{147,153,931,949}; two of these trials targeted patients after ESWL for residual stone fragment growth or new stone formation. In three nonrandomized, non-placebo-controlled studies, one retrospective study, and one non-RCT, potassium citrate reduced the risk of clinical stone events (see Table 38.12).

One study has shown no difference in stone formation.¹⁵⁴ This study included a total of 50 hypocitraturic calcium nephrolithiasis patients treated with 90 mEq of potassium alkali/day over 3 years. One possibility for the lack of efficacy may have been because of the small size.

In three nonrandomized, non-placebo-controlled studies, alkali treatment showed a significant decrease in new stone events.^{157,158,929} One retrospective study comparing three groups of patients with medullary sponge kidney (MSK) received potassium citrate or no treatment. The results showed that treatment with potassium citrate was effective in reducing renal stones compared with nontreated groups.⁹³⁰ In one nonrandomized but placebo-controlled study, 503 subjects

Table 38.12 Major Clinical Pharmacotherapeutic Trials of Calcium and Noncalcium Nephrolithiasis

Trial	Reference	Treatment	No. of Subjects	Design	Outcome
Thiazide diuretics	Laerum ⁹¹⁷	Hydrochlorothiazide vs. placebo	50	RCT	Decreased new stone formation and prolonged stone-free interval
	Ettinger ⁹¹⁶	Chlorthalidone vs. magnesium hydroxide vs. placebo	124	RCT	Chlorthalidone more effective than magnesium hydroxide or placebo in reducing stone events
	Ohkawa ⁹²²	Trichlormethiazide vs. no treatment	175	RCT	Decreased calciuria and stone formation rate
	Borghi ⁹²³	Diet vs. diet + indapamide vs. diet + indapamide + allopurinol	75	RCT	Diet + pharmacotherapy better than diet alone
	Yendt ⁹²⁴	Hydrodiuril	33	NNT	Decreased number of stone events or invasive and noninvasive procedures
	Coe ⁵⁰⁰	Trichlormethiazide	37	NNT	Decreased new stone formation
	Coe ³⁶³	Trichlormethiazide vs. allopurinol vs. both	222	NNT	Decreased new stone formation
	Yendt ⁹²⁵	Hydrochlorothiazide	139	NNT	Decreased new stone formation or stone growth
	Backman ⁹²⁶	Bendroflumethiazide	44	NNT	Decreased new stone formation
	Maschio ⁹²⁷	Hydrochlorothiazide + amelioride vs. both + allopurinol	519	NNT	Decreased new stone formation
Alkali treatment	Pak ⁹²⁸	Hydrochlorothiazide	37	NNT	Decreased new stone formation
	Pak ¹⁵⁷	Potassium citrate vs. pretreatment in calcium and uric acid stone formers	89	NNT	Decreased stone events
	Preminger ¹⁵⁸	Potassium citrate	9	NNT	Decreased new stone formation
	Pak ⁹²⁹	Potassium citrate	18	NNT	Decreased stone events
	Fabris ⁹³⁰	Potassium citrate vs. no treatment	65	NNT	Decreased stone rate in those treated with potassium citrate
	Barcelo ¹⁵³	Potassium citrate vs. placebo	57	RCT	Decreased new stone formation and increased urinary citrate
	Hofbauer ¹⁵⁴	Diet + sodium potassium citrate vs. diet	50	RCT	No difference in stone formation
	Ettinger ¹⁴⁷	Potassium magnesium citrate vs. placebo	64	RCT	Decreased new stone formation
	Soygur ⁹³¹	Potassium citrate vs. no treatment post-shock wave lithotripsy	110	RCT	Decreased stone recurrence
	Kang ⁹¹⁹	Mix of potassium citrate, thiazide, allopurinol vs. no treatment after percutaneous nephrolithotomy	226	NCT	Decreased stone recurrence
Allopurinol treatment	Ettinger ⁹³²	Allopurinol vs. placebo	60	RCT	Decreased stone events
	Coe ³⁶³	Thiazide vs. allopurinol vs. both	202	RCT	Decreased stone events vs. pretreatment
Febuxostat	Goldfarb ⁵⁶⁷	Febuxostat vs. allopurinol or placebo	99	RCT	Febuxostat decreased 24-hour urinary uric acid excretion more significantly than allopurinol; no change in stone size or number
Other treatment	Dahlberg ⁹³³	D-Penicillamine	89	R	Decreased stone event and dissolution of stones
	Pak ⁹⁰⁰	D-Penicillamine or α -mercaptopyrionylglycine vs. conservative medical therapy	66	R	Both drugs equally effective in reducing stone events
	Chow ⁹³⁴	D-Penicillamine or α -mercaptopyrionylglycine vs. conservative medical therapy	16	NNT	Decreased stone event
	Barbey ⁹³⁵	D-Penicillamine or α -mercaptopyrionylglycine vs. conservative medical therapy	27	R	Decreased stone events
	Williams ⁹³⁶	Acetohydroxamic acid vs. placebo	18	RCT	Decreased stone size
	Griffith ⁹³⁷	Acetohydroxamic acid vs. placebo	210	RCT	Decreased stone growth
	Griffith ⁹³⁸	Acetohydroxamic acid vs. placebo	94	RCT	Decreased stone growth

N, Number of patients; NCT, nonrandomized controlled trial; NNT, nonrandomized, non-placebo-controlled trial; R, retrospective; RCT, randomized controlled trial.

Modified from Sakhae K, Maalouf NM, Sinnott B. Clinical review. Kidney stones: 2012: pathogenesis, diagnosis, and management. *J Clin Endocrinol Metab.* 2012;97(6):1847–1860.

Table 38.13 Commonly Used Drugs in the Treatment of Hypercalciuric Calcium Nephrolithiasis

Drug	Recommended Dosage(s)	Comments
Hydrochlorothiazide	50 mg/day, 25 mg bid	A single dose is preferred because twice-daily dosage may cause nocturia, discomfort, and noncompliance.
Chlorthalidone	25 mg/day, 50 mg/day	Both dosages lower urinary calcium by the same degree; long-acting; may cause hypokalemia and secondary hypocitraturia
Indapamide	1.2 mg/day, 2.5 mg/day	This treatment may have fewer side effects than hydrochlorothiazide, including lower incidence of hypokalemia and hypotension.
Amiloride	5 mg/day	Potassium sparing; lowers urinary calcium but to a lesser degree than hydrochlorothiazide
Amiloride-hydrochlorothiazide	5 mg-50 mg/day	Maintains the hypocalciuric effect of thiazide while averting the development of severe hypokalemia
Trichlormethiazide	2 mg/day, 4 mg/day	Not marketed in the United States

were treated with a mixture of potassium citrate, thiazide, and allopurinol, compared with no treatment after percutaneous nephrolithotomy, for a mean duration of 41 months; there was a decreased stone formation rate, from 1.89 to 0.46 stones/year.⁹¹⁹ One caveat in these studies was that alkali treatment that was given to patients with urinary citrate excretion ranged from normal to low-normal and low urinary citrate.^{147,153,154}

Alkali treatment is relatively safe, with minor GI side effects, although certain susceptible individuals can have considerable gastric distress. The side effects have to be weighed against its efficacy in recurrent calcium oxalate stones,^{147,153,154} and uric acid stones,^{499,929} as well as in patients with residual stones after shock wave lithotripsy.⁹³¹ One primary potential side effect that has been gaining attention is the risk of CaP stone formation. A rise of the urinary pH above 6.7 favors the generation of monohydrogen phosphate. Despite this concern, in two nonrandomized, non-placebo-controlled trials^{158,930} in patients with dRTA and MSK with preexisting high UpH, alkali therapy was shown to be effective in reducing recurrent calcium stone formation. Larger scale prospective RCTs are still necessary to determine the effects of alkali treatment on CaP stone incidence.

Although sodium bicarbonate may offer the same degree of urinary alkalinization when used in an equivalent dose to potassium alkali, it may increase the risk of calcium stone formation because of sodium-induced hypercalciuria and promotion of monosodium urate-induced CaOx crystallization.^{499,929} The initial recommended dose for alkali is 30 to 40 mEq/day. In practice, 24-hour urine will be measured to follow the increase in citraturia and titrate the alkaline dose. The goal is to maintain the UpH higher than 6.1 and less than 7 to avoid the potential complication of calcium stone formation. A 24-hour UpH, however, may not reflect diurnal variation in UpH and periods of high urinary acidity.⁹⁵⁰ Therefore, higher doses of alkali treatment administered at night may ameliorate abnormally acidic urine in those with recurrent uric acid kidney stones.⁹⁵⁰ Because unduly acidic urine is the predominant feature of patients with UA nephrolithiasis, alkali therapy with potassium citrate is first-line treatment in these patients.⁴⁷ Treatment with allopurinol can be considered when there is very high urinary UA levels, and alkali treatment is unsuccessful in patients with inflammatory bowel disease, bowel resection, and recurrent UA stones, despite adequate urinary alkalinization.

Alkalinization is also helpful in cystinuria because cystine solubility increases with increasing pH. Generally, it is recommended that alkalinization should be conducted conservatively, not to exceed a pH of 6.5 to 6.7, because highly alkaline urine may increase the risk of CaP stone formation. Defective acidification and high UpH was previously described in patients with mixed cystine and calcium stones.⁹⁵¹ It is often difficult to determine the optimal dose of alkali to be administered to each individual patient.^{565,952} Therefore, it is advisable to monitor patient responses with direct measurements of urinary supersaturation with respect to cystine.⁵⁶⁷ Both potassium citrate and sodium bicarbonate have been shown to be equally effective in alkalinizing urine in this population.⁹⁵³ Sodium alkali can be used in patients with renal insufficiency to avoid hyperkalemia. The dosage of alkali is typically divided equally three or four times daily.

Combined Pharmacologic Treatment

The effects of combined thiazide and alkali treatment were shown not be significantly different from thiazide monotherapy in reducing the risk of kidney stone recurrence in one poor-quality study.⁹⁵⁴ Furthermore, stone recurrence was not shown to be different between thiazide combined with an allopurinol versus thiazide treatment alone.⁹²³

Xanthine Oxidase Inhibitors

Allopurinol is used for the treatment of hyperuricosuric calcium stone formers rather than uric acid stone formers. One RCT in hyperuricosuric calcium stone formers, comparing allopurinol versus placebo,⁹³² has demonstrated significantly decreased stone events with allopurinol treatment. Another RCT in recurrent hyperuricosuric calcium stone formers, comparing thiazide versus allopurinol and combined thiazide and allopurinol, showed significantly reduced stone events with the combination treatment³⁶³ (see Table 38.12). The efficacy of allopurinol alone in the treatment of hyperuricosuric CaOx stone-forming patients with multiple metabolic abnormalities is less evident.⁹⁵⁵ The effective dose is 300 mg/day, which can either be administered as a single dose or divided into three equal doses throughout the day. Side effects of allopurinol are reported as uncommon, including a skin rash in 2% of treated patients and more severe but even rarer life-threatening hypersensitivity reactions, including acute interstitial nephritis and Steven-Johnson

syndrome.⁹⁵⁶ Adjustments are needed in patients with impaired kidney function because allopurinol is primarily excreted by the kidney.

A nonpurine xanthine oxidase inhibitor analogue, febuxostat, has been approved for the treatment of hyperuricemia associated with gouty arthritis. In a retrospective study, the use of febuxostat in patients with gout with allergies to allopurinol was shown to be a safe alternative.⁹⁵⁷ In a 6-month, double-blinded RCT, hyperuricosuric calcium stone formers with one or more calcium stones detected by CT were treated with 80 mg/day of febuxostat, 300 mg/day of allopurinol, or placebo. Febuxostat reduced 24-hour urinary UA significantly more than allopurinol but there was no change in stone size or number over this period of time.⁹⁵⁸ Therefore, no conclusion comparing the drugs can be drawn until a longer study is conducted to determine the incidence of stone events (see Table 38.12). One advantage of febuxostat is that it is principally metabolized by the liver, thereby rendering dose adjustments in those with impaired kidney function unnecessary.⁹⁵⁹

Magnesium Treatment

In only one study in calcium stone formers, treatment with magnesium lowered stone recurrence but not significantly compared with placebo.⁹¹⁶

Cystine Chelation Therapy

The treatment of cystine stones can be challenging. Conservative management with hydration and urinary alkalization are the first steps in the management of a cystinuric patient but, more often than not, they do not suffice. Thiol derivatives that act as chelating agents are commonly used in patients with severe cystinuria (>1000 mg/day). These drugs reduce a single cystine molecule into two cysteines and form a highly soluble disulfide compound of the drug with the cysteine molecules.⁹⁶⁰ The effect of a disulfide compound (D-penicillamine) on stone events was first described by Dahlberg and coworkers.⁹³³ Subsequently, α -mercaptopyrionylglycine (tiopronin) was marketed for use in cystinuric patients. In two studies, pharmacologic treatment with D-penicillamine or tiopronin was shown to be superior to hydration and alkalization alone in reducing stone events^{934,935} (see Table 38.12). However, to date, no RCTs have demonstrated the superiority of pharmacologic treatment over placebo in cystinuric patients.

A few studies have suggested that captopril, which is also a thiol derivative, might also be effective in reducing urinary cystine excretion,^{961,962} but its efficacy has not been confirmed by others.^{935,963} The potential role of inhibitors of cystine crystal growth has been explored. The most effective inhibitors tested were L-cystine dimethyl ester (L-CDME) and L-cystine methyl ester (L-CME). However, this effectiveness has only been tested in experimental models and not yet in cystinuric populations.⁹⁶⁴

Tiopronin has a lower incidence of side effects compared with D-penicillamine, and it is therefore used more often.⁹⁰⁰ The side effects include GI irritation, abnormalities in hepatic enzymes, pancytopenias, skin irritation and disorders, and proteinuria.⁹⁰⁰ Effective therapy should be aimed at maintaining a urinary cystine concentration at 250 mg/L, although there is variability among individual patients. Therefore, assessment of urinary supersaturation with respect to cystine solubility is important. However, to date, there has been no

clinical study exploring whether such measurement will avert the complications of cystine stone formation.^{565,567,952} One important caveat is that urinary cystine measurement does not differentiate between a drug-cystine complex and unbound cystine. Furthermore, urinary measurement of cystine may not change with drug treatment, which may lead to misguided attempts to increase the dose of thiol drugs to lower urinary cystine levels. Therefore, it is advisable to monitor the patient's response to treatment via urinary supersaturation, which may guide dosing of chelating agents better.

PHARMACOTHERAPY OF INFECTION-RELATED STONES

According to the guidelines of the AUA, surgical treatment is still the best management for infectious stones (see later). However, medical treatment can be used in those with significant comorbidities who are not amenable to surgery and also to prevent stone recurrence after successful stone removal. These treatments include antibiotics, dissolution therapy, urease inhibitors, urinary acidification, dietary modification, and other supportive measures.

Antibiotics

Antibiotics are important during both pre- and perioperative periods to prevent urinary sepsis from surgical procedures. Antibiotics should be directed against specific microorganisms established by positive urine cultures.

Dissolution Therapy

Boric acid, permanganate, and other solutions have been instilled into kidneys to dissolve infectious stones.⁹⁶⁵ However, dissolution treatment has fallen out of favor because of its cost, duration of hospitalization and, most importantly, significant associated risks. The increased mortality following internal irrigation led to a ban in the United States by the U.S. Food and Drug Association (FDA) on the use of this treatment approach.

Urease Inhibitors

Urease-producing microorganisms, by converting urea to ammonia, play a key role in supersaturation of urine with respect to ammonium magnesium phosphate (struvite), which usually coexists with calcium carbonate apatite stones, because both stone types are favored in an alkaline environment.⁹⁶⁶ Although inhibition of bacterial urease has been shown to retard stone growth and prevent new stone formation, it cannot eradicate existing stones nor the underlying infection. However, when combined with antimicrobial therapy, urease inhibition provides palliation for patients who cannot undergo definitive surgical management.^{575,967}

The urease inhibitor acetohydroxamic acid (AHA) is the only FDA-approved urease inhibitor that can be used orally to inhibit urease enzyme activity.^{937,938} This drug is cleared by the kidney and can penetrate bacterial cell walls. It also functions synergistically with antibiotics. Three RCTs have shown significant reduction in stone growth with AHA compared with placebo.⁹³⁶⁻⁹³⁸ AHA can confer serious GI, neurologic, hematologic, and dermatologic side effects in 20% of patients, but these tend to resolve on discontinuation.⁹³⁶⁻⁹³⁸ The starting dose of AHA is 250 mg by mouth twice a day. AHA is also contraindicated in patients with chronic kidney disease and serum creatinine concentrations higher than

2.5 mg/dL, which increase the risk of toxicity and poor urinary concentration.^{937,938}

Urinary Acidification

Because urine alkalinity (pH >7.2) promotes the formation of struvite and carbonate apatite stones, urinary acidification has been used to control stone burden in this population. L-Methionine, which imposes an acid load (H₂SO₄), can lower UpH. An in vitro simulation of decreasing UpH from 6.5 to 5.7 increased the dissolution rate of struvite stones.⁹⁶⁸ It was suggested that oral intake of L-methionine, 1500 to 3000 mg/day, can be used in humans for the dissolution of infectious stones, although there has been no trial to prove this yet.⁹⁶⁸

SURGICAL MANAGEMENT OF INFECTION-RELATED STONES

Struvite staghorn calculi generally require surgical removal and, if not properly treated, may require nephrectomy.⁹⁶⁹ The goal is eradication of stone burden and prevention of stone regrowth, renal damage, and persistent infection. A significant number of patients treated with open surgical stone removal have a recurrence following surgery because of recurrent urinary tract infections. Rather than open surgical stone removal, percutaneous nephrolithotomy can completely remove struvite stones 90% of the time,^{970,971} with a recurrence rate approaching only 10% in kidneys rendered stone-free.⁹⁷² A retrospective analysis of 43 patients with pure or mixed struvite stones from Duke University Medical Center from 2005 to 2012 has found stone recurrence in 23% of patients, with stable renal function over a median follow-up of 22 months (range, 6–67 months). ESWL with ureteral stenting alone will result in stone-free rates of 50% to 75%.⁹⁷³ Retrograde ureteroscopy with holmium:YAG laser stone disruption can fragment almost all minor staghorn stones, with a recurrence rate of 60% at 6 months.⁹⁷⁴ The AUA suggests a combined approach of percutaneous nephrolithotomy and shock wave lithotripsy.

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 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

1. A 27-year-old female referred for the management of kidney stones. She passed several stones and required 5 shock wave lithotripsy bilaterally for the last 6 years. She has no other past history or family history of illness. She denies any specific medical treatment for stones. The stone analysis showed 85% apatite and 15% brushite. Twenty-four hour urine showed volume 2.05 L, pH 6.72, calcium 255 mg/day (normal <200 mg/day), sodium 177 mEq/day, potassium 48 meq/day, magnesium 117 mg/day, phosphorus 1292 mg/day, uric acid 691 mg/day (normal <700 mg/day), oxalate 31 mg/day (normal <40 mg/day), citrate 564 mg/day (normal >320 mg/day), ammonium 42 mEq/day (normal <40 meq/day) and sulfate 46 mEq/day (normal <40 mEq/day). Urinary creatinine was appropriate for her body weight. The best treatment at this point will be

- Thiazide and alkali
- Thiazide and potassium chloride
- Oral citrate acid
- a & b
- a & c

Answer: b

2. A 30-year-old woman has history of nephrolithiasis since the age of 18. She admits to multiple surgical procedures including shock wave lithotripsy and percutaneous nephrolithotomy. She denies any history of gout and primary hyperparathyroidism. Family history is positive for kidney stones in her father. Stone analysis showed pure calcium phosphate. Laboratory evaluation showed serum sodium 137 mEq/L, potassium 4.5 mEq/L, chloride 110 mEq/L, total CO₂ content 20 mEq/L, serum total calcium of 9.8 mg/dL, phosphorus 3.4 mg/dL, and PTH of 67 pg/mL (normal 10–65). Twenty-four hour urine showed total volume 3.1 L, pH 6.90 (normal <6.02), calcium 300 mg/day (normal <200 mg/day), sodium 160 mEq/day (normal <200), oxalate 25 mg/day (normal <40 mg/day), citrate 20 mg/day (normal >320 mg/day), sulfate 30 mEq/day (normal <40 mmol/day). The most likely cause of kidney stones is

- Idiopathic hypercalcuria
- Chronic diarrhea
- Proximal renal tubular acidosis
- Distal renal tubular acidosis

Answer: d

To fully establish the diagnosis in the above case one should perform

- Urinalysis and urine culture
- Ammonium chloride load showing no change in serum CO₂ content and urinary pH
- Ammonium chloride load showing fall in serum CO₂ content without any change in urinary pH
- None of the above

Answer: c

The most appropriate initial treatment for the patient will be

- Thiazide diuretics
- Potassium chloride
- Potassium citrate
- Sodium citrate

Answer: c

3. A 22-year old woman spontaneously passed stones on six occasions and underwent three shock wave lithotripsies in last 6 years. Due to an overweight problem, she has been following a specific dietary regimen. Her diet history includes limited dairy products and low carbohydrates but includes a heavy intake of red meat, chicken, and fish. Stone analysis revealed pure calcium oxalate stones. Laboratory examination showed serum sodium 139 mEq/L, serum potassium 4.0 mEq/L, serum chloride 108 mEq/L, total CO₂ content 25 mEq/L, total serum calcium 9.7 mg/dL, phosphorus 2.4 mg/dL, PTH 50 pg/mL (normal 10–65). Twenty-four hour urine showed total volume 2.7 L, pH 5.90 (normal <6.02), sodium 130 mEq/L, calcium 457 mg/day (normal <200 mg/day), citrate 218 mg/day (normal >320 mg/day), sulfate 70 mEq/day (normal <40 mEq/day), ammonium 66 mmol/day (normal <35 mmol/day).

The most likely diagnosis is

- Distal RTA
- Chronic diarrhea state
- Excess dietary salt intake
- Excess protein intake

Answer: d

The first line initial treatment for the patient will be

- Thiazide diuretics
- Dietary salt restriction
- Potassium citrate
- Restriction in animal protein diet

Answer: d

4. A 45-year-old male presented with recurrent calcium oxalate stones since age 14. He denies history of bowel disease or any intestinal symptoms. His 24-hour urine results showed total volume of 0.85 L, pH 5.25 (normal <6.02), calcium 50 mg/day (normal <200 mg/day), oxalate 200 mg/day (normal <40 mg/day), citrate 15 mg/day (normal >320 mg/day), sodium 65 mEq/day (normal <200 mEq/day), uric acid 510 mg/day (normal <800 mg/day).

The most likely diagnosis at this point is

- High dietary oxalate intake
- Chronic diarrhea and fat malabsorption
- High protein consumption
- Primary hyperoxaluria

Answer: d

What is the first diagnostic test?

- Restrict dietary oxalate intake and repeat 24-hour urine for biochemical stone risk
- Restrict animal protein intake and repeat 24-hour urine for biochemical stone risk
- Quantify stool fat content
- Analyze 24-hour urine for glycolate and L-glycerate
- Genetic testing for glyoxylate and hydroxypyruvate reductase mutations

Answer: d

Which one is the least?

- Dietary oxalate restriction
- Potassium alkali treatment
- High fluid intake
- Vitamin B6

Answer: a